CRIMSON Power Management IC

A Major Qualifying Project (MQP) Report
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Abstract

This Major Qualifying Project provided integrated circuit design experience through the development of a Power Management Integrated Circuit (PMIC) with on-chip programmability, monitoring and control. The PMIC uses various power management methods with external powering such as a Micro-Solar Cell Manager and Linear Battery Charger in combination with a Boost Converter. For the rest of the chip, the LDO, BGR and Bias Network provide internal power. To monitor, the chip will measure current and temperature with an 8-bit SAR ADC filtered by an active Sallen Key LPF and sampled by a 32kHz RTC. These readings will be controlled by a Watchdog and FSM to determine whether the PMIC is consuming excessive amounts of power. To encode each programmable section, a SPI Chain will be implemented to send Serial Data from an FPGA.

We created the Conversion and Regulation Integrated Management SolutiON (CRIMSON), which is an intelligent PMIC design with full user configurability. This project provided WPI seniors the opportunity to work with state-of-the-art design principles and to gain competitive analog IC design experience. This design will be manufactured by TSMC in June 2026 using the TSMC 180nm CMOS 3v3/1v8, 1P6M process.

Acknowledgements

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1 Introduction

The Conversion and Regulation Integrated Management SolutiON (CRIMSON) is a power management IC developed for two main purposes. The first is to provide power for other CHIPS Design Studio projects, such as the Secure CMOS Image Sensor developed by Hubert Liu, Francesco Marrocco, Kelu Liu, Yuran Xue, and Philip Miu in 2025 [13]. The second is to be as testable and configurable as possible, for use in a future WPI electronics testing course.

Both goals were met through this project, resulting in a highly configurable chip that can be used to process, store, and regulate power while taking all computational load of monitoring or controlling power off the sink. This report serves as both an explanation of the work done on the chip and documentation for future WPI chip designers.

Sections 2 provides a dive into the background research performed before determining the direction we wanted to take with CRIMSON. It is evident that there are many PMICs available on the market, but not many that achieve the level of flexibility that we have with CRIMSON.

Section 3 provides a comprehensive overview of the CRIMSON system. There is a primary use case, coined the CRIMSON Intended Usage (CIU), that is broken into two sub-architectures: Power and Monitoring. These sub-architectures compose the top-level CRIMSON architecture. One level below the sub-architectures is the Configurability (CFG) Layer, which is a series of programmable interconnects with targeted input/output ports to allow for full user customization.

Section 4 discusses the design options that were considered for each block. It presents the background research each designer completed before determining their final circuit architecture.

Section 5 discusses the architectural integration, design, and results of each block. This section is critical for readers interested in learning about the core functionality of each block.

Finally, Section 6 discusses social considerations, ethical and professional responsibilities, and engineering standards that were used throughout CRIMSON's design.

2 Literature Review

This chapter explores the prior chip-level research completed to assist us in defining CRIMSON's use case, architectural implementation, and unique capabilities. We surveyed market trends and design articles from major semiconductor manufacturers.

2.1 Trends in Power Management for IoT and Edge Systems

Smart, wireless IoT devices permeate all aspects of our daily lives and are crucial in aiding our interactions with the physical environment. They can be found in applications ranging from activity monitoring, home security, and entertainment devices all the way to large scale environmental or utility monitoring [8]. Their increasing use requires them to evolve to meet different challenges. An important one being how they are powered.

Trends have emerged in the powering of IoT and edge systems. One of these is the use of PMICs. Integrating power management on a single chip provides the advantage of being small size and lightweight. Low voltage operation and local processing in PMICs also increase energy efficiency. Standard PMICs often include sensor interfaces, energy conversion, processing, and communication [16].

Another is using LSB-first SAR ADC for low activity signals. Voltage and current signals are used to monitor crucial nodes and must be converted to the digital domain. Using a LSB-first SAR ADC architecture results in reduced power consumption from a reduction in switching [16].

IoT and edge systems can also be self-powered through the incorporation of an energy harvest system. Common implementations include harvesting light energy through solar cells, thermoelectric energy harvesting through thermoelectric devices, and mechanical vibration harvesting through piezoelectric power generators [16].

2.2 Current Commercial PMIC Devices

Similar devices to the PMIC we designed currently exist on the market. These include the Texas Instruments bq25120A battery management IC and the Nordic Semiconductor nPM1100 power management IC.

The bq25120A targets wearable and IoT applications. It contains a linear LiPo battery charger, buck converter, and out LDO. It can charge batteries at a programmable charge current between 5 mA

and 300 mA and up to a programmable voltage between 3.6 V and 4.65 V. The buck converter has a programmable output range between 1.1 V and 3.3 V with a maximum output current of 300 mA. The output LDO has a programmable voltage output between 0.8 V and 3.3 V with a maximum output current of 100 mA [4].

The nPM1100 targets a general embedded systems application. It contains a LiPo battery charger and buck converter. The battery charger has programmable output current between 20 mA and 400 mA up to a voltage between 4.1 V and 4.35 V. The buck converter has a programmable output voltage between 1.8 V and 3.0 V with a maximum output current of 150 mA [15].

3 System Architecture

The Conversion and Regulation Integrated Management SolutiON (CRIMSON) is a power management integrated circuit (PMIC) designed for edge and IoT applications. One of the major challenges of edge/IoT devices is the ability to perform operations in real-time with limited computing resources. Power management is critical to the reliability of these devices and requires active monitoring, which consumes a significant portion of the already limited computing power. CRIMSON addresses this challenge by actively monitoring its own power nodes and performing intelligent power delivery to other systems within the device. It benefits from its ability to harvest solar power and provide programmable power outputs, while powering itself at the same time. CRIMSON aims to alleviate the complexity of power management by providing a self-contained power package to future developments within the edge/IoT device environment. The architecture was designed using several PMIC design concepts from [7].

3.1 Applications

CRIMSON has a vast range of applications. We have designed it for a specific use-case that harvests solar power to replenish a rechargeable battery, which provides power for the whole device through several programmable outputs. For use in edge devices, an example application is defined in Figure 1 where a bird tracking device is implemented using CRIMSON, a solar cell, battery, microcontroller, and RF module for wireless communication. Figure 2 depicts the critical aspects of the CRIMSON architecture with the necessary off-chip components and connections for our intended use-case. CRIMSON's powering capabilities are detailed in Power Architecture. CRIMSON's self-monitoring capabilities are enabled by Monitoring Architecture.

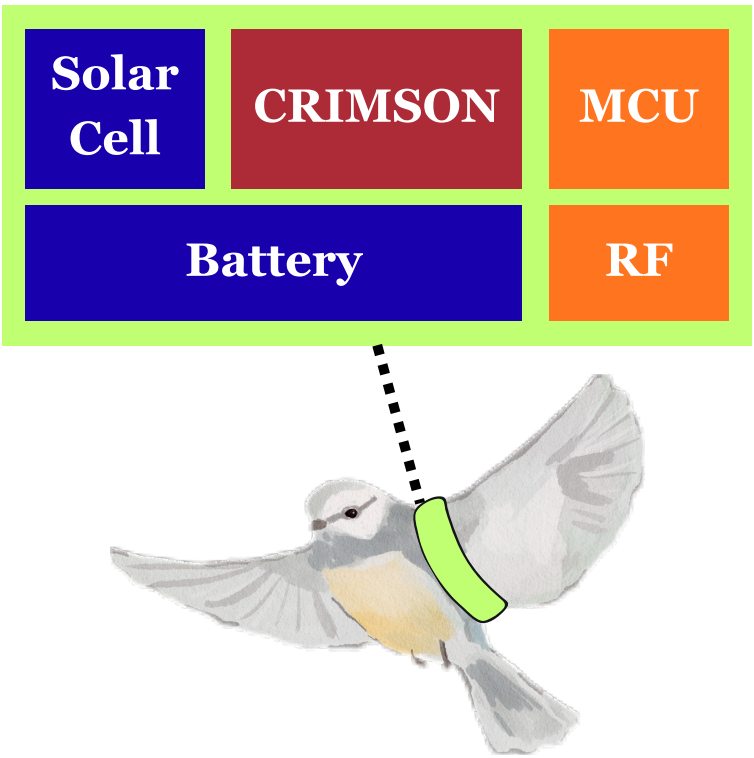

Figure 1: The CRIMSON Bird Tracking Example Use Case

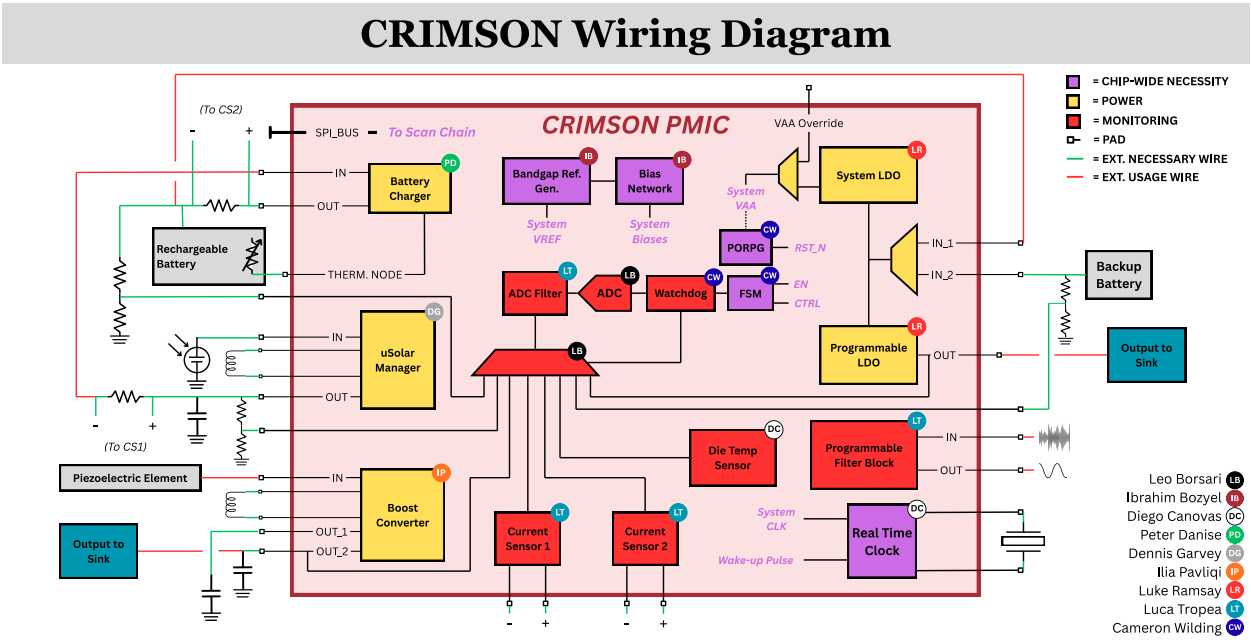

Figure 2: The CRIMSON Wiring Diagram

3.2 Configurability Layer

CRIMSON's ability to excel in a vast range of devices is enabled by the on-chip hierarchy termed the Configurability (CFG) Layer. This network consists of programmable interconnects that allow the user the ability to configure each block's connections. The user can choose to use some or all the intended architecture, or they can choose to fully isolate each block. The CFG Layer is depicted in Figure 3. The CFG Layer is comprised of the sub-circuit core and a series of interconnects that are enabled/disabled by a dedicated chain of elements called the SPI Scan Chain. The SPI Scan Chain allows the user to program each block using a standard SPI interface with a latch enable pin that drives the Scan Chain values to the blocks all at once after the Chain has been filled. Each block's Scan Chain input and output pins are connected to another block's Scan Chain input and output pins to create a chip-wide Scan Chain controllable with just five external pads. The CFG Layer was also designed to make CRIMSON easy to test. This design was created with support from the CHIPS Design Studio, which will be hosting future MQPs focused on testing manufactured designs. CRIMSON will be used in these projects, as well as future testing courses at WPI.

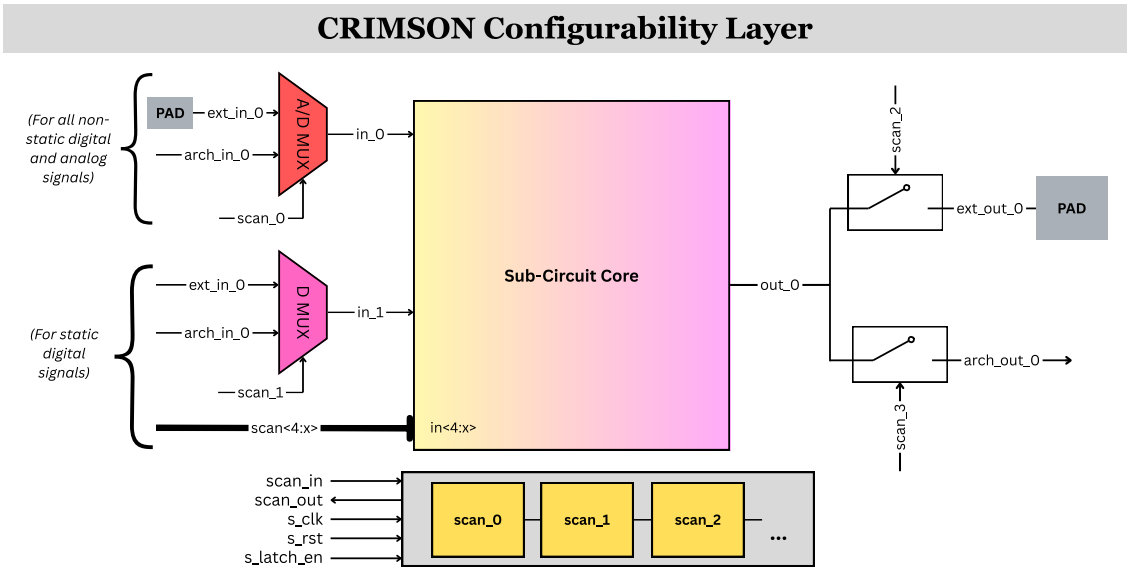

Figure 3: The CRIMSON Configurability Layer

3.3 Power Architecture

The CRIMSON Power Architecture (depicted in Figure 4) is what allows CRIMSON to harvest energy to charge batteries and provide power. Starting from left to right in the diagram, the Solar Generator receives energy from the solar cells and drives current into an output capacitor. The Battery Charger is

externally connected to the output of the Solar Generator and uses the raw power to recharge the battery. The battery is then connected to the main power input. It is at this stage that the user may decide to provide a different power source and elect not to use the Solar Generator and Battery Charger in this fashion.

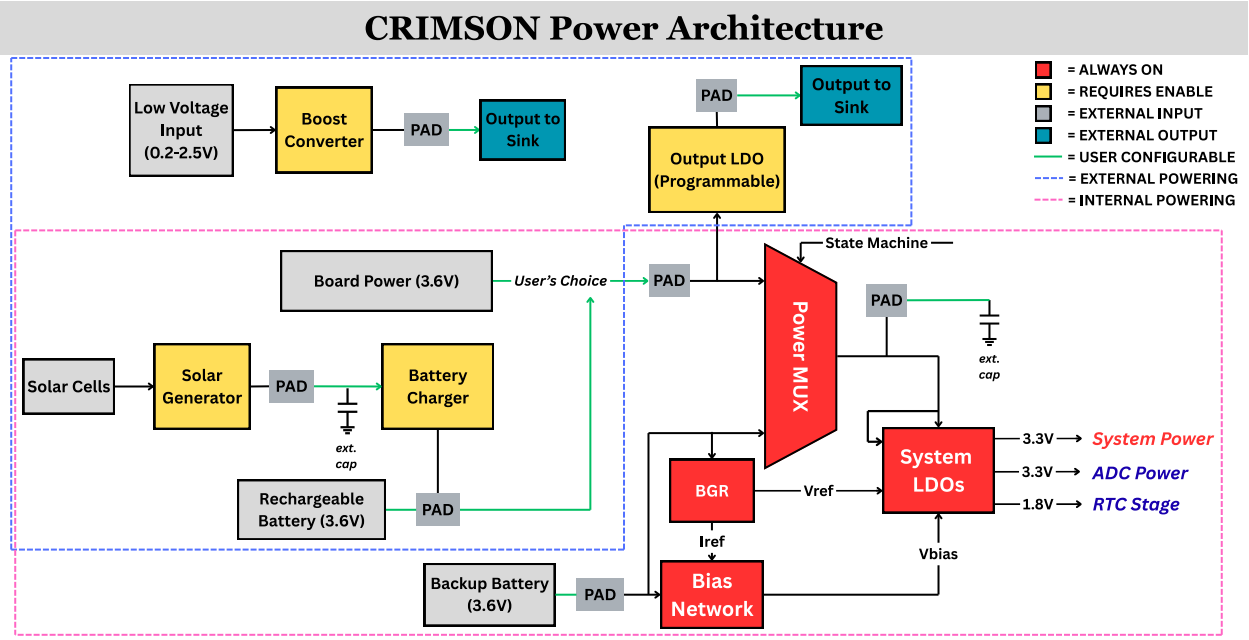

Figure 4: The CRIMSON Power Architecture

The main power input drives the Programmable Output LDO and the first input of the Power MUX. The Power MUX has a secondary input: the Backup Battery. The Backup Battery is necessary to the intelligence of CRIMSON as it provides a constant power source for the State Machine to determine if the input power at the main power input is adequate. When the power is adequate, the State Machine will switch its power input to run off the main power input to preserve Backup Battery lifetime. The output of the Power MUX provides power to the System LDOs, which provide power lines for the ADC and RTC. There is an external capacitor connected to the output of the Power MUX to prevent power loss when switching between power inputs. The Bias Network and Bandgap Reference Generator are always connected to the Backup Battery because they need to always be active to ensure the Monitoring Architecture remains in reliable operation.

The Boost Converter in the top left of the Power Architecture is a standalone block that can harvest energy from a second low voltage input to provide a second power output to the device.

3.4 Monitoring Architecture

The CRIMSON Monitoring Architecture (depicted in Figure 5) affords CRIMSON the ability to self-regulate. Starting from left to right in the diagram, there are eight monitoring inputs that are connected to the large Reading Selector MUX. The Output LDO and Boost Converter blocks are monitored to ensure they provide adequate output voltage to the device. The Die Temperature Sensor is monitored to ensure CRIMSON is not overheating. The Backup Battery is monitored to ensure that its power delivery is still healthy and that it can continue to act as a reliable backup. The Solar Generator is monitored for its voltage and current output to ensure that it can drive enough current to charge the battery charger. The Rechargeable Battery is monitored to ensure that it is operating at a suitable output voltage to power CRIMSON and the device together. The Resistor Dividers are necessary to step down the voltage from the power blocks to 3.3V to ensure the ADC can read the voltages as the ADC operates on a 3.3V power supply rail. The Reading Selector Mux reads all eight inputs cyclically to provide constant, routine monitoring of all critical nodes.

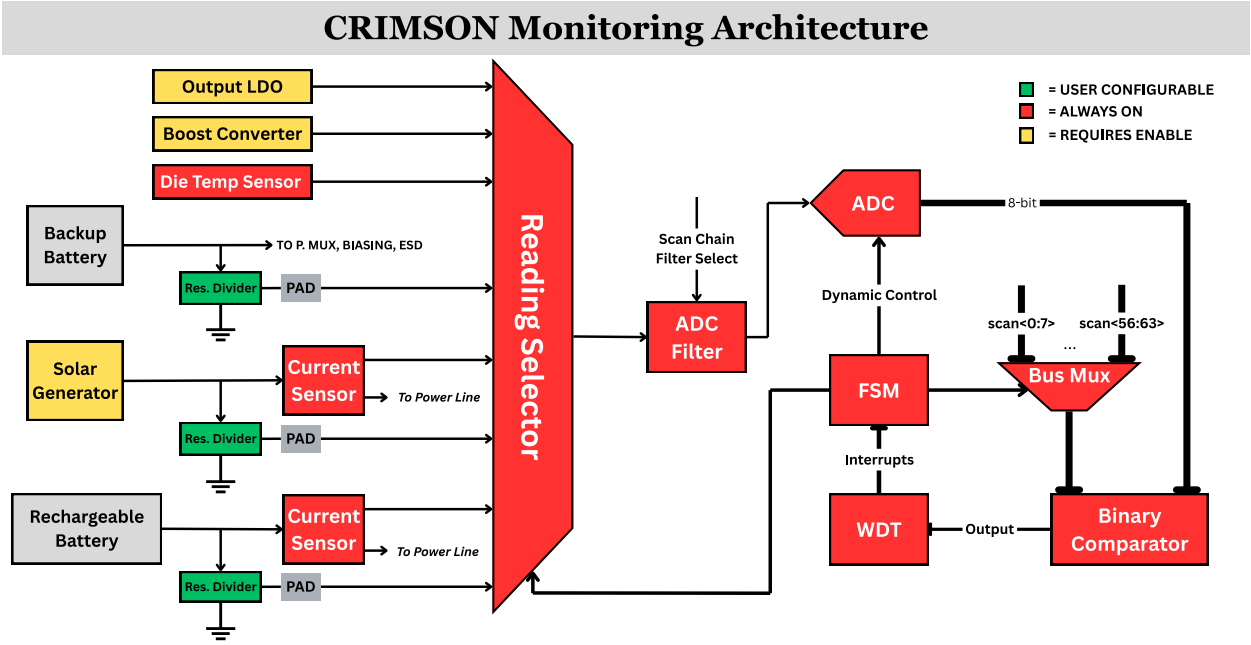

Figure 5: The CRIMSON Monitoring Architecture

The output of the Reading Selector is connected to the ADC Filter block, which ensures the frequency of the signal to be sampled by the ADC does not violate the Nyquist Sampling Rate. CRIMSON operates at a nominal 32KHz. The output of the ADC connects to the Digital Controller, which performs a binary comparison operation to determine whether the ADC reading indicates the monitored node is performing at desired operating conditions. The Watchdog sends interrupts to the State Machine in the

event of a block operating outside of its nominal conditions. The State Machine will perform the necessary operations to bring CRIMSON back to nominal conditions. More details on specific operations are available in the respective Design Implementation subsections.

4 Design Options

This chapter discusses possible implementations for each of the major sub-blocks. Many architectures can be used to achieve similar results; each section discusses the options the block designers considered when choosing their implementation strategy.

4.1 Low Dropout Regulators Options

Several design options were considered during the development of the LDO for this PMIC. The two major differences between LDO architectures are the choice of PMOS or NMOS pass transistor and the programmable resistor network structure. An NMOS-pass LDO can provide faster transient response and stronger drive capability due to the higher electron mobility of NMOS devices. However, it typically requires a gate voltage higher than the input supply voltage. As a result, an additional charge pump or boosted bias circuit is often needed, increasing design complexity and power consumption. A PMOS-pass LDO offers simpler gate drive requirements, since only the gate needs to be pulled low to increase conduction. This allows easier low-dropout operation without requiring boosted voltages. However, PMOS devices have lower carrier mobility, which often requires a larger transistor size to achieve comparable drive strength and dropout performance [3] [26]. For the programmable resistor network, two common approaches are parallel and series resistor structures. A parallel resistor network can provide a larger number of programmable output combinations while maintaining compact routing. However, this approach can lead to increased power consumption due to additional current paths, leakage, and switch parasitics. A series resistor network uses cascaded resistor segments with selectable taps. This method can simplify resistor matching and improve layout symmetry, but may require larger total area and provide fewer output combinations depending on the tap structure [25].

4.2 Micro-Solar Manager Options

The VI curve of solar cells acts like a voltage source at low currents, but at higher currents, the voltage declines exponentially until it reaches 0. Their power curve can be summarized by their *open-circuit voltage* and *short-circuit current*, two extremes of the curve. At low currents, a large increase in current will only result in a small drop in voltage, yielding a net power increase. At higher currents, a small increase in current will result in a large voltage drop, resulting in a net power drop. The maximum extractable power along the VI curve is called the Maximum Power Point. It is the responsibility of the solar cell manager to

find and operate the solar cell at this point in order to maximize energy harvesting. Because of a real-world solar cell's moving maximum power point. It is necessary to find and track the maximum power point continuously in real time. A multitude of strategies already exist to accomplish this. Some of these include Perturb and Observe, Incremental Conductance, Lookup Table, but the one implemented is Fractional VOC. Fractional VOC functions upon the fact that for a given solar cell, its maximum power point is around the voltage, which is a known fraction of the open circuit voltage.[19] A secondary solar cell experiencing the same environment must be connected, resulting in a larger device footprint and a sacrifice of power per unit of solar. Despite the drawback, this tracking method is computationally simple and optimal to implement in a simple PMIC.

4.3 Real-Time Clock Options

The operation of a real-time clock (RTC) relies on maintaining a stable and accurate reference frequency, typically generated by a 32.768kHz crystal oscillator. The crystal and oscillator circuit form a resonant system whose frequency is determined by the physical properties of the crystal and the load capacitances connected to it. During startup, the oscillator must provide sufficient loop gain to overcome the crystal's losses and initiate oscillation. Once oscillation begins, the amplitude gradually increases until a steady state is reached where the loop gain settles to unity. At this point the oscillator produces a stable periodic waveform that can be used as a time reference. Because RTC systems prioritize extremely low power consumption, the oscillator circuitry is typically designed to operate with minimal bias current while still maintaining reliable startup and stable oscillations. The resulting clock signal is then buffered and used to drive digital divider circuits, which reduce the frequency to generate precise timing intervals such as seconds, minutes, and hours for timekeeping applications.

4.4 Data Converters Options

The Data Converters consisted of two blocks: an Analog to Digital Converter (ADC) and a Digital to Analog Converter (DAC). The ADC is used to monitor signals on the chip; the DAC is used to provide a programmable bias for the Boost Converter.

4.4.1 Analog to Digital Converter Options

The Analog to Digital Converter (ADC) on chip needed to be flexible to accurately sample a wide range of input frequencies. Several architectures were considered: Sigma-Delta, Integrating, Successive-

Approximation Register, and Flash [17]. The Sigma-Delta ADC excels in low-frequency applications due to its high resolution. It is often used in highly accurate, low-bandwidth applications such as audio and sensor interfaces. The Integrating ADC is one of the oldest architectures; it uses an RC network to integrate the input signal over a set time constant and counts how long the input signal integrates before reaching its threshold. The Integrating ADC is typically used in precision instrumentation. The Successive-Approximation Register ADC performs a mixed-signal binary search to converge upon the digital equivalent of the input value. This architecture is used in medium to high-speed applications requiring low power and a small footprint. The Flash ADC is the fastest architecture in this list and compares the input signal against $2^N - 1$ reference voltages using a comparator bank and encoder. This architecture is used in applications where speed is the most critical requirement.

4.4.2 Digital to Analog Converter Options

The Digital to Analog Converter (DAC) needed to provide an accurate, steady bias voltage to the Boost Converter. Several architectures were considered: String, R-2R Ladder, Binary-Weighted, and Current [11] [12]. The String DAC divides a reference current throughout a resistor string of size 2^N. This architecture is highly accurate and inherently monotonic. The R-2R Ladder is a network that produces a voltage sum; the number of "rungs" is equal to N. The Binary Weighted scales each bit in powers of two; this is a fast architecture but managing area and matching becomes difficult at larger resolutions. The Current DAC is the most common way to produce a bias voltage; it is often used in RF and high-frequency applications where output current is critical to driving loads or converted with trans impedance stages.

4.5 Linear Battery Charger Options

The role of an integrated battery charger is to send a constant charging current to the battery and terminate charging when a certain voltage is reached. The battery charger should also incorporate safety features. There are multiple ways to accomplish this within an integrated circuit. Current mirror based designs such as ours are a common strategy. However, other exist such as PWM based designs the use external capacitors and inductors to form a boost converter. An example of this design is the LT1571 battery charger integrated circuit. The current output by the boost converter is used to charge the battery. The output current is sensed through a sense resistor and the PWM is adjusted to change to a voltage that will create the desired charging current outputted to the battery [23]. This method requires a greater use of external components as well as more internal circuitry. A current mirror based design was chosen for its

simplicity and lack of external components.

4.6 Digital Controller Options

This section examines the potential implementations of the Digital Controller on CRIMSON, which is primarily an analog chip. A controller is necessary for intelligent functions.

4.6.1 Watchdog Options

To create interrupts for the State Machine using the Watchdog, we needed to find a way to compare the ADC output to preset values, to tell when an ADC reading needed to be reacted to. This left us with two real choices of comparator designs: the ripple comparator and the tree comparator. Ripple Comparators work by comparing the two most significant bits of two numbers, then passing down their outputs to the next bit comparison. This can take a significant amount of time, as each bit has to wait for each before it to finish. The alternative was a Tree comparator, which divides the numbers into groups, then compares those groups against each other, and then compares the output of all of the groups. This is significantly faster, as all bits are being compared at the exact same time, but uses up significantly more space. Originally, CRIMSON was designed to run at a 1 MHz clock, so the faster compare speed of the Tree was needed to complete a compare in 1 clock cycle. However, as we slowed down the clock to the current 30 kHz, the extra speed became unnecessary and a waste of space, so the Ripple was chosen.

4.6.2 State Machine Options

For the State Machine, there were two main options: a Mealy State Machine, and a Moore State Machine. A Mealy State Machine, as shown in Figure 6, changes its output not only on the current state, but also on the current inputs. A Moore State Machine, as shown in Figure 7, only changes its outputs based on the current state. While a Mealy State Machine can react faster to an input change, and requires less states then a Moore State Machine to achieve the same job (taking up less space), the Mealy has one fatal flaw: glitches. The Moore State Machine completely synchronous, only making output changes once a clock cycle. However, the Mealy can change output between clock cycles, leading to unexpected glitches, where the system reacts to one input that happened during a single clock cycle before another. This behavior is not an enormous deal in fully digital systems, but in mixed signal designs like this, non-synchronous behavior could lead to blocks turning off when they are not supposed to, or odd race conditions.

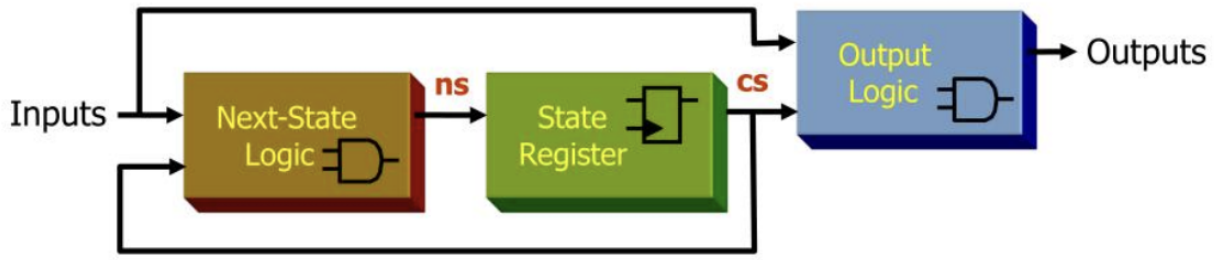

Figure 6: Mealy State Machine

Figure 7: Moore State Machine

4.7 Programmable Active Filters Options

Three programmable filter blocks were developed for the CRIMSON chip. These blocks include a dedicated ADC anti-aliasing filter and two general-purpose filter blocks, with one utilizing a Sallen-Key topology and the other a Gm-C topology.

4.7.1 ADC Anti-Aliasing Filter

For optimal performance, the chip's 8-Bit SAR ADC uses active filtering on its input to condition the sampled signal to prevent aliasing and LSB jitter from high-frequency noise. The input signal must be bandlimited to prevent components greater than the Nyquist frequency ($F_s/2$) from entering the ADC, which would otherwise cause aliasing. This requires a low-pass filter with a cutoff frequency set at or just below half of the sampling rate of the ADC, and in order to maintain signal accuracy and amplitude, an ideal anti-aliasing filter should feature a unity gain passband with minimal variation [22, 6].

Because the monitoring architecture (Figure 5) utilizes a Reading Selector MUX to select the internal channel to monitor, multiple design options were considered. A filter could be placed on each channel of the MUX input, which allows for high-speed switching and settling time, but at the cost of higher

complexity and area due to multiple instances of the filter. Alternatively, only a single filter can be placed on the output of the MUX, significantly reducing cost, though it induces a higher settling time, reducing throughput [22]. Because the ADC will sample relatively slowly at 3.2KSps for internal signals, a single filter on the output was chosen as the most efficient design since the settling time requirement is not a concern at this speed.

Because of this multiplexed configuration, the filter may see input steps as large as full scale (0V to 3.3V) as it switches between channels [22], meaning that this filter must support a full power bandwidth with a 3.3V amplitude to ensure the ADC does not sample intermediate signals caused by the filter's output slewing slower than the sampling period.

The filter topology that best meets these requirements of a maximally flat, unity gain passband and rail-to-rail full power bandwidth, is a 2nd order Sallen-Key lowpass topology using a rail-to-rail op-amp 9. The Sallen-key topology allows for the Butterworth response with a flat passband, and while higher orders offer improved performance, 2nd order was an ideal middle ground for power consumption and design complexity, given the speed and resolution of the ADC.

Furthermore, because the ADC supports a low-speed internal monitoring rate of 3.2Ksps and custom higher speed sampling rates up to 100KSps, the filter is also capable of selecting cutoff frequencies from the default 1.6kHz up to 50kHz in 5kHz increments using a programmable resistor block.

4.7.2 General Purpose Sallen-Key and Gm-C Filter Blocks

The general-purpose filter blocks are designed for a wide range of configurability, featuring user-programmable cutoff frequencies and the ability to cascade individual stages into higher-order low-pass, high-pass, bandstop, and bandpass configurations, allowing for a variety of external signals to be filtered as needed. These filters were designed to target initial provided specs of 10μW per block and a frequency target of 50Hz to 100Khz. To simplify integration, the supporting structure of these programmable filter blocks was kept identical, where the digital control and signal routing are shared across both blocks, only the filter cores themselves utilize the two different topologies.

Switched-Capacitor vs. Continuous-Time Tradeoff

Initially, switched-capacitor filters were considered due to their high PVT stability and precise tuning capabilities[21] by changing the clock frequency of the switched capacitor resistor, where $R_{eq} = \frac{1}{f_{clk} \times C}$. However, they were ruled out because of the introduction of switching noise and additional design complexity,

requiring analysis and design in the discrete-time domain [20]. Instead, continuous-time architectures, Gm-C and Sallen-Key were selected to simplify the design process and maintain a continuous-time signal.

Gm-C Filter Block and Capacitor Multiplier

The Gm-C filter topology was chosen for its wide range and ease of tunability, since the cutoff frequency can be adjusted linearly by simply modulating the bias current of the OTA with an I-DAC, (where filter cutoff $F_c = \frac{G_m}{2\pi \times C}$ and $G_m = \frac{2I_{bias}}{V_{od}}$), in addition, it has the ability to operate at much lower power compared to op-amp filters. To meet the low-frequency requirement of 50 Hz, a capacitor multiplier circuit was initially implemented to scale the effective capacitance of the grounded capacitors used in the low-pass topology, which reduces the small signal cutoff frequency.

However, late in the design process and after significant development time of this circuit, the capacitor multiplier circuit was removed for several reasons. The multiplier added four additional op-amps per low-pass filter stage, adding significantly higher power consumption and introducing a number of close-proximity poles and zeros, complicating the phase response. This caused instability at high frequencies in some test cases. Additionally, the multiplier significantly worsened linearity, limiting the usable signal amplitude to a few 10's of mV before distortion occurred, more constrained than the already limited Gm-C filters themselves, requiring the input signal amplitude to be kept very small to avoid distortion. Ultimately, the capacitor multiplier circuit was removed to preserve the block's function as a general-purpose filter, since the reduced linear range and potential instability, especially when filters were cascaded, would have significantly limited its use cases, with the tradeoff of losing the ultra-low cutoff frequency configuration.

The Gm-C low-pass topology that is used in the design of this block is a 2nd-order topology referenced from a figure within "Widely tunable low-pass gm-C filter for biomedical applications" [14]. This low-pass design was then modified into a high-pass topology, and its equations were extracted. Figure 8 shows these topologies and their equations.

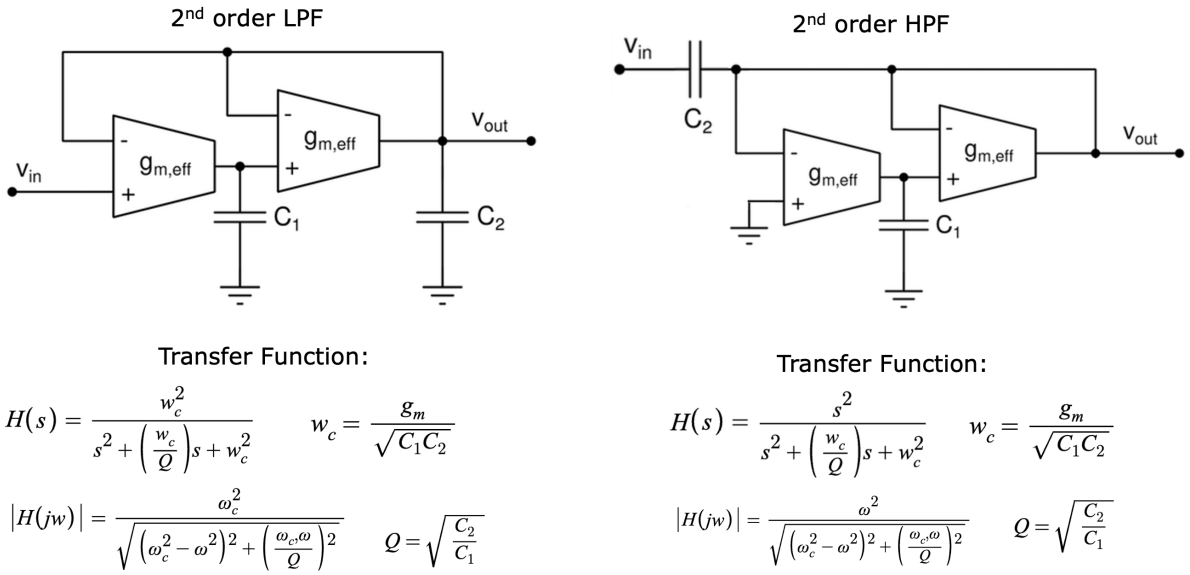

Figure 8: Gm-C low-pass and high-pass topologies

Sallen-Key Filter Block

To complement the Gm-C type filter block, another programmable filter block utilizing the Sallen-Key topology was also integrated. Rather than instancing another Gm-C block to create a second usable filter block on chip, the Gm-C filter cells were substituted for Sallen-Key filter cells. This was done to maximize the chip's versatility. While the Gm-C block offers a wider tunability range, the Sallen-Key block was implemented to provide filtering for higher amplitude signals with the tradeoff of a more limited cutoff frequency tuning range. The Sallen-Key filters are programmed using a binary-weighted parallel resistor block to allow for linear steps in programmable cutoff frequency. Figure 9 shows the low-pass and high-pass Sallen-Key circuit topologies used within this block.

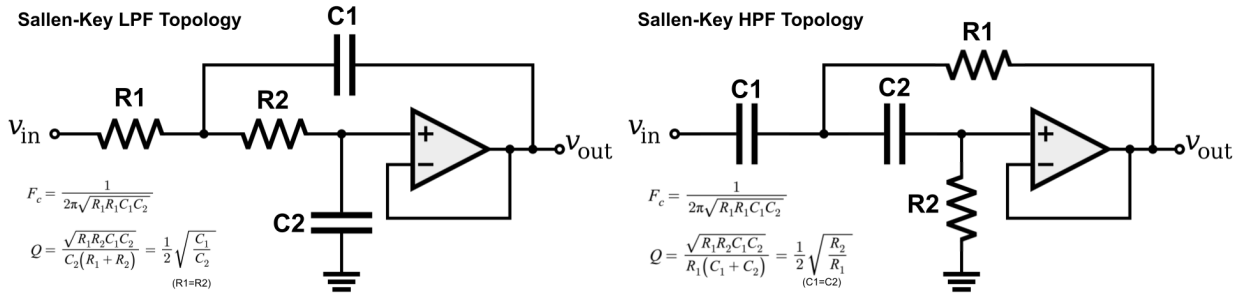

Figure 9: Sallen-key low-pass and high-pass Topology and Equations

4.8 Buck-Boost Converter Options

A boost converter is a type of DC-DC converter in which an input voltage is increased (boosted) at the output. A conventional boost converter takes an input voltage source, and uses it to charge and discharge an inductor, which is controlled by a switching transistor connected in parallel between the inductor and a diode, controlled by a pulse width modulated (PWM) signal at the gate of the transistor. When the transistor is on, the inductor begins to store current, as the transistor is on, creating a direct path to ground, so the inductor is able to hold on to its current. When the transistor is off, this creates an open circuit to ground, so that current can only go through the diode. In this case, the diode is forward biased, which is then connected to a load capacitor and resistor. The limitations of this topology are that it is intended to be made with discrete components, as opposed to it being integrated, along with it having limitations in programming a constant output for a wide range of inputs, including input voltage values greater than what the output would be. The second problem could be fixed by using a buck-boost converter, or a similar topology, but that increases complexity and does not fix the first problem.

A different architecture that addresses both these problems is the architecture presented by Kanago et al[10]. Here, a circuit made up of 5 transistors, an inductor, and two capacitors is used. Due to IC sizing constraints, the inductor and the capacitors need to be off-chip, but the transistors can be on-chip, which solved the first problem. The first transistor in this circuit acts as the switching transistor, the next two act as a diode, and the last two set the output. The output in this circuit is set by Vth+Vbias, where Vth is the threshold voltage of the transistor, and Vbias is the voltage at the gate of each output transistor. This is independent from the input voltage, which allows a fixed output voltage for a variable input voltage, solving the second problem.

4.9 Current Sensor Options

To support the self-regulation and power management capabilities of the CRIMSON IC, on chip current sensors are required to monitor the solar supply and the chip's current draw.

4.9.1 High-Side vs. Low-Side Sensing

The design of the current sensing circuitry began with a comparison between low-side and high-side sensing. Low-side sensing places the sense resistor between the load and ground, allowing a low common-mode input voltage to the sense amplifier, allowing for easier single-supply design, but it lifts the load's direct

connection from ground, potentially causing reference issues. With high-side sensing, the sense resistor is placed between the positive supply and the load, allowing the load to remain grounded, but this places the common-mode voltage of the sense amplifier input near the power supply voltage, meaning the sense amplifier must have a high enough common-mode voltage spec. Additionally, high-side sensing allows the detection of load-to-ground short circuits [18]. High-side sensing was chosen to ensure that all circuit blocks within the chip remain directly grounded.

4.9.2 Topology Choice

Several designs within Analog Devices' AN-105 application note [18] were considered. The circuit design titled "Positive Supply Rail Current Sense" (Design 10) was chosen, as it allows implementation with generic components rather than ADI specific products, allows for configurable gain, and can be implemented with CMOS devices, not requiring BJTs which many other designs in the application note require. Figure 10 shows this topology and its gain equations.

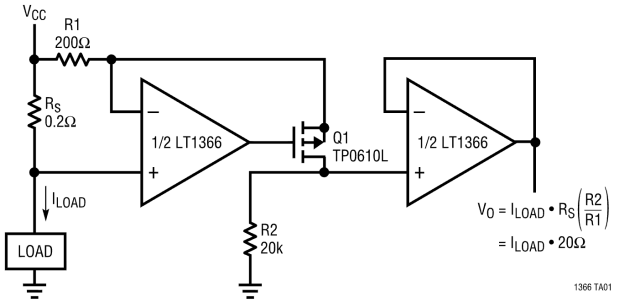

Figure 10: Positive Supply Rail Current Sense Topology Presented by Analog Devices in AN-105[18]

4.10 Temperature Sensor Options

Many integrated temperature sensors are commonly included in power management circuits, microprocessors, and environmental monitoring systems to ensure safe and efficient operation. Temperature sensors operate by exploiting electrical properties that vary predictably with temperature. In integrated circuits, temperature is commonly measured using devices whose voltage or current characteristics change with thermal energy, such as diode-connected transistors or resistive elements. One widely used approach relies on the base–emitter voltage of bipolar transistors, which decreases with increasing temperature at a roughly linear rate. By comparing two voltages generated at different current densities, a voltage proportional to absolute temperature (PTAT) can be produced. This temperature-dependent signal can then be amplified,

digitized, or used directly for thermal monitoring and control. Accurate temperature sensing is important for system reliability, power management, and performance optimization, since many electronic components exhibit temperature-dependent behavior.

5 Design Implementations

This chapter explores the implementations of each major sub-block of CRIMSON. For each block, the information consists of the architectural integration, the sub-circuits and applicable sub-architectures, the results, and a brief testing overview.

5.1 Low Dropout Regulators Implementation

The low-dropout regulator (LDO) block in this design was implemented as a multi-output power management system consisting of four separate regulators. The design uses dedicated LDOs so that each subsystem can receive a regulated supply voltage that is appropriate for its current and performance requirements. The four LDOs that were designed for this integrated circuit are a programmable output LDO, a system LDO, an ADC LDO, and an RTC LDO.

5.1.1 Internal Output LDOs

Three internal low-dropout regulators were implemented to provide dedicated regulated supply rails for on-chip subsystems. The regulators include the System LDO, ADC LDO, and the RTC LDO. Rather than powering all of the internal sub blocks from a single shared regulator, separate LDOs were used to reduce interference between the sub blocks, make sure all blocks are able to have a regulated supply under their maximum current needs, and to provide subsystems with the needed supply voltage. Although these regulators serve different purposes, they were all designed using the same core LDO topology with the primary difference being the resistor divider used to set the final output voltage.

Each internal regulator was designed as a PMOS-type LDO consisting of an error amplifier, NMOS pass transistor, compensation capacitor and resistive feedback network. A PMOS pass device was chosen to have a lower dropout voltage and to avoid any additional circuitry for boosting the gate voltage. The error amplifier, which compares the reference voltage to the divided output voltage and adjusts the PMOS gate accordingly. This system regulates the output through negative feedback. The figures below show the circuits that were implemented.

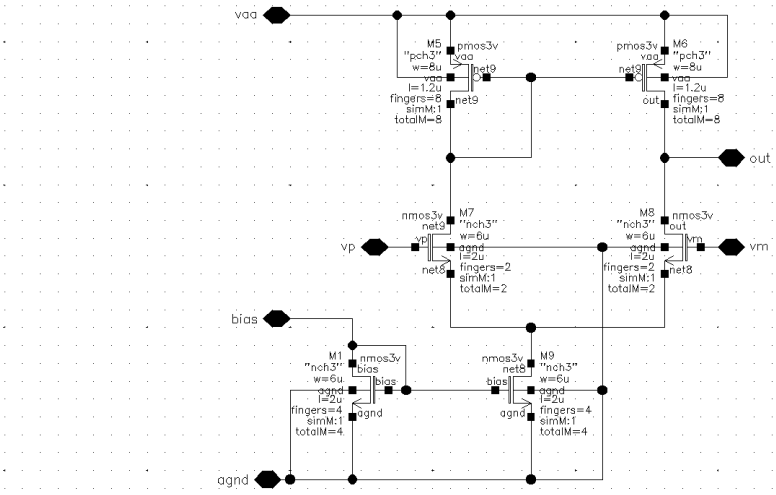

Figure 11: LDO NMOS Differential pair

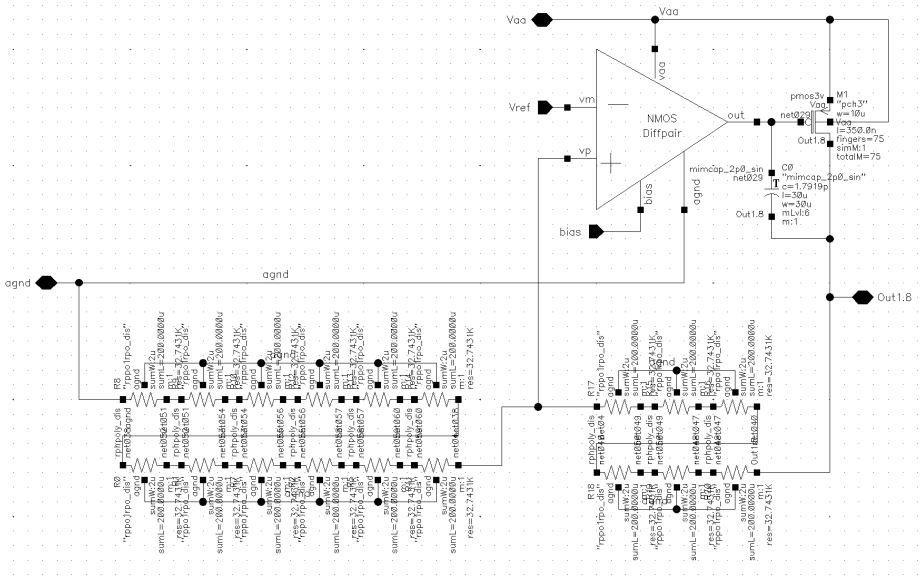

Figure 12: 1.8 V RTC LDO Schematic

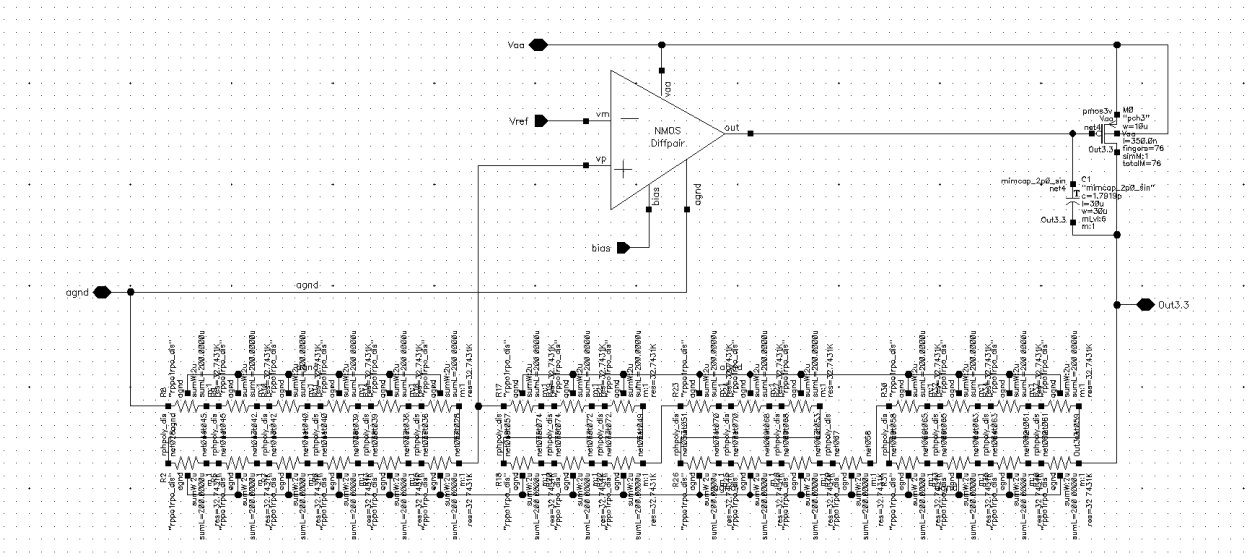

Figure 13: 3.3 V ADC and System LDO Schematic

An NMOS differential pair was selected for the error amplifier due to its strong transconductance and adequate gain in low power systems. When comparing NMOS and PMOS transistors, NMOS's typically have higher electron mobility resulting in larger transconductance and small signal gain. This gives the amplifier an edge when comparing smaller differences in the reference voltage and the feedback voltage. In addition, the NMOS input pair integrates well with the PMOS pass transistor due to it being able to effectively pull the PMOS gate low.

The System LDO and ADC LDO were both designed for a fixed 3.3 V output using a 400 kΩ bottom resistor and a 690 kΩ top resistor in the feedback resistor divider. Although both regulators produce the same output voltage, due to the ADC expected to draw significantly higher current than the rest of the system, it was given its own to prevent an excessive voltage drop on the main system supply.

The RTC LDO was designed to generate a fixed 1.8 V output using the same architecture as the ADC and System with modified resistor values. For this regulator, the bottom resistor remained at 400 kΩ while the top resistor was reduced to 200 kΩ. This LDO is being used lower the power consumption of the RTC, as supplying it with 3.3 V would greatly lower its power efficiency.

Using the same LDO structure across all three internal rails simplified design, verification, and layout since only the feedback network required modification between the different systems.

5.1.2 Programmable Output LDO

In addition to the internal fixed-output regulators, a programmable output LDO was implemented to provide a flexible external supply capable of supporting multiple voltage levels. This regulator was designed to generate selectable outputs of 1.2 V, 1.8 V, 2.5 V, and 3.3 V. This allows a single LDO to serve circuits with different supply requirements.

The programmable output LDO uses the same core topology as the internal regulators, consisting of an error amplifier, PMOS pass transistor, compensation capacitor, and a feedback network. Continuing the same architecture ensured the same regulation behavior across all of the LDOs in the system.

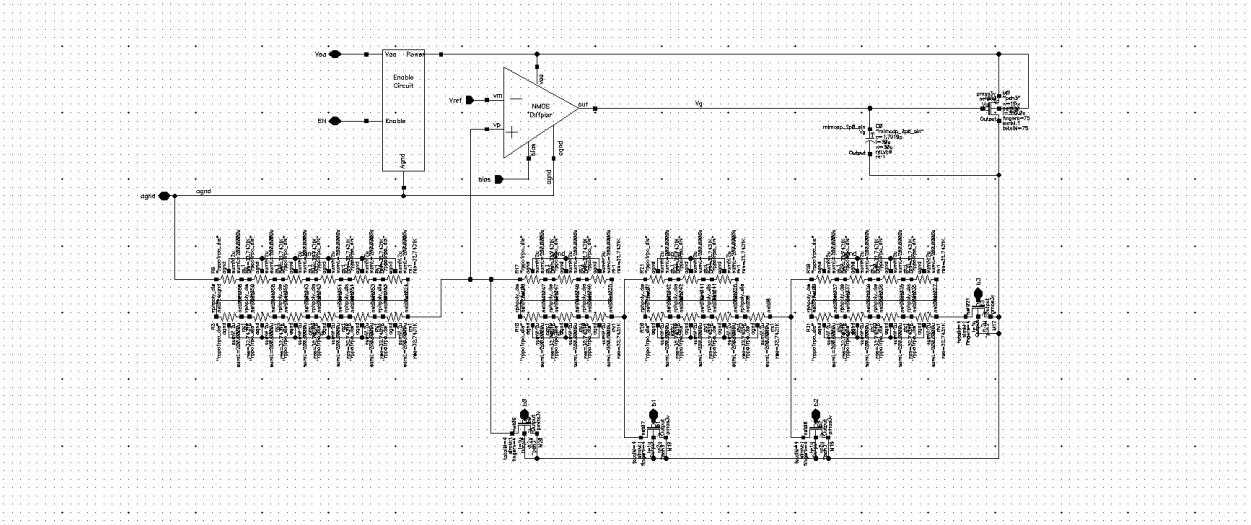

Figure 14: Programmable Output LDO Schematic

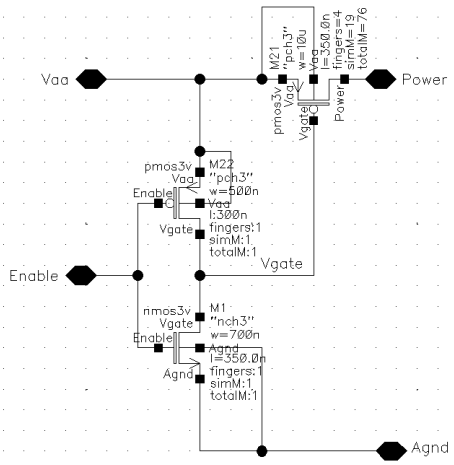

Figure 15: Programmable Enable Schematic

When comparing the system LDO's to the output LDO, the same error amplifier and PMOS pass transistor was used. The resistor feedback network is similar to the System and ADC LDOs as it has a 400kΩ bottom resistor and a 690kΩ top resistor. The key difference between them is that the in the output LDO the top resistor has 4 PMOS transistors that take a digital signal controlling where the current exits which changes the resistor value. When the system wants to output 1.2 V, the top resistor is bypassed, at 1.8 V the top resistor is 200kΩ, at 2.5 V the top resistor is 437kΩ, and at 3.3 V the top resistor is the full 690kΩ. Another Key difference between the output LDO and the internal LDO's is the enable circuitry. This was added to the circuit to give the PMIC the flexibility and to only utilize it when necessary to reduce power consumption.

5.1.3 Simulations and Results

Following the schematic implementation of the internal and output LDOs, simulations were performed to evaluate their performances under certain operating conditions. This results section focuses on the key characteristics used to verify proper LDO operation, including line regulation, load regulation and error amplifier behavior. The differential pair was put through several AC simulations. The results are shown in the figure below.

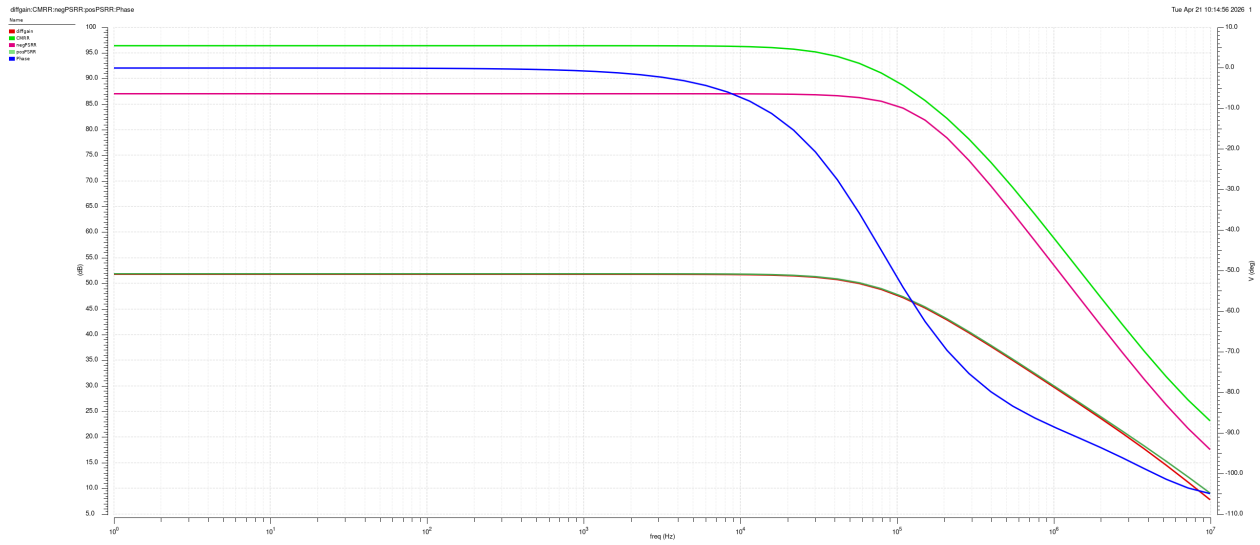

Figure 16: AC Simulation Results

The figure above shows the following characteristics; CMRR in light green, Phase in blue, negative PSRR in pink, positive PSRR in green, and differential gain in red. It is important to note that the postive PSRR and differential gain traces cross over, making it difficult to see the red trace. Based on the figure,

below is a table showing the results.

Spec	Performance
CMRR	96 dB
+PSRR	52 dB
-PSRR	87 dB
Differential Gain	51 dB
Bandwidth	82 kHz
UGBW	7.2 MHz
Phase Margin	76°

Table 1: LDO Diffpair Performace

The line simulations goal is to observe the LDOs output when the load goes from it lowest point to its highest point quickly. The red trace shows the two resistors being placed in and out of a parallel connection, showing the switch from low to high and high to low load. The output is shown in the blue trace. In the line simulation, the goal is for the output to not change when their is a variation in the supply voltage. The red trace shows the changing Vaa and the blue trace shows the output.

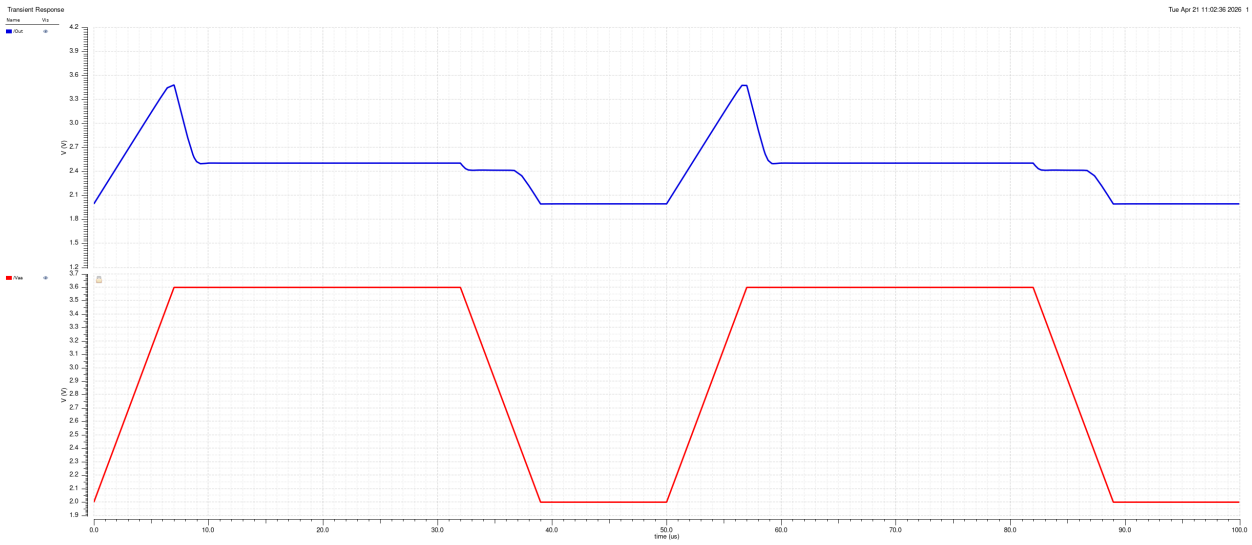

Figure 17: LDO Line Simulation

Figure 18: LDO Load Simulation

The line simulation figure was ran on the output LDO when programmed to output 2.5 V, however the behavior across all of the LDOs is the same. As Vaa increases the output follows until Vaa stops increasing where then the output is able to settle to its voltage level of 2.5 V. Once Vaa begins to drop the output voltage initially drops, however, it stays consistent until reaching the dropout voltage threshold. That threshold represents how much voltage headroom it needs between Vaa and the desired output to accurately produce the output voltage. Once that threshold is met, the output follows the Vaa. This is typical LDO line performance.

The load simulation was ran on the Output LDO set to output a voltage of 3.3 V, however, similarly to the line simulation the behavior is the same across all of the LDOs. When the red signal is low the effect load drops to the minimum that it was tested which was 30.3 μA. When the red signal is high, the effective load raises to the maximum that was tested which was 8 mA. These values were chosen because they are the absolute maximum and minimum loads that each LDO will be placed under in this application. As shown in the figure, there is a brief spike during the sudden shift in load, but the voltage quickly settles back to its desired output with very little oscillation.

Initially, there were proposed specs that the LDO needed to meet in order to properly supply the different blocks in the system. Those proposed specs and my results are shown below.

Specification	Desired	Achieved	Met?
Output Range	1.0 V to 3.3 V	1.2 V to 3.3V	Yes
Programable Outputs	TBD	4 Outputs	Yes
Temperature Range	$-10°C$ to $70°C$	$-10°C$ to $70°C$	Yes
Power Delivery	0.1 mW to 10mW	0.1 mW to 26.5mW	Yes
Power Consumption	$< 10\,\mu W$	$< 17\,\mu W$	No
Layout Size	TBD	$255\,\mu m \times 101\,\mu m$	Yes

Table 2: LDO Performace

5.1.4 Configurability Layer

To support testing and flexible system integration, a configurability layer was added around the LDO network using multiplexers, scan controlled selection logic and probe pad access. This layer allows for internal bias and reference signals to be selected from either on-chip sources or external pads. This ultimately makes validation post tape out possible while also giving the option to use an off chip device in the case one of the LDOs were to fail. Below is the configurablilty layer for all four of the LDOs.

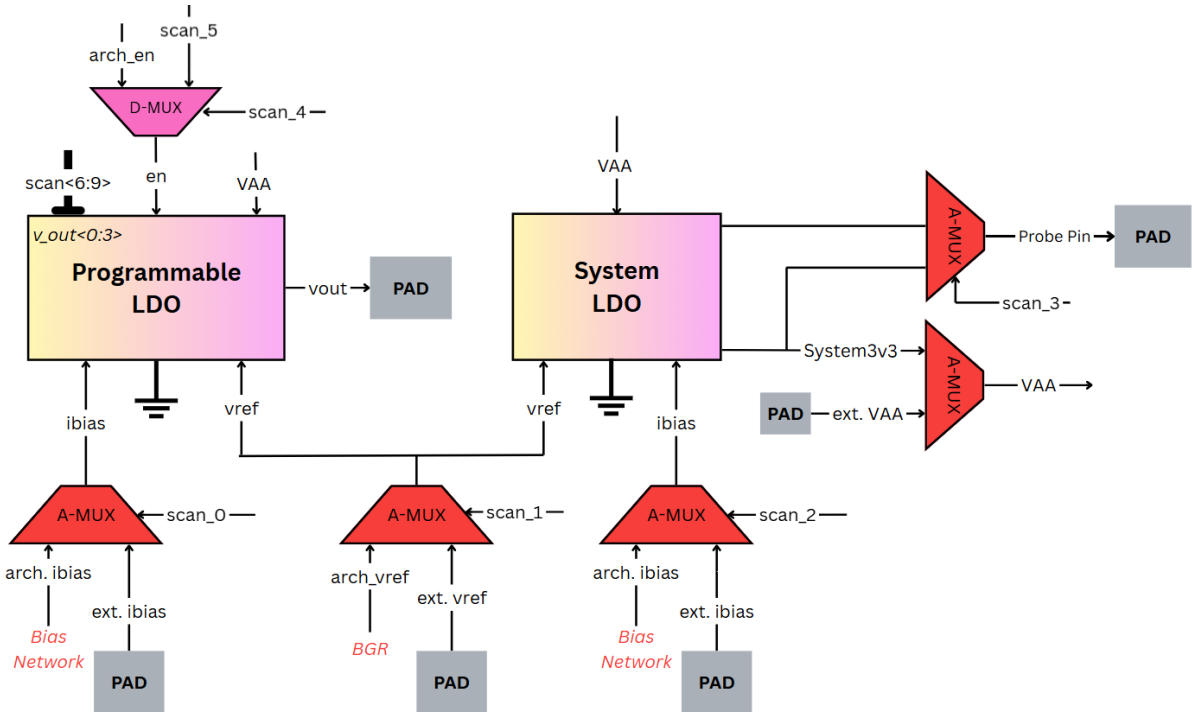

Figure 19: Programmable and System LDOs: Output and System Configurability Layer

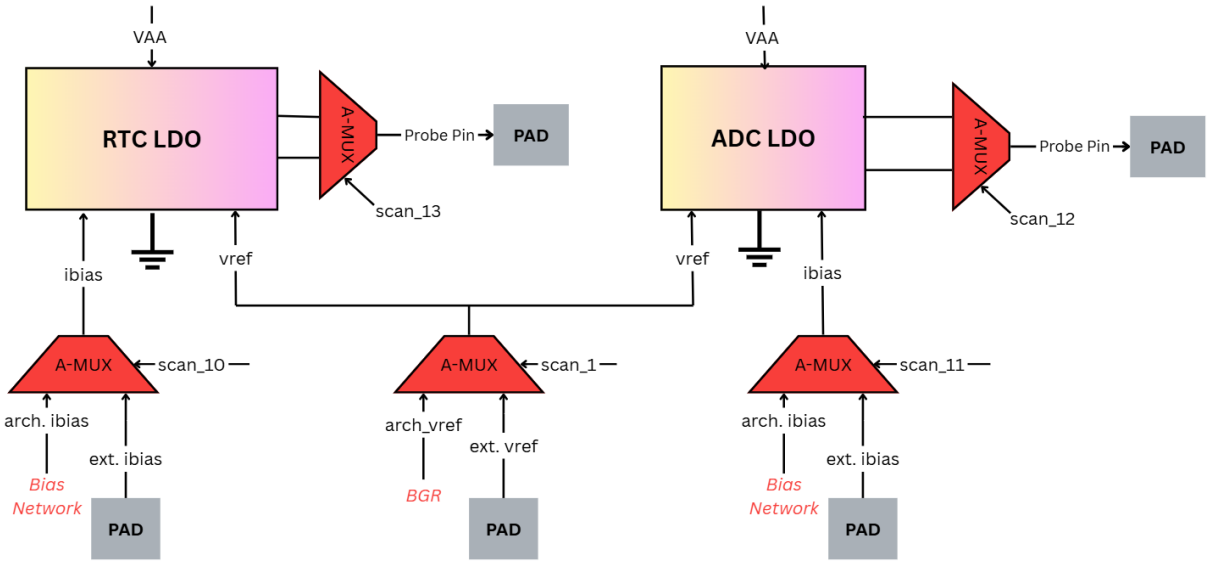

Figure 20: RTC and ADC LDOs: Output and System Configurability Layer

5.2 Micro-Solar Manager Implementation

The μSolar block is responsible for managing the operating point of the connected solar cell along its power curve and converting its output to one usable by the rest of the chip or chip implementer. A top-level diagram of the μSolar Manager architecture can be found in Figure 21.

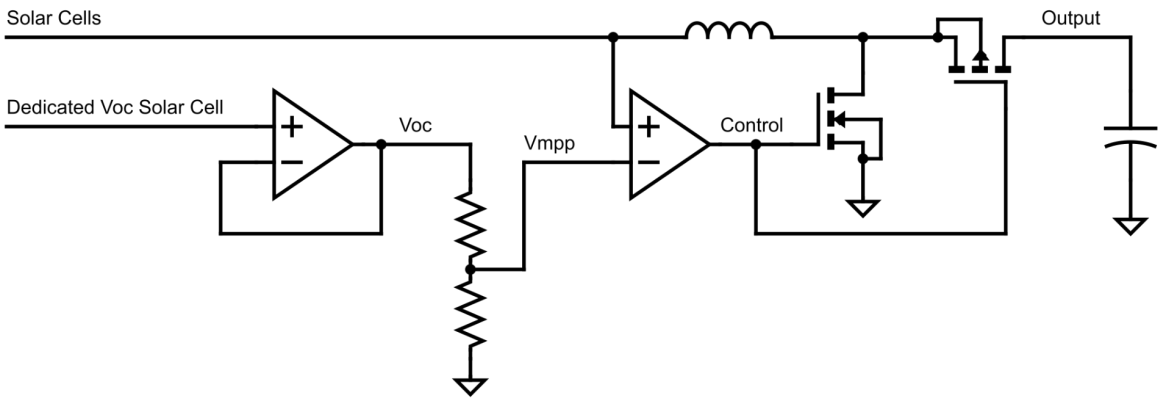

Figure 21: μSolar Block System Diagram

It comprises two main parts: a fractional VOC voltage tracker and a boost converter.[5] The

fractional VOC voltage tracker generates the voltage that the solar cell must operate near to ensure maximum power extraction, and the boost converter allows us to modulate the apparent load to the solar cell. During operation, the open circuit reference voltage is generated by buffering the dedicated VOC solar cell.[10] This is done to draw close to no current from the solar cell in order to preserve its measured open circuit voltage and prevent a voltage disturbance from loading the cell. The buffered voltage is divided by a voltage divider in accordance with the ratio tailored to the solar cell in use, relying on Fractional VOC MPPT. The voltage division ratio can be left to the default with on-chip resistors, or a user can adjust the ratio by placing another resistor on an external "voc_fraction_adjust" pin, which acts as a resistor in parallel. The maximum power point voltage is then compared against the solar cell voltage. If the solar cell voltage is higher, the control signal transitions to high, the NMOS will turn on, and energy begins to be stored in the inductor. As the inductor stores energy, its voltage will drop and cause the solar cell voltage to drop. Once this drops below the maximum power point voltage, the control signal will transition to low, the NMOS will turn off, and the PMOS will turn on. This causes the energy stored in the inductor to flow to the output and be stored in an external capacitor. This cycle continues and oscillates around the maximum power point. Its oscillation is dictated by the current of the solar cell, inductor size, and output capacitance. Because of this comparator-based control, no clock signal is needed, and this boost converter is "self clocked."

The configurability layer implementation for the μSolar block is similar to the framework used for other blocks: the configurability layer can toggle the block-level enable, toggle biasing to an external source, but also, unique to this block, it can toggle the block's output to other components interchip. The configurability layer diagram for the μSolar block can be found in figure 22.

Figure 23 displays the μSolar manager testbench. It contains two basic solar cell models, a boost inductor, and an output storage capacitor. The pads interconnecting the off-chip and on-chip circuits are also included to accurately model the behavior between the off-chip components and the IC. Figure 24 shows the behavior of the solar cell model with a small output capacitor and no load to quickly demonstrate its charging capability. The control signal is visible, and the rise in voltage can be observed. Figure 25 demonstrates the steady state behavior with a load and a large output capacitance. The final performance metrics of the block, as well as a comparison to the initially proposed specs, can be found in Table 3

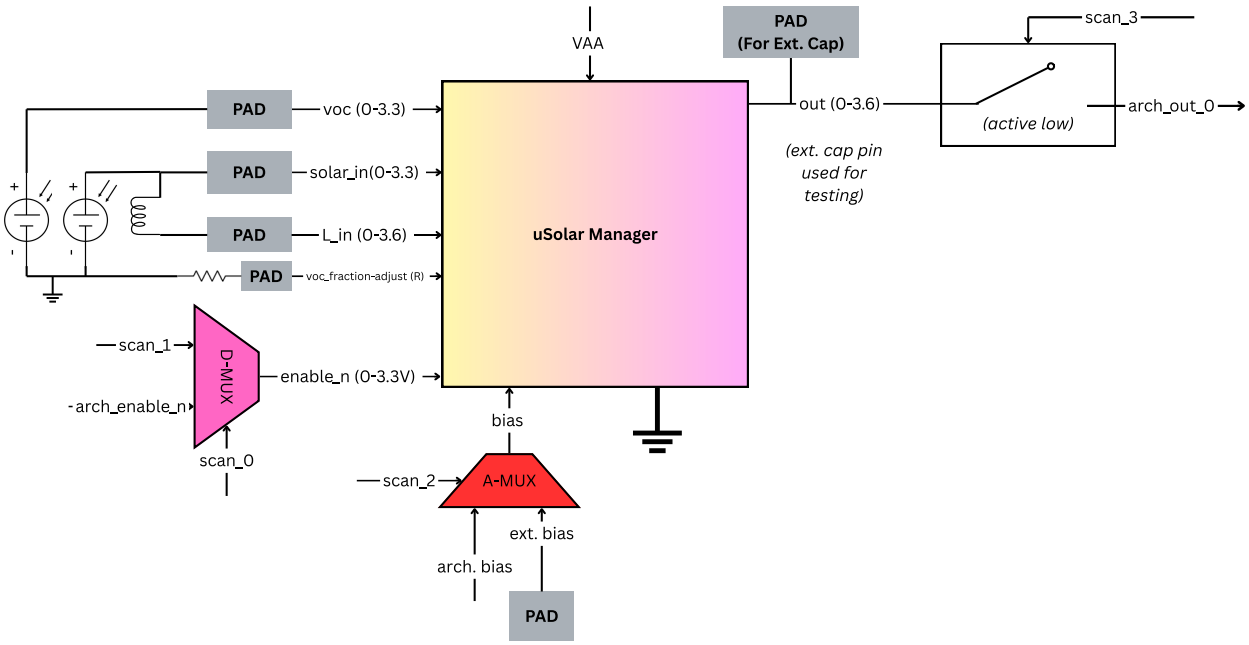

Figure 22: μSolar Configurability Layer Diagram

Spec	Target	Performance	Met?
Introduction	0.1V-3V	0.12V-3V	No
MPPT	Present	Fractional VOC	Yes
Output Voltage Range	2.5V to 3.3V	0-3.6V	Yes
Temperature Range	-10°C to 70°C	Functional	Yes
Quiescent Power Consumption	<50μW	33uW	Yes
Efficiency	>75%	92.5%	Yes
Layout Size	Not Bounded	106μm x 53μm	Yes

Table 3: μSolar Performance Results

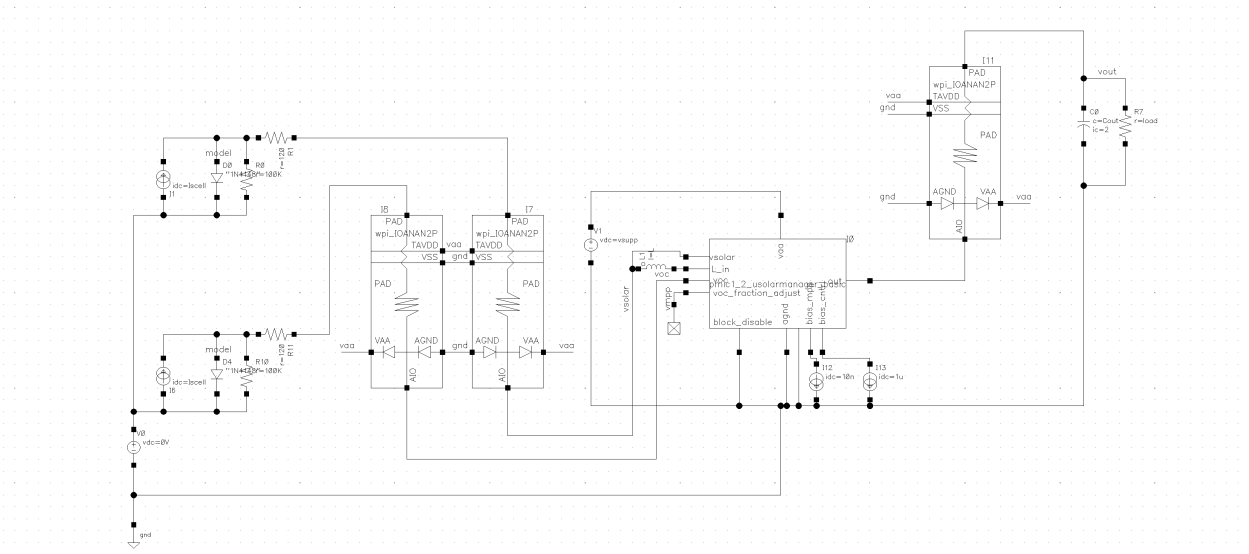

Figure 23: μSolar Top Level Testbench Setup

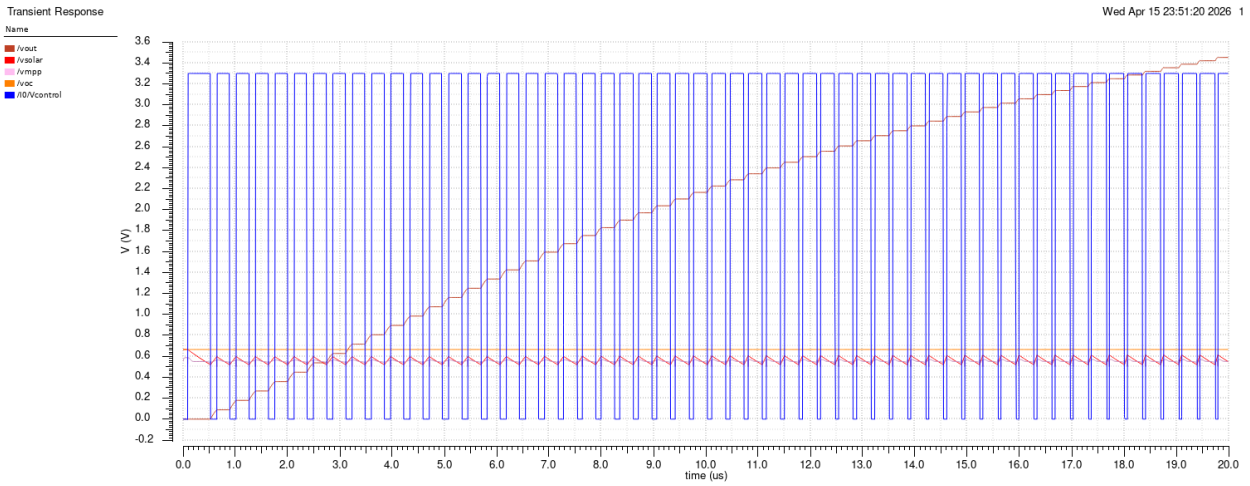

Figure 24: μSolar Testbench Results

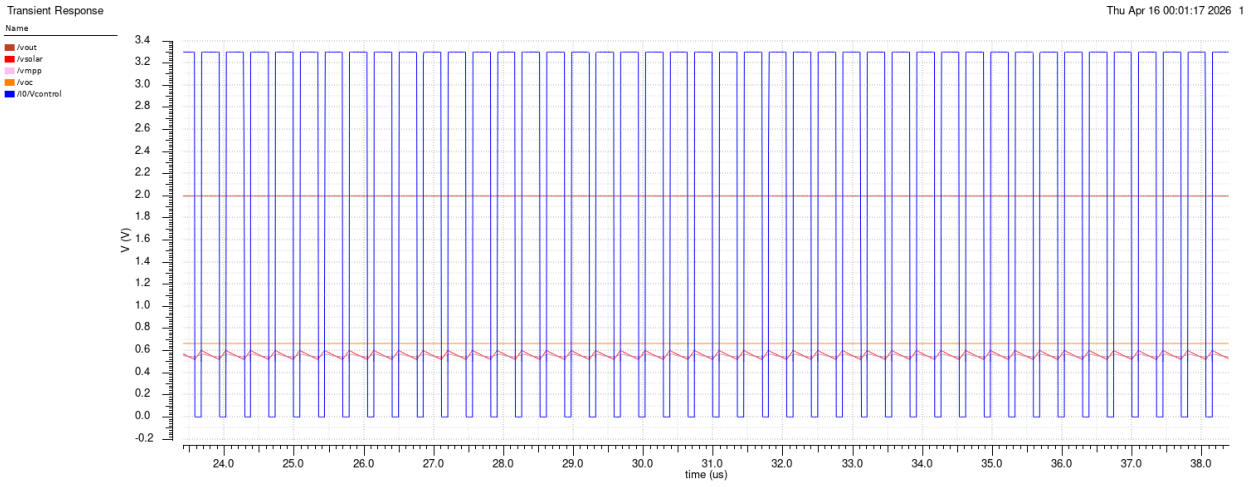

Figure 25: μSolar Steady State Loaded Testbench Results

5.3 Real-Time Clock Generator Implementation

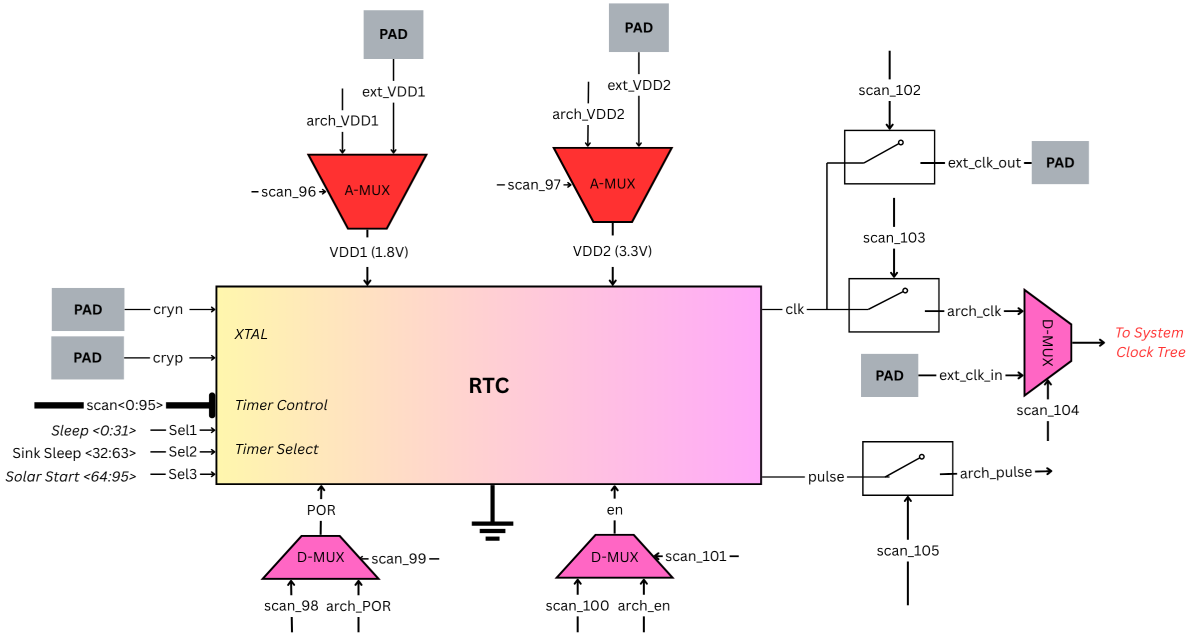

Figure 26: RTC Configurability Layer

To create a timer, a clock must be provided to be able to generate periodic pulses. By using an external source such as a crystal, it must be oscillated to generate a clock. The pads for cryn and cryp are connected to both ends of the 32kHz crystal feeding into XTAL, where clk is the rail-rail square wave clock. Two voltage sources will be needed for the RTC, VDD1 will start up the oscillation for low power and VDD2 will shift up the signal to the proper voltage. The clock will be sent directly as an output sink and to the clock tree, although if the clock is not received then an external clock can be used as an input source. The RTC contains multiple digital inputs that are used to determine the timer's period. To determine the period, input data is fed into the Timer Control of the RTC, which are three 32bit numbers that have different values and can be selected by the digital control unit. The Power on Reset (POR) will restart the counter's value to allow the timer to start at zero seconds while the Enable (EN) will allow the RTC to generate the timer's pulses. There will be overrides via the Scan Chain to disable the clock and timer output feeding into the clock tree, in case the timer or clock is being deactivated.

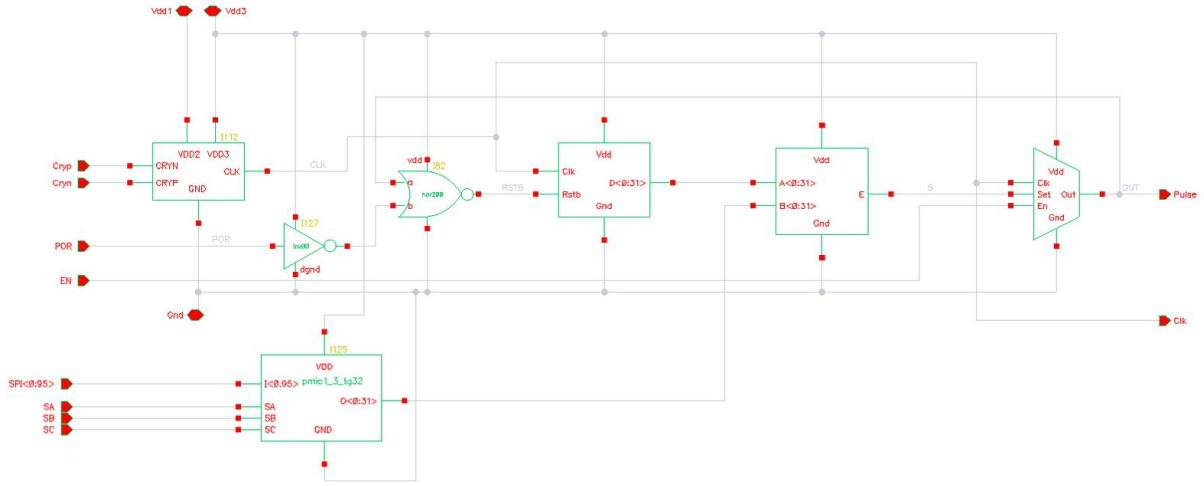

Figure 27: RTC Top Level Design

The primary functional components of the system are the oscillator, counter, and comparator. The oscillator generates a stable reference clock signal, 32kHz, which is then passed to a counter stage, in which the high-frequency signal is converted into a much lower frequency timing signal. The comparator stage monitors the counter output and compares it to a predefined reference value corresponding to the desired timing interval. When the counter reaches this programmed value, the comparator generates a signal indicating that the specified time period has elapsed. At the output of the system, a clock gate is implemented to act as a latch, ensuring that the timer pulse is generated only during a controlled portion of the clock cycle. This gating mechanism guarantees that the output pulse occurs while the clock is high, preventing multiple triggers caused by the rising or falling edges of the clock signal. By synchronizing the output event in this way, the design improves timing reliability and ensures stable interaction with downstream digital logic.

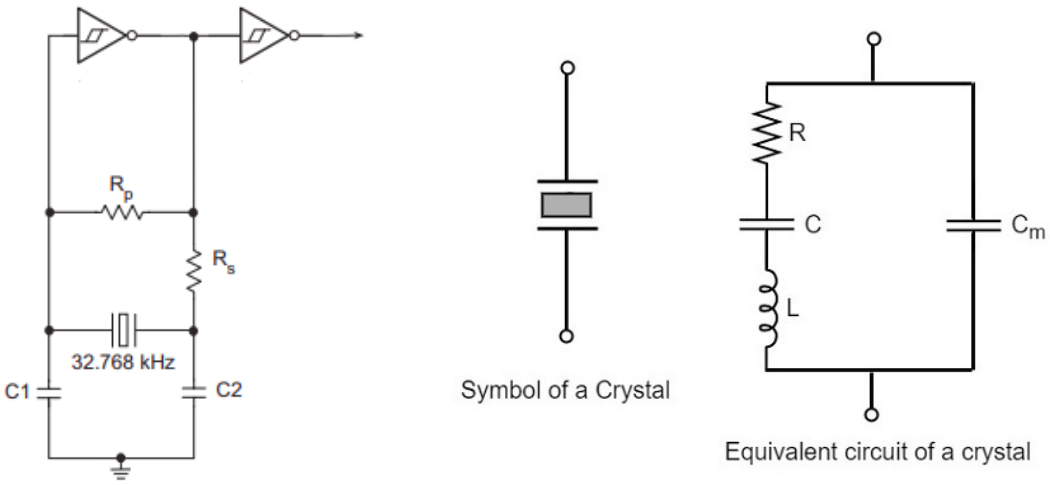

Figure 28: Pierce Oscillator with XTAL Equivalent Model

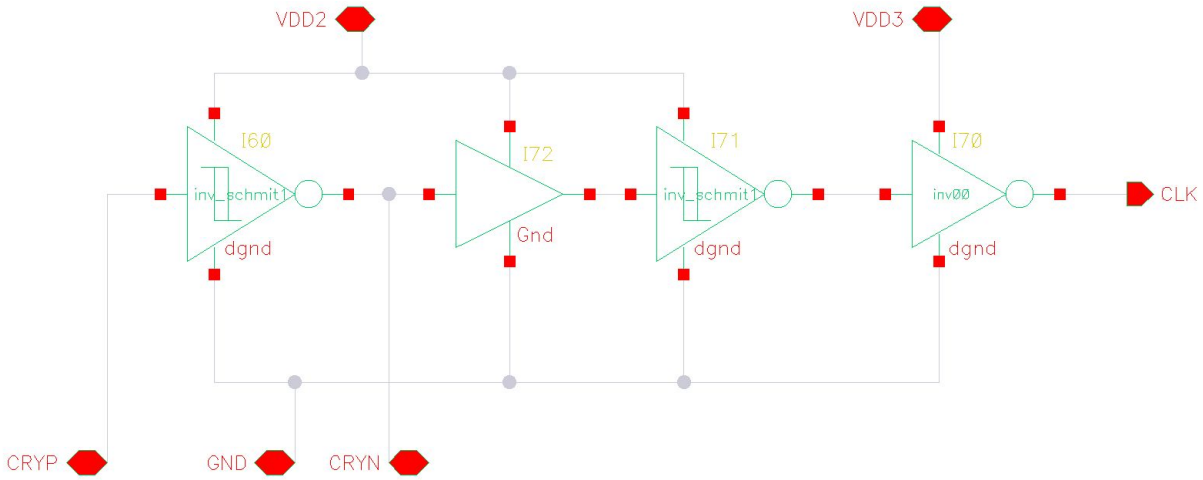

Figure 29: On Chip Oscillator

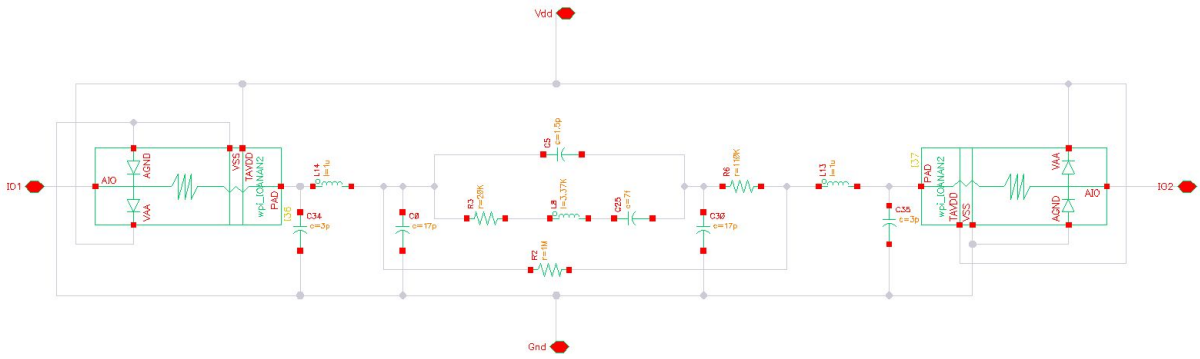

Figure 30: Off Chip Crystal Model

The oscillator was implemented using two Schmitt trigger stages. The first Schmitt trigger functions as the primary amplification stage, providing the necessary loop gain to sustain oscillation from the crystal. The second Schmitt trigger acts as a buffer, isolating the oscillator core from the load and producing a cleaner square-wave output for the subsequent digital circuitry. A tapered buffer was inserted between these two stages to reduce dynamic power consumption by gradually increasing the drive strength rather than switching a large load directly. While this approach provided a small reduction in power consumption, further improvements were achieved by lowering the supply voltage of the oscillator core. To significantly reduce power consumption, the oscillator was operated at a reduced supply voltage of 1.8V instead of 3.3V. Since the downstream digital logic operates at 3.3VV, a level shifting stage was required to translate the oscillator output to the higher voltage domain. This level shifting was implemented using a simple inverter or buffer configured as a level shifter. As a result, the low-power 1.8V oscillator signal could be reliably converted to a 3.3V clock signal, allowing the remainder of the system to operate at standard logic levels while maintaining reduced overall power consumption in the oscillator stage.

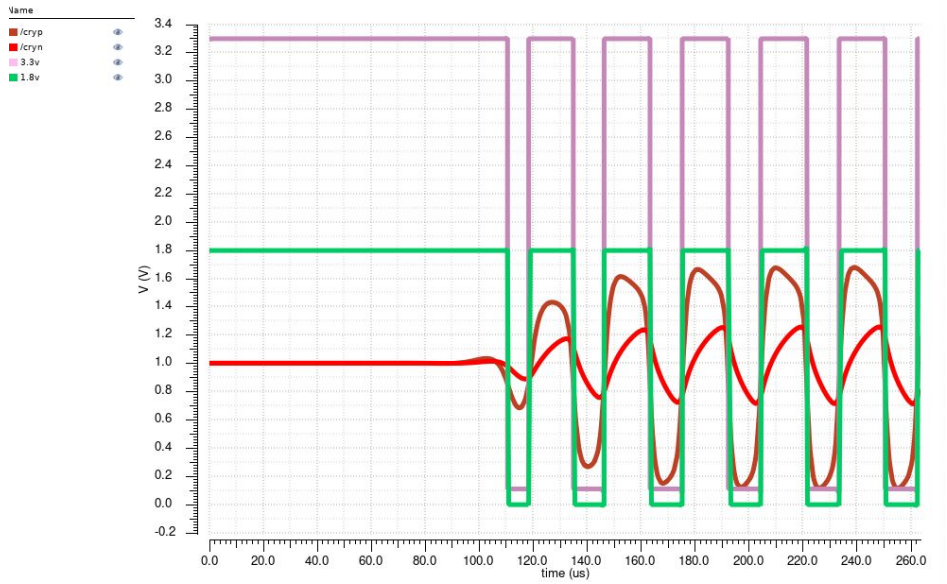

Figure 31: Digitization of Oscillation

The waveform shown illustrates the startup behavior of the crystal oscillator. Initially, the crystal requires a short period of time to begin oscillating and reach a steady-state periodic waveform. During this startup phase, the oscillation amplitude gradually increases as energy builds within the resonant network. Once the crystal oscillation becomes sufficiently strong and approaches the supply rail VDD1, the first Schmitt trigger activates. Due to the hysteresis characteristic of the Schmitt trigger, the small sinusoidal oscillation produced by the crystal is converted into a stable digital transition. This transition propagates

through the following stages, triggering the second Schmitt trigger and producing a clean square-wave output.

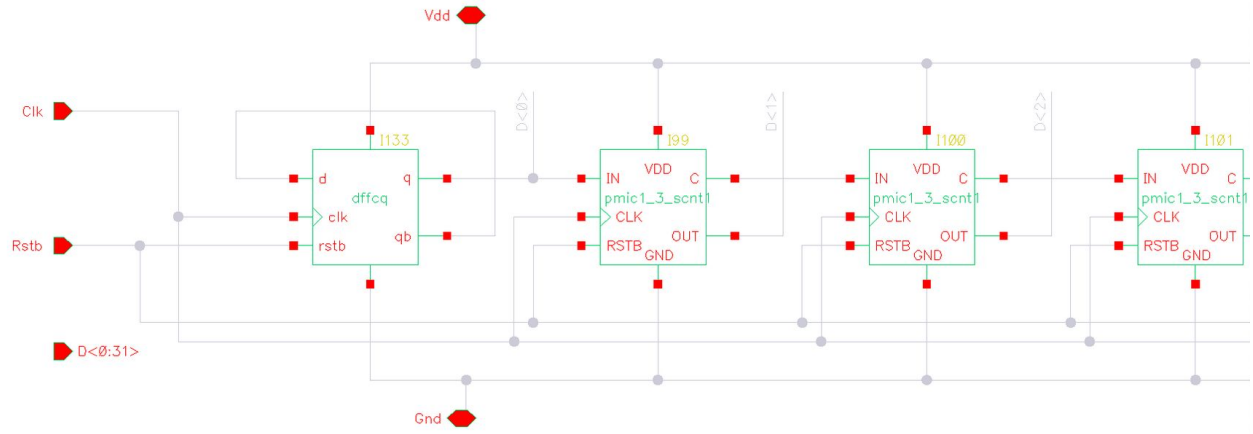

Figure 32: Synchronous Counter

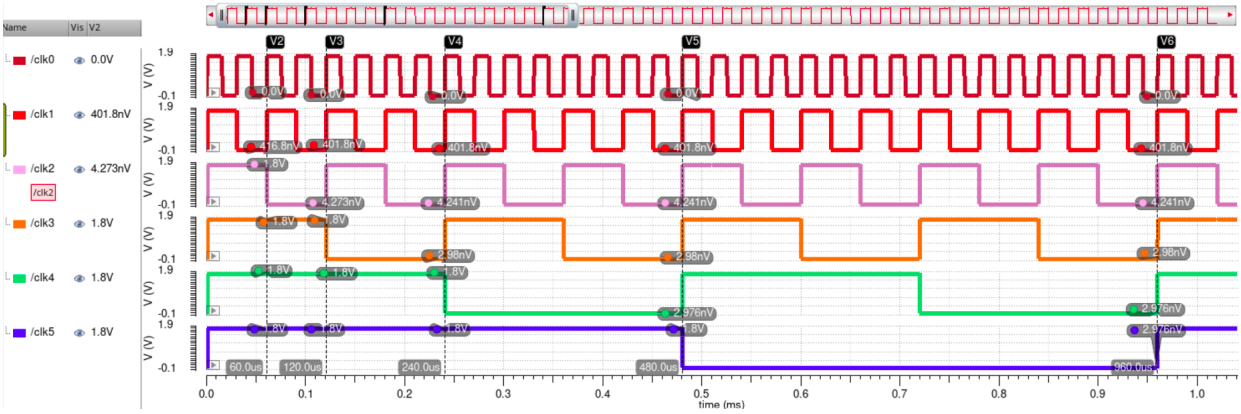

Figure 33: Frequency Division

The counter plays a critical role in the RTC system, as it enables the tracking of time by dividing the incoming clock signal into progressively lower frequencies. Each clock pulse produced by the oscillator represents a fixed time interval, and by counting these pulses the system can measure longer periods of time. The maximum period that the timer can represent is determined by the number of clock pulses that the counter can accumulate before resetting or triggering an event. This functionality is implemented using a chain of digital registers, typically realized as flip-flops connected in series. Each register divides the input clock frequency by a factor of two, producing an output signal that toggles at half the frequency of the previous stage. As more registers are added to the chain, the total division ratio increases exponentially. Specifically, a chain of N registers divides the input frequency by a factor of 2 to the power of (N). In this design, five registers are connected in series, resulting in a total frequency division of 32. Consequently, five

intermediate signals are produced, corresponding to successive frequency divisions of the input clock (2, 4, 8, 16, and 32). These divided clock signals can then be used by the RTC to generate longer timing intervals and support accurate timekeeping.

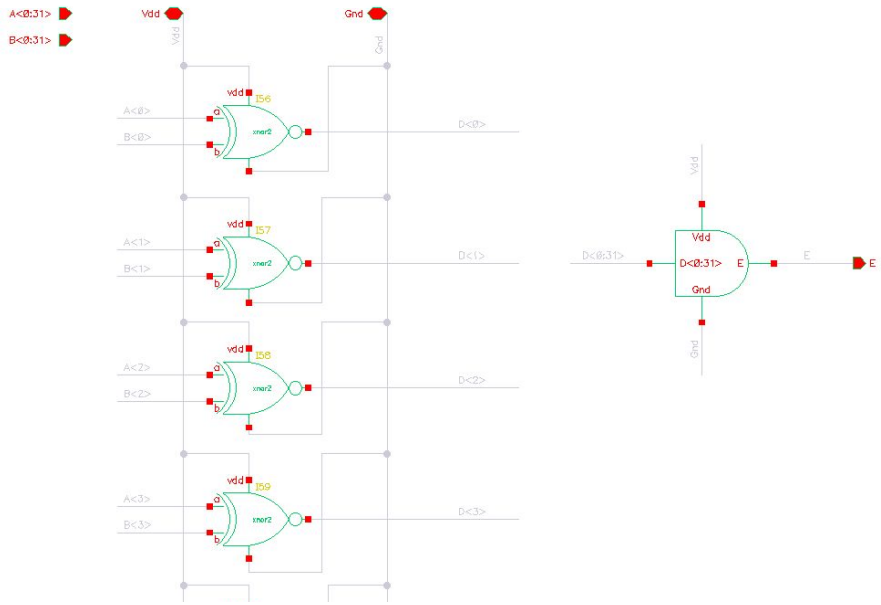

Figure 34: Comparator Design

The digital comparator is responsible for determining when the current counter value matches the externally programmed 32-bit input value. This comparison allows the RTC to detect when the desired timing interval has been reached. When the counter output becomes equal to the external reference value, the comparator signals the control logic to trigger the set/reset operation. At this moment, the timer generates an output pulse and the counter is reset, allowing the timing process to begin again for the next interval. The comparison operation is implemented using a bitwise equality check between the counter output and the external input value. Each corresponding pair of bits is connected to an XNOR gate, which produces a logic HIGH only when the two input bits are equal. The outputs of all XNOR gates are then combined using a large multi-input AND gate. This AND gate produces a HIGH output only when every bit of the counter matches the corresponding bit of the external input value. When this condition occurs, the comparator confirms that the two values are identical, signaling the RTC to generate the timing pulse and restart the counting cycle.

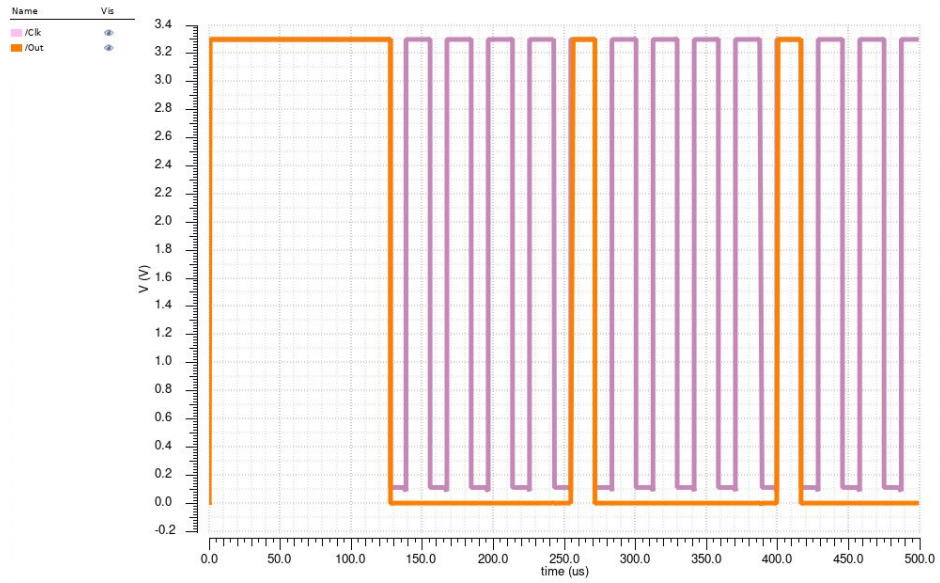

Figure 35: Timer Pulse vs Clock Output

As a result, the RTC is capable of generating pulses at well-defined time intervals, allowing the system to produce a range of lower-frequency timing signals derived from the original 32.768kHz oscillator. These divided clock signals can then be distributed to other sub-blocks within the integrated circuit, enabling them to operate at appropriate sampling or timing frequencies without requiring additional clock generation circuitry. This architecture allows the RTC to serve as a central timing reference for multiple components within the system. Overall, the design successfully met the functional and performance specifications of the RTC, including reliable oscillation, accurate clock division, and proper signal interfacing between voltage domains. However, the power consumption of the circuit exceeded the desired target, indicating that further optimization of the oscillator and buffering stages may be required in future design iterations.

Specification	**Desired**	**Achieved**
Timer Range	1ms - 1hr	30μs - 36.4hr
Temperature Range	-10° - 70°	Achieved
Power	$< 50\mu$W	429.3μW
VAA Range	2.5-3.3V	1.8-3.3V

Table 4: The RTC Specs Comparison

5.4 Data Converters Design Implementation

An 8-bit Successive-Approximation Register (SAR) architecture was chosen for the Analog to Digital Converter (ADC). For the Boost Converter Digital to Analog Converter (BC-DAC), a 6-bit String architecture was selected. In this section, the implementation and results will be explored.

5.4.1 Analog to Digital Converter: Architectural Integration

The SAR architecture was chosen for its inherent flexibility and small footprint. The goal of CRIMSON is to be as flexible as possible. SAR ADCs can often achieve a wide range of sampling frequencies, which allows us to use the ADC with the nominal 32KHz CRIMSON clock, or drive an external clock to sample higher frequency/dynamic signals [17]. With this goal in mind, the ADC was built using several reference designs, the most notable of which is [2].

To integrate the ADC into CRIMSON, the ADC Configurability Layer (depicted in Figure 36 was designed. Starting with the control signals at the top, the clock can be provided by the clock tree or it can be controlled directly by using the Scan Chain. This is an important measure as the clocking of the SAR ADC directly controls its convergence on the final digital value. The *start* and *en* signals are driven by the Finite State Machine (FSM), which has its own CFG Layer (in a later section) that allows for isolation of its output signals. The rst_n signal is driven by the PORPG, which also has its own CFG Layer. The *modesel* signal is driven architecturally, or it can be driven by the Scan Chain (like the clock). When the *modesel* signal is high, it allows the ADC to shift data out serially on the *sdata* line during clock cycles; when it is low, the ADC will perform its conversion. The power to the ADC can be provided architecturally from the ADC LDO, or from the external VAA pad (which bypasses all system LDOs). A similar structure is seen at the bias input, except there is a dedicated ADC bias pad. The voltage input to the ADC must pass through the ADC filter, which has its own bypass capabilities within its CFG layer, allowing for testing/full isolation of the ADC. The output bits $b0 - b7$ are captured by the Watchdog, but they are also driven to the Scan Chain to monitored when performing a full Scan Chain readout.

To operate the ADC, the start bit must be set. The conversion will take eight full clock cycles after the start bit has been set. On the ninth clock cycle, the ADC reading is final and is ready to be captured by the FSM and/or Scan Chain. The ADC will not reset until the rst_n signal is set and released; during this time, the ADC output can continue to be read. The ADC requires clock cycles to continue its operation, even during rst_n and *start* pulses. Thus, clock gating is used to put the ADC into low power mode when it is disabled.

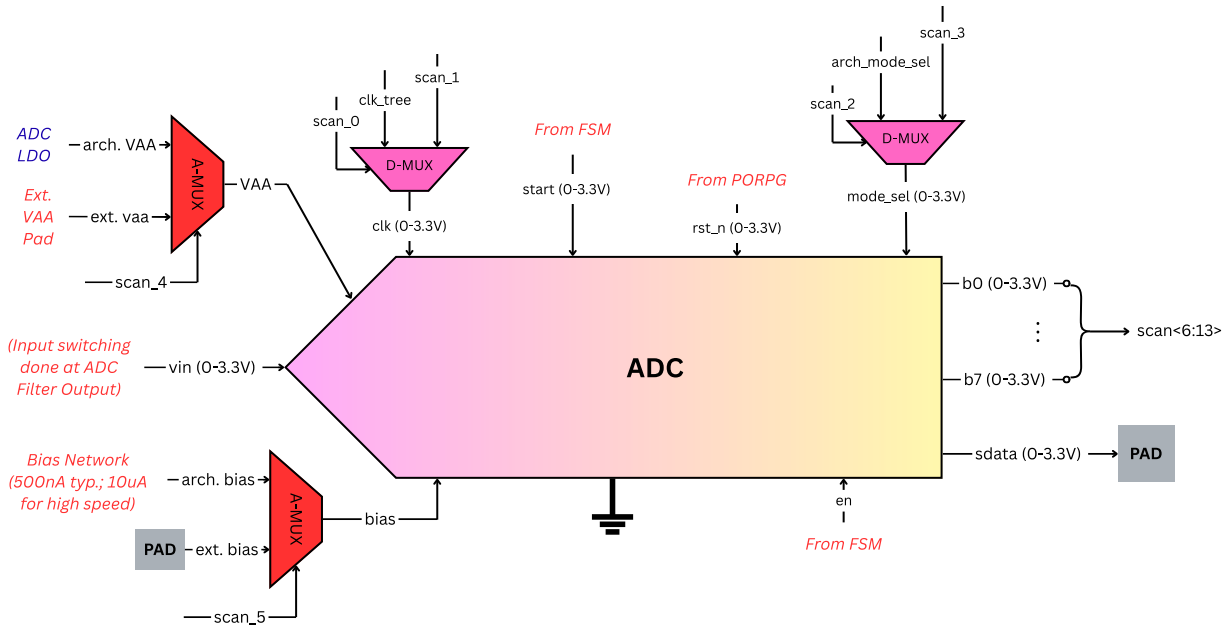

Figure 36: The ADC Configurability Layer

5.4.2 Analog to Digital Converter: Top Level Design

The top level schematic capture of the ADC is depicted in Figure 37. The SAR ADC consists of several sub-circuits, starting with a Sample and Hold (S/H) circuit that triggers on the *start* signal. The S/H circuit consists of a transmission gate and a 1pF sampling capacitor. The sampled signal drives the positive input of the comparator, which triggers on the negative edge of the clock. The output of the comparator is provided to the SAR Logic, which stores the readings, and drives the internal DAC, which drives the negative input of the comparator. The system uses clock gating to minimize power consumption when disabled.

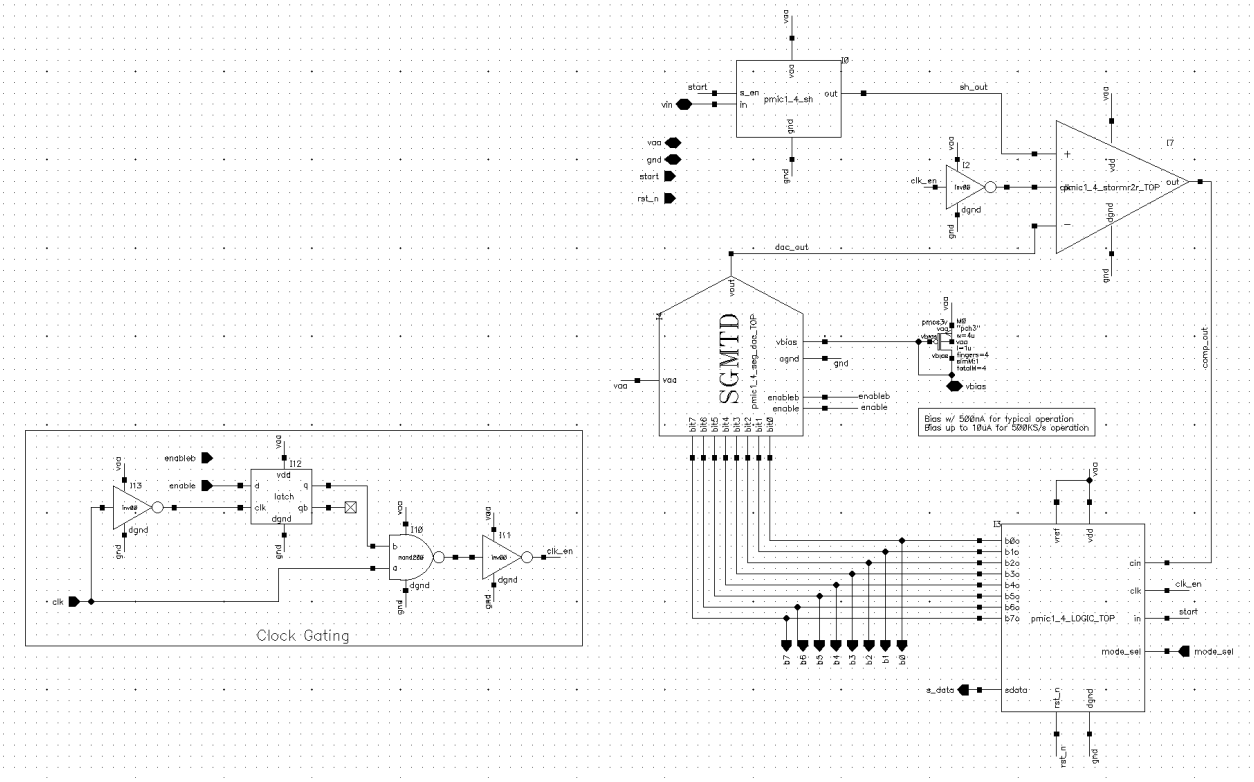

Figure 37: The ADC Top Level Schematic

5.4.3 Analog to Digital Converter: Comparator

The Comparator is a Rail-to-Rail StrongArm comparator, which is designed using a reference design in [27]. Figure 38 depicts the schematic of the comparator core. The outputs of the core (vx and vy) are latched to provide strong HIGH and strong LOW outputs. The comparator needs to be rail-to-rail so that CRIMSON can monitor the full range of possible input signals. The StrongArm architecture ensures the comparator is fast, reliable, and low power.

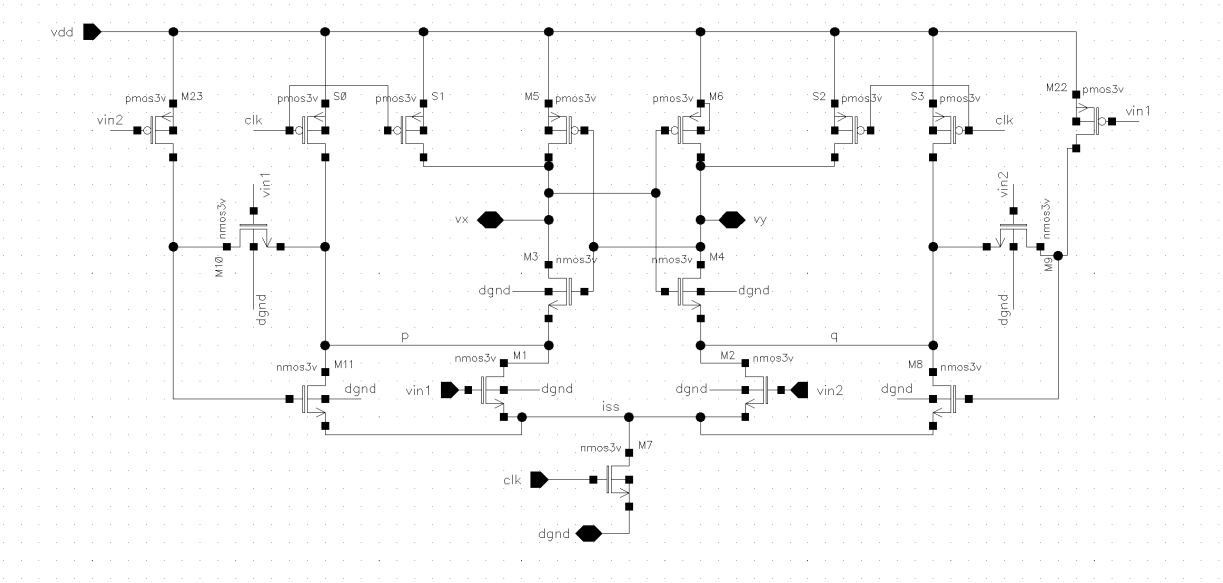

Figure 38: The ADC Comparator Core Schematic

5.4.4 Analog to Digital Converter: Logic

The Logic is recreated in TSMC180nm from the reference design [2]. The Logic consists of three major building blocks: the non-overlapping clock generator, the digital shift register, and the SAR logic block. For more information, the logic implementation is expanded upon further in the reference design. The timing diagram of the ADC control signals is depicted in Figure 39. Note, *modesel* corresponds to RST, and *start* refers to IIN in the Timing Diagram. In the reference design, $PRST$ acts as the ADC reset and is active high, whereas CRIMSON's ADC reset is active low.

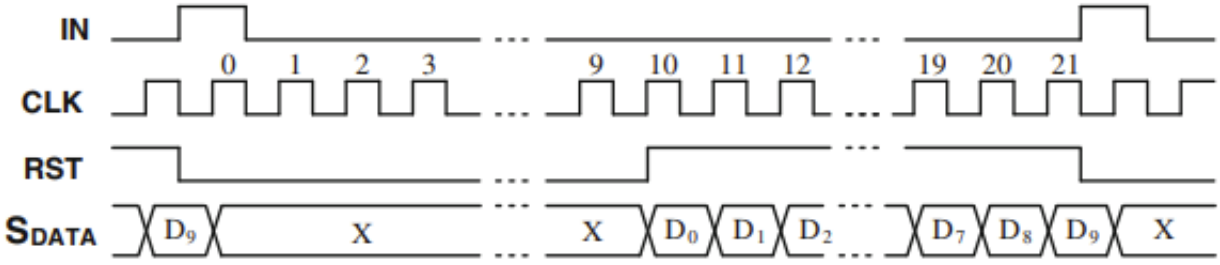

Figure 39: The ADC Logic Timing Diagram [2]

5.4.5 Analog to Digital Converter: DAC

The internal DAC is necessary to allow the SAR ADC to converge upon its final value. Typically, capacitive DACs are used in SAR architectures for their low-power, fast switching nature. However, because

capacitive DACs are size inefficient and subject to leakage and thermal noise, a capacitive DAC is unsuitable for the flexible nature of CRIMSON. Thus, a Segmented String DAC structure was chosen to allow for a much wider range of clocking frequencies and for a smaller layout footprint. In segmented DAC design, one can opt for several DAC sub-architectures, which can allow for smaller footprints and combined characteristics of each sub-architecture. The String architecture was chosen for both segments to optimize accuracy and settling speed [12]. A concept diagram of a Segmented DAC is depicted in Figure 40.

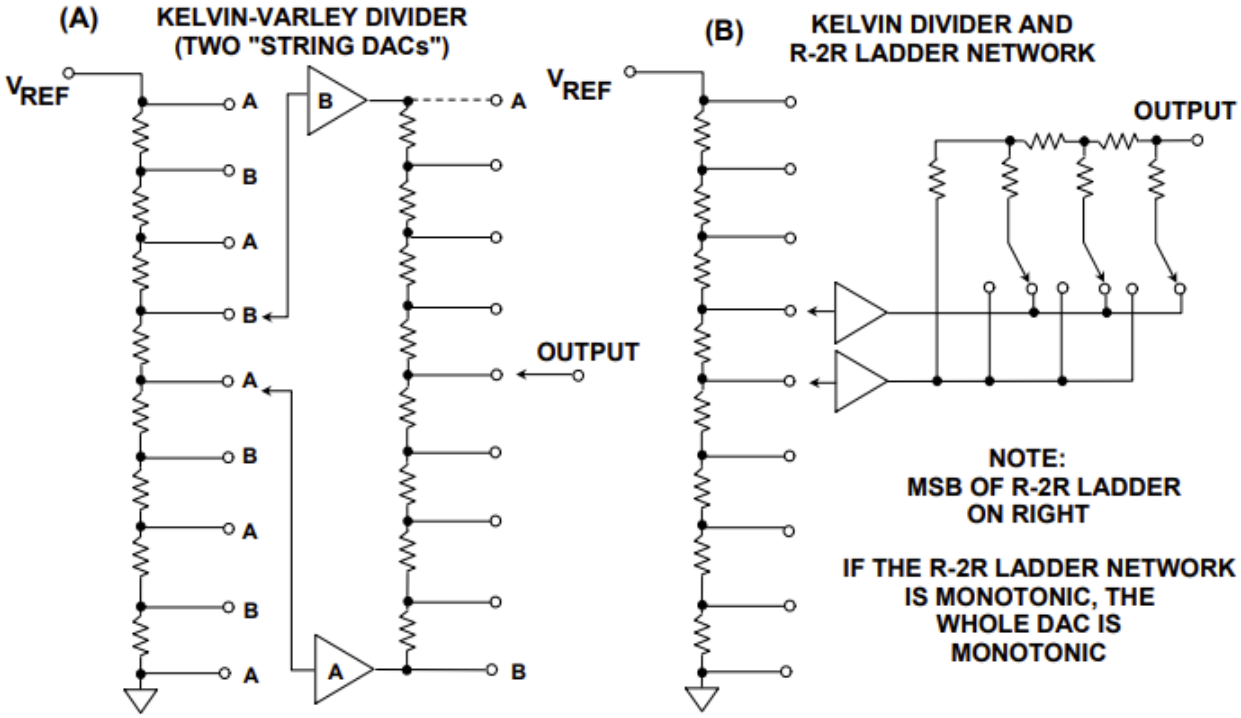

Figure 40: The Segmented DAC Concept Diagram

The Segmented String DAC consists of two String DACs; one is programmed by the four MSBs, the second is programmed by the four LSBs. The MSB String DAC has its high reference voltage set by the system voltage and its low reference voltage set by the system ground. It has two outputs, which set the high and low reference voltages for the LSB String DAC. Rail-to-rail opamp buffers sit between the two Strings to prevent resistive coupling. A third buffer sits at the output of the LSB String DAC to minimize loading effects at the input of the Comparator. Figure 41 depicts how the two String DACs are connected; Figure 42 depicts the implementation of the MSB String DAC; Figure 43 depicts the implementation of the LSB String DAC.

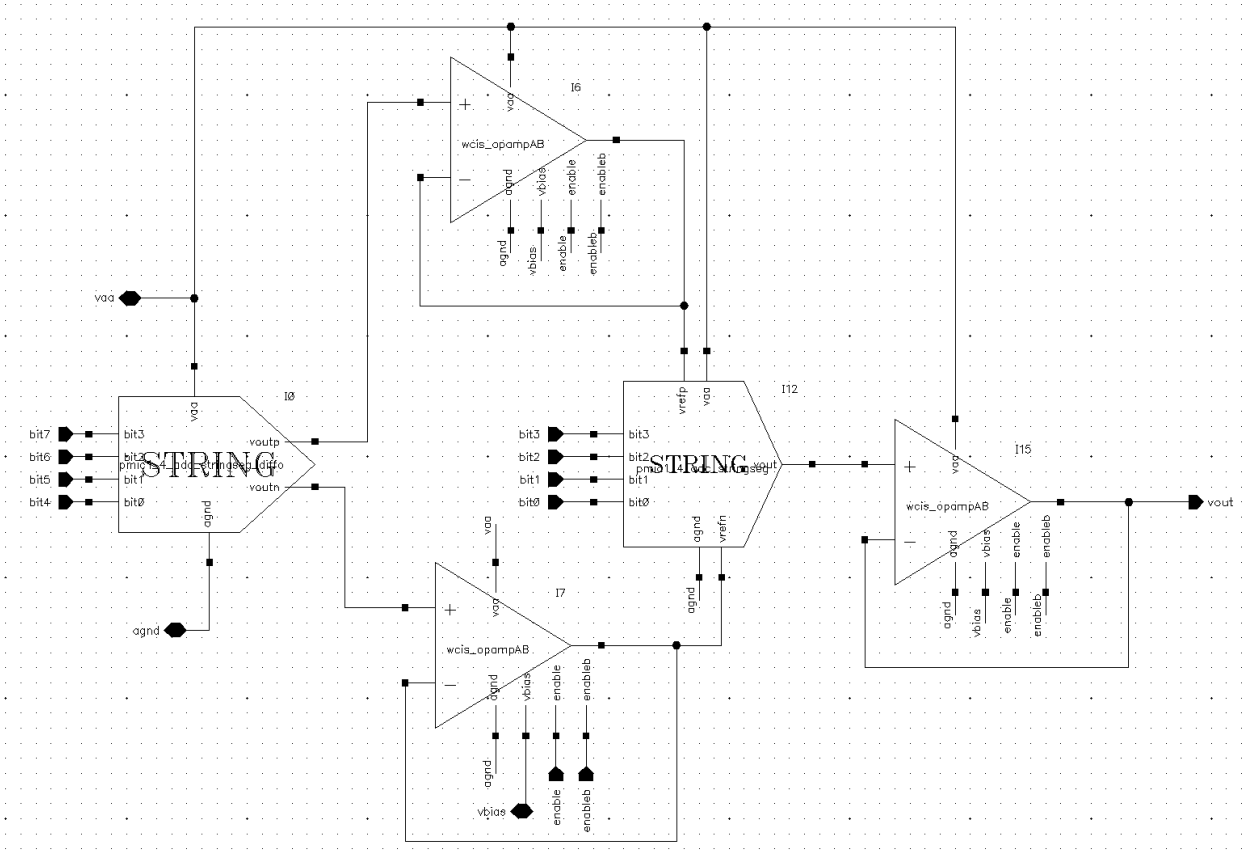

Figure 41: The Segmented DAC Top Schematic

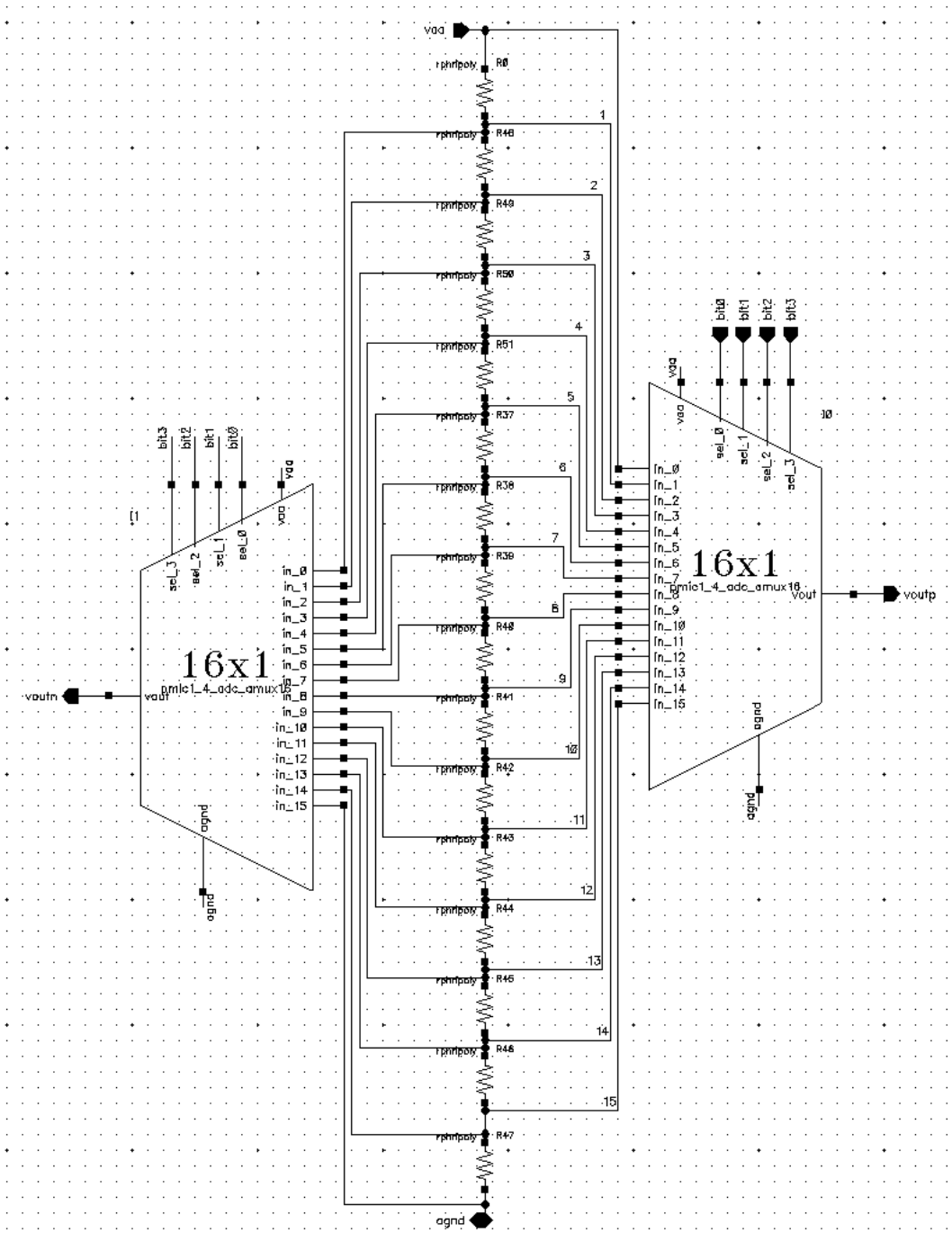

Figure 42: The MSB String DAC Schematic

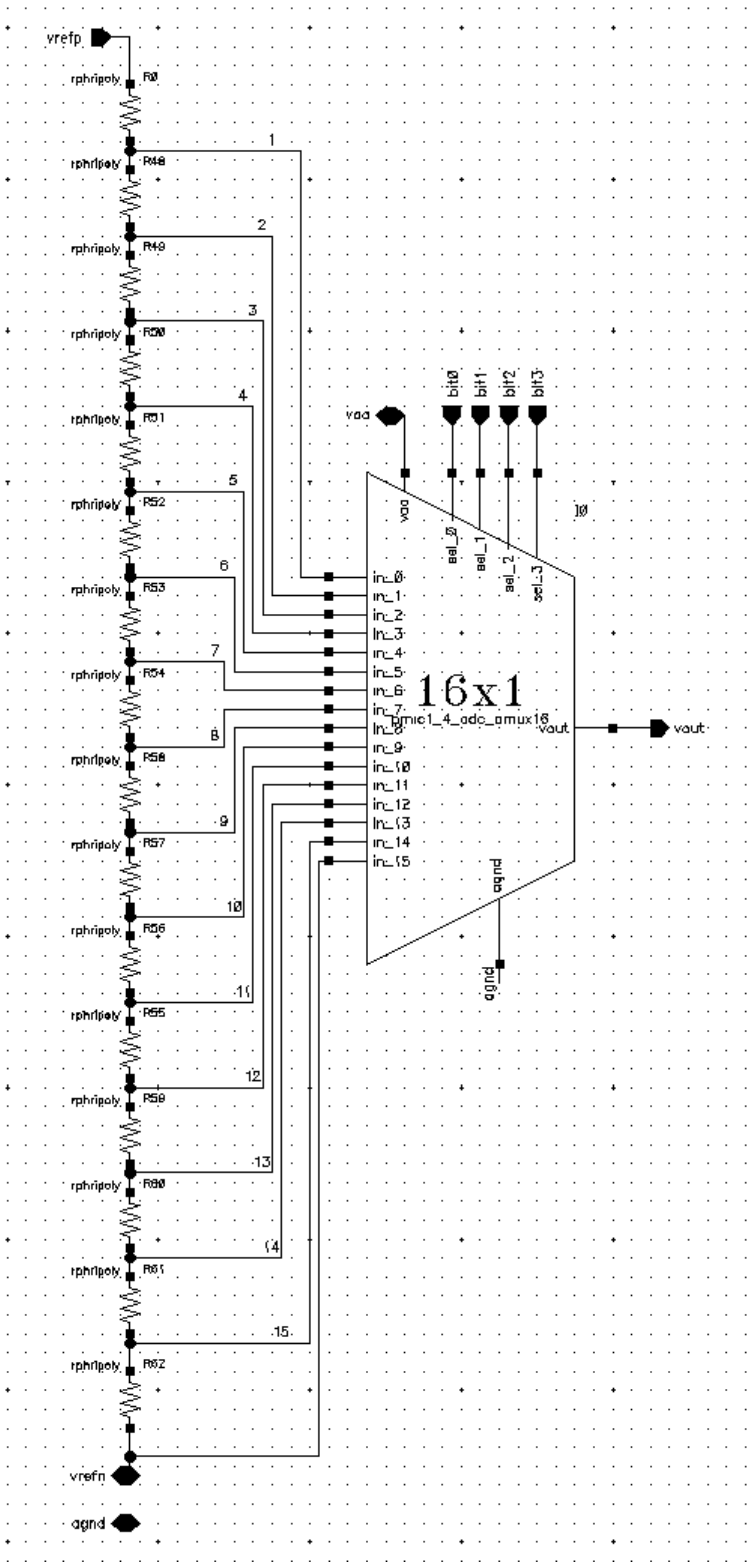

Figure 43: The LSB String DAC Schematic

5.4.6 Analog to Digital Converter: Results

The ADC was given several specs to meet. Table 5 provides the given versus the achieved specs. Figure 44 depicts the ADC in action. The input signal to the ADC in this simulation is 1.65V, which should give an ADC output of 10000000. The ADC has a final reading of 01111111, which is 1 LSB too low. However, it is consistent, as the second reading also gives 01111111, showing deterministic outputs. The signal dac_out in Figure 44 shows the convergence of the ADC on the sampled signal. Additionally, the current spikes that occur during the computation are seen in the bottom plot of Figure 44.

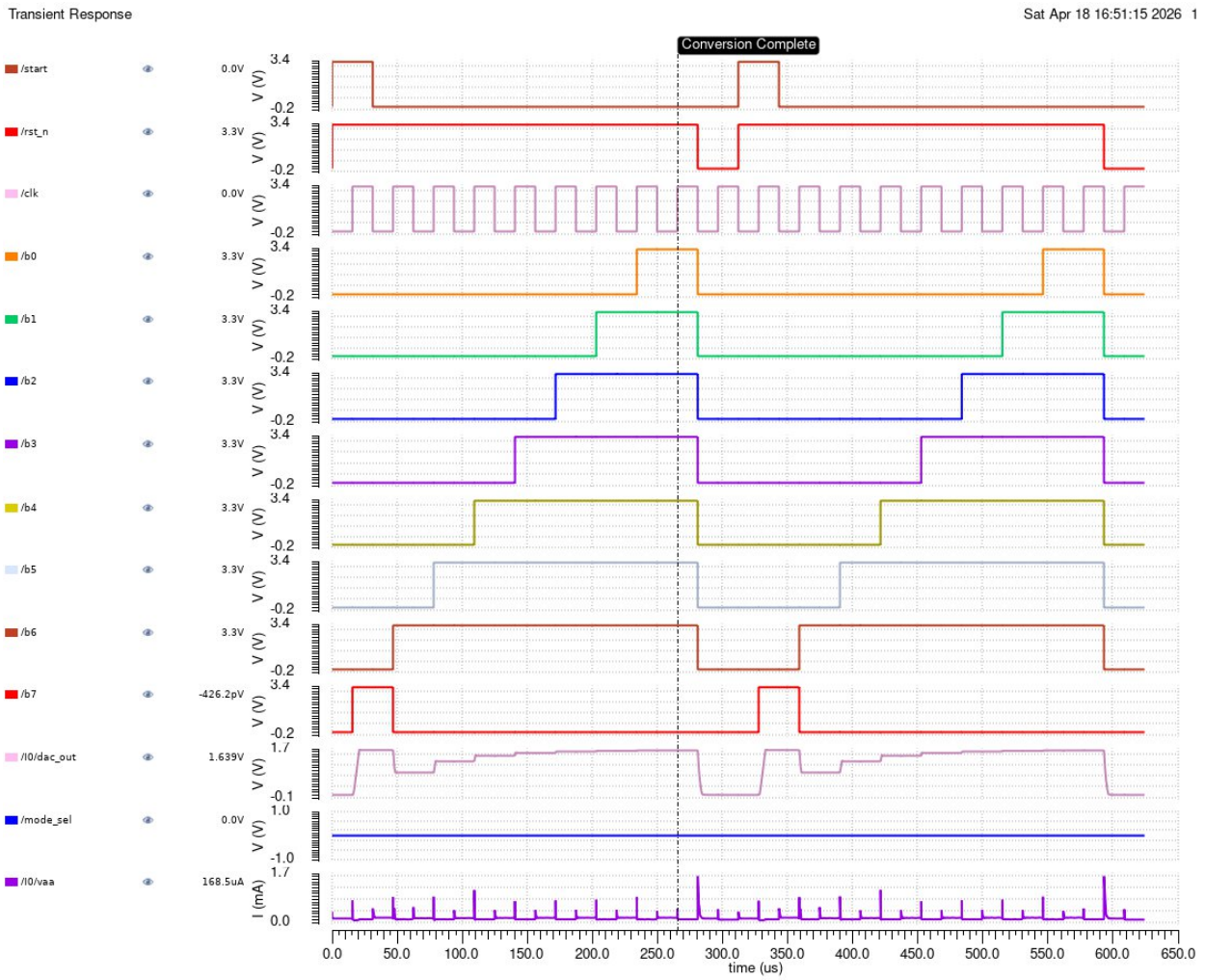

Figure 44: The ADC Results Graph

Specification	Desired	Achieved
Input Range	0.1-3.0V	0-3.3V
Sampling Rate	100S/s-100KS/s	100S/s-500KS/s
Temperature Range	-10° - 70°	-20° - 100°
Max Error	+/-1LSB	+/-0.5LSB
Power	$< 100\mu$W	442.2μW
VAA Range	2.5-3.3V	2.875-3.3V

Table 5: The ADC Specs Comparison

As intended, the ADC has a massive range of sampling rates, and achieves full rail-to-rail capabilities. It is also generally robust in terms of temperature tolerance, and is highly accurate thanks to the Segmented String DAC. The power consumption and VAA range specs, however, are not met. The power can be improved with smaller, less power hungry opamps. These opamps were pulled from the previous CHIPS Design Studio MQP, and do provide rail-to-rail DAC performance, but are not optimized for power consumption. This is also why the VAA range spec is not met; the opamp can no longer operate in saturation with a VAA range below 2.875, and thus cannot act as a true analog buffer stage with high current drive capabilities. Future works may include designing a rail-to-rail buffer capable of driving the String DACs.

To test the ADC, one should refer to Figure 36 for how to fully isolate the block, drive all control signals using the Scan Chain, and provide external power and bias as appropriate. For full automation, ensure the ADC outputs are grabbed by Scan Chain bits 6-13, cyclically scanning in control signals and scanning out the output signals to be processed by a script. For hands-on testing, use the *sdata* pad to monitor the ADC outputs and be sure to set $mode - sel$ high to switch the ADC into "shift-out" mode.

5.4.7 Digital to Analog Converter: Design

The standalone Digital to Analog Converter (BC-DAC) was designed to give the Boost Converter maximum flexibility while being space and power efficient. Since the BC-DAC need only to provide a static bias voltage, switching speed is not a concern. The optimal architecture for this DAC would be an R-2R Ladder; however, with a resolution of 6-bits and needing to be highly accurate, a 6-bit R-2R ladder required a longer lead time than was available for the design. Additionally, the Boost Converter was given lots of chip area, which afforded us the resources to use the much more accurate String architecture instead. Figure 83 (BC CFG) shows how the BC-DAC is integrated into CRIMSON. The BC-DAC schematic capture is

depicted in Figure 45. It consists of a string of 64 high-resistive poly layer resistors (of minimum sizing) with four 16:1 analog multiplexors, combined with a 4:1 analog multiplexor. The result is a highly accurate, compact DAC.

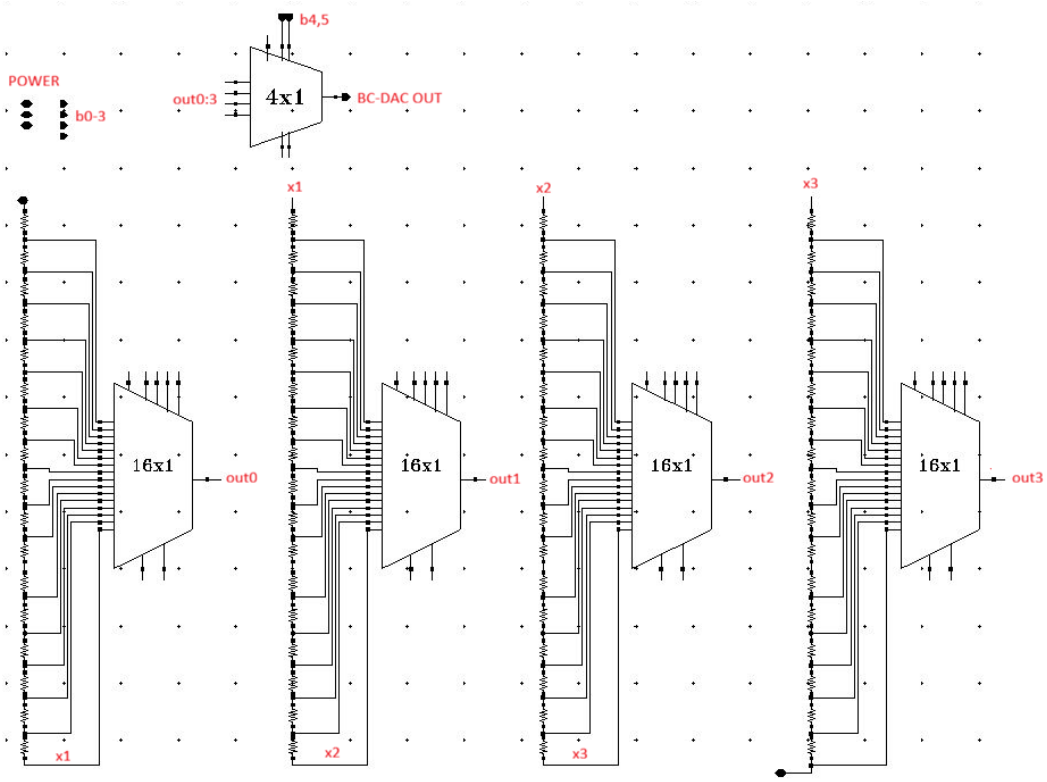

Figure 45: The BC-DAC Top Schematic

5.4.8 Digital to Analog Converter: Results

Since the BC-DAC is a sub-block of the Boost Converter, it was most important for the BC-DAC to be accurate and low power. The desired versus achieved output values for each bit of the BC-DAC is depicted in Table 6. Each error is well below the 50% LSB error margin, making this DAC highly accurate. The average power consumption of the BC-DAC is 70.46μW. This can be reduced by increasing the size of the resistors. For the sake of size, the resistor sizing was minimized, but can be enlarged in less area-confined applications. A graph of the BC-DAC results is in Figure 46.

Bit Code	Ideal Output	Achieved Output	Error
000001	0.0515625V	0.04853V	-0.00303V
000010	0.103125V	0.09706V	-0.006065V
000100	0.20625V	0.19412V	-0.01213V
001000	0.4125V	0.43676V	0.02426V
010000	0.825V	0.825V	0V
100000	1.6V5	1.64945V	-0.00055V

Table 6: The BC-DAC Errors Per Major Bit Code

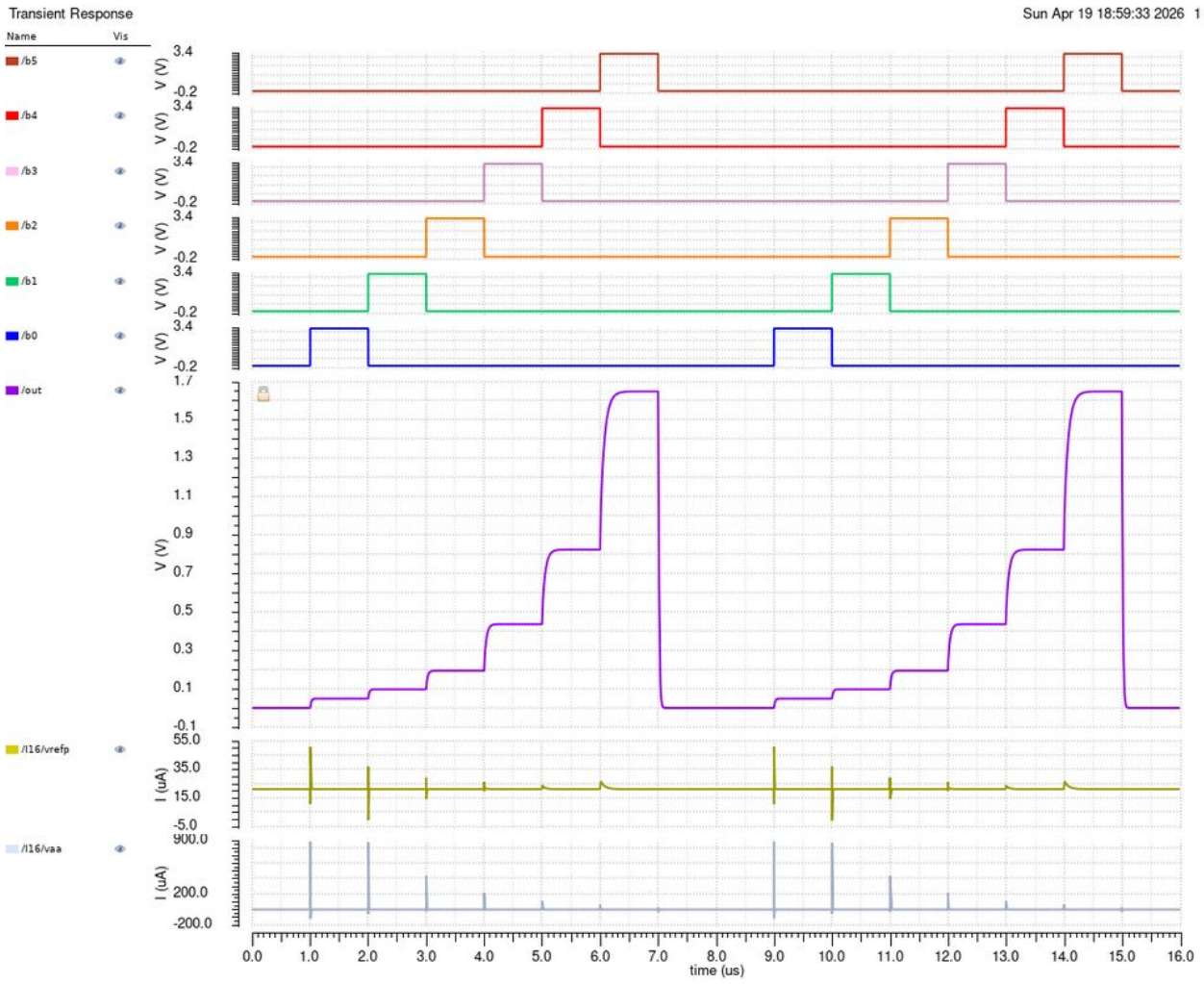

Figure 46: The BC-DAC Transient Simulation

There is no direct method for testing the BC-DAC as, again, it is part of the Boost Converter (a larger, testable block). To ensure the BC-DAC is operational, one can scan in bias Scan Chain values for

the Boost Converter and monitor its output dependent on the biasing. The output of the Boost Converter should roughly equal $V_b + V_t$ where V_t is the threshold voltage of the Boost Converter output transistors.

5.5 Linear Battery Charger Implementation

The linear battery charger consists of two main blocks: the CCCV (constant current/constant voltage) charger and the thermistor interface. CCCV charging is when the battery is charged at a constant current until a certain voltage is reached, then the battery will continue to charge at a constant voltage. This voltage is right under the max charge voltage of the battery, which in this case is 3.6 volts. The thermistor interface measures a beta = 3560K, 2.1kΩ nominal thermistor located directly on the battery. The interface will send a reduce charging current signal, terminate charging current signal, or no signal depending on the battery temperature in accordance with JEITA charging profiles [9].

5.5.1 CCCV Charging Stage

The design of the CCCV charging stage is based off the Analog Devices LTC4054 battery charging IC [24]. The CCCV charging stage is centered around a PMOS current mirror with a ratio of 1 to 100. The voltages on both sides of the current mirror are kept equal by a differential pair in negative feedback that outputs to a PMOS on the reference side. This enforces constant current operation. There is a second differential pair which establishes the relationship between a programming resistor and the output current. It does this through measuring the voltage across the programming resistor. The selected programming resistor is chosen through NMOS switches and a logic network that takes in the selected current of the scan chain as well as the reduce charge signal. The charger terminates charging through pulling the gates of the current PMOS high.

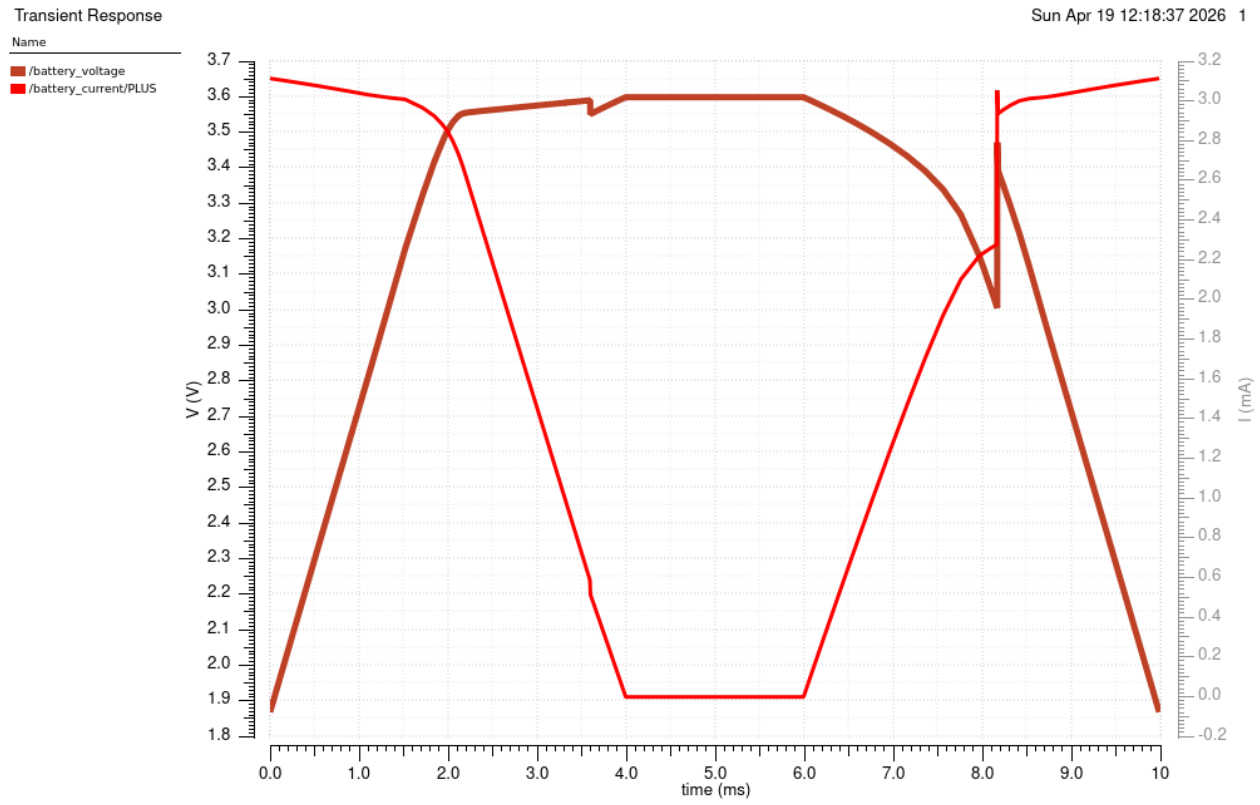

Figure 47: Example Charging Profile with Current Set to 3 mA

The example charging profile shows a transient simulation of a battery charging. From 0 ms to 2 ms the battery voltage increases while current stays at a constant 3 mA. This is constant current operation. From 2 ms to 4ms the current decreases while the battery voltage stays around a constant 3.6 V. This is constant voltage operation. At 3.5 ms the charger terminates charging. Charging resumes when the battery voltage falls to 3.0 V at 8 ms.

5.5.2 Thermistor Interface

The thermistor interface pulls a constant current through the thermistor. It then uses a series of comparators and voltage references generated from a resistor ladder to determine what temperature range the battery is in. It then sends the corresponding reduce charge or terminate charge signal to the battery.

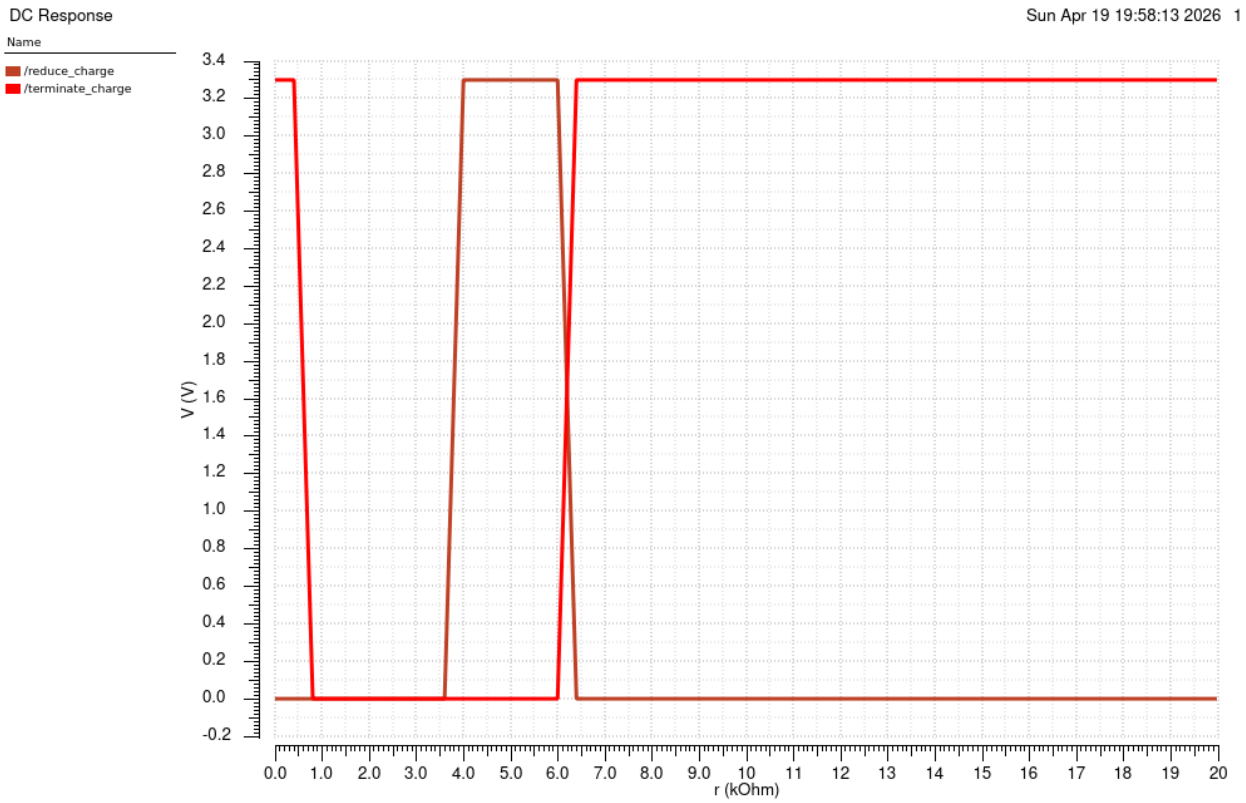

Figure 48: Thermistor Interface outputs per Thermistor Resistance

The thermistor output simulation shows how the resistance of the thermistor will affect the charging of the battery. When the terminate charge signal is high, the battery temperature is less the 0°C or greater than 60°C. When the reduce charge signal is high the battery is between 0°C and 10°C.

5.6 Digital Controller Implementation

To control the function of the chip, several digital control blocks were needed. The Power On Reset Pulse Generator (PORPG) creates a system-wide reset pulse for the chip after power stabilizes, allowing for all digital blocks to start in a known reset state. The Watchdog turns ADC readings into interrupts for the state machine using two ripple comparators to find if the reading is between an upper and lower bound. The ADC Controller generates signals needed to operate the ADC, providing current channel, a start signal, and a finished reading signal. The State Machine is a Moore state machine that controls enable signals for the entire chip, as well as determining timer lengths and from where the chip draws power.

5.6.1 Power On Reset Pulse Generator (PORPG)

A Power On Reset Pulse Generator (PORPG) is a circuit that exists to create a reset pulse for all sequential logic blocks on the chip after full power has been reached. The PORPG must also reset the system if a brownout (i.e. a temporary power loss) is observed. This is to prevent any logical glitches that may occur by operation below threshold voltages, and to ensure all logic starts in a known state. Our PORPG is a two part circuit based on a reference design in Ay, 2010[[1]]. The first part of the circuit, shown in Figure 49, is the logic, that produces a pulse when the two input voltages are different, using the two CMOS inverters to safely control the inputs to the XOR gate.

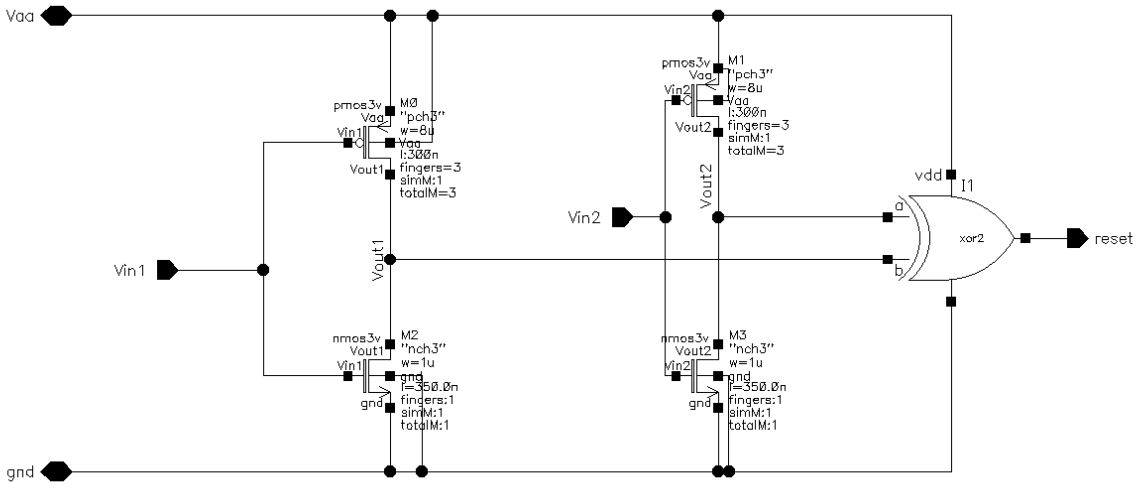

Figure 49: PORPG Logic Element

The second part of the circuit is the repeatable delay element, shown in Figure 50. This element

creates a delay in the rise of the output voltage Vout as compared to Vin. These elements can be chained together to customize both how long after power is reached that the reset starts (by adding delay elements before V1 is taken), and also how long the reset pulse is (by adding delay elements between V1 and V2). In Figure 51, our POR-PG circuit can be seen, using 3 cascaded delay elements. The delay between full power and the reset pulse is controlled by Element 1 and 2, and the length of reset pulse is controlled by Element 3, which is between V2 and V3, the input voltages being used for the logic circuit.

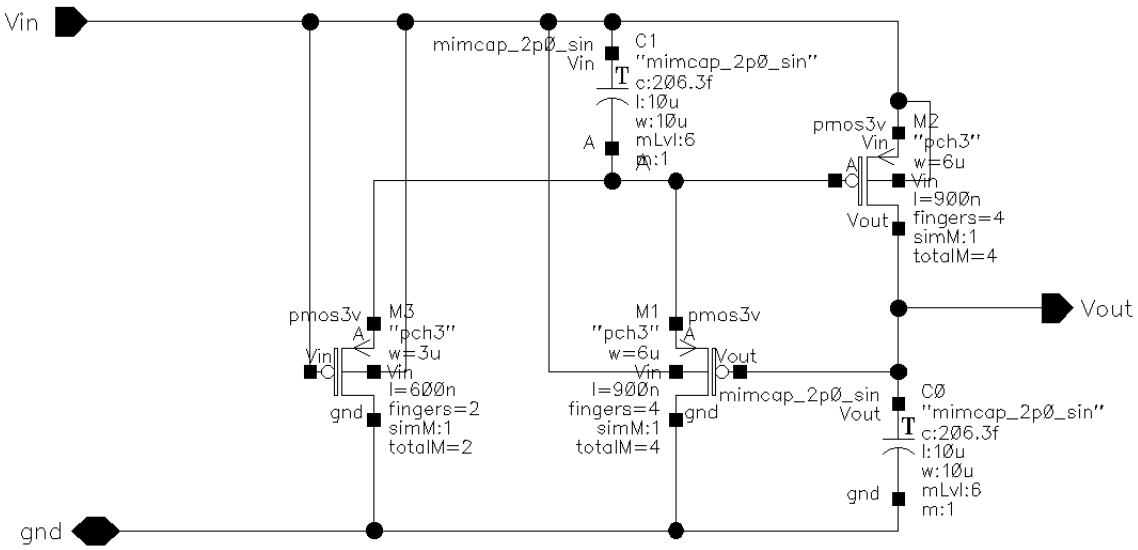

Figure 50: PORPG Delay Element

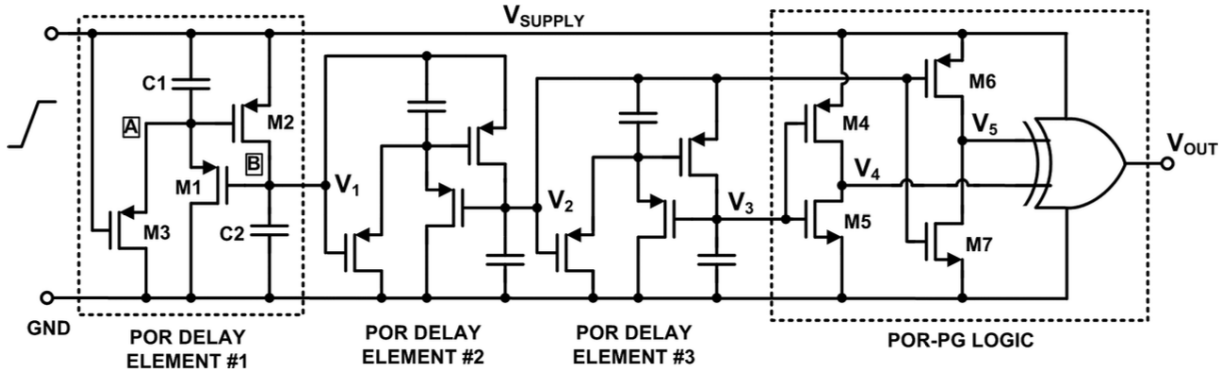

Figure 51: PORPG Complete Design

Below, in Figure 52, is a transient simulation of the circuit over a millisecond, showing the response to a 0.1 ms rise time in the power rail, and then a brownout where power drops to 1.35V. In both cases, the circuit properly responded with a reset pulse after a delay that would allow the system to regain full power.

You can also see how in1 and in2 (V2 and V3) are delayed from each other, caused by Element 3. The reset pulse time is different between the two pulses, which is the main weakness of this circuit: the delay and pulse length are dependent on the rise time of the voltage. However, in this system this will not cause an issue, as the reset on our logic elements is asynchronous, and the pulse lengths this circuit provides is more than enough for our logic.

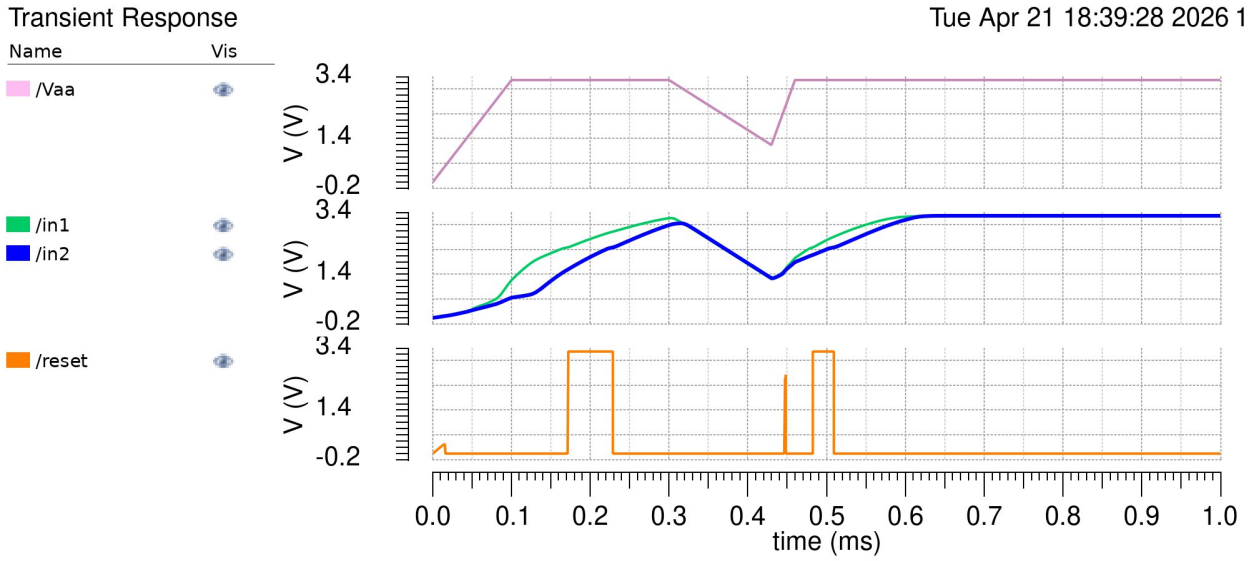

Figure 52: PORPG Transient Simulation

5.6.2 Watchdog Top Level

The top-level design of the Watchdog is depicted in Figure 53. The Watchdog uses two ripple comparators, one for the upper bound of the ADC value, which outputs true if the ADC value is less than or equal to the upper bound set in the Scan Chain, and one for the lower bound, which outputs true if the ADC value is greater than or equal to the lower bound in the Scan Chain. These are then AND'd together to create the signal "adcgood", which goes high if the ADC value is between the preset range.

To switch the upper and lower bounds against which the ADC signal is compared, the current ADC channel signals (adccha2, adccha1, and adccha0) are used to turn on and off the transmission gates connected to the scan chain bits. As there are 8 channels and each channel needs an 8-bit value to compare against, there are 64 scan chain bits per comparator, or 128 in total. By holding these values in the scan chain, changing signal thresholds is quite easy.

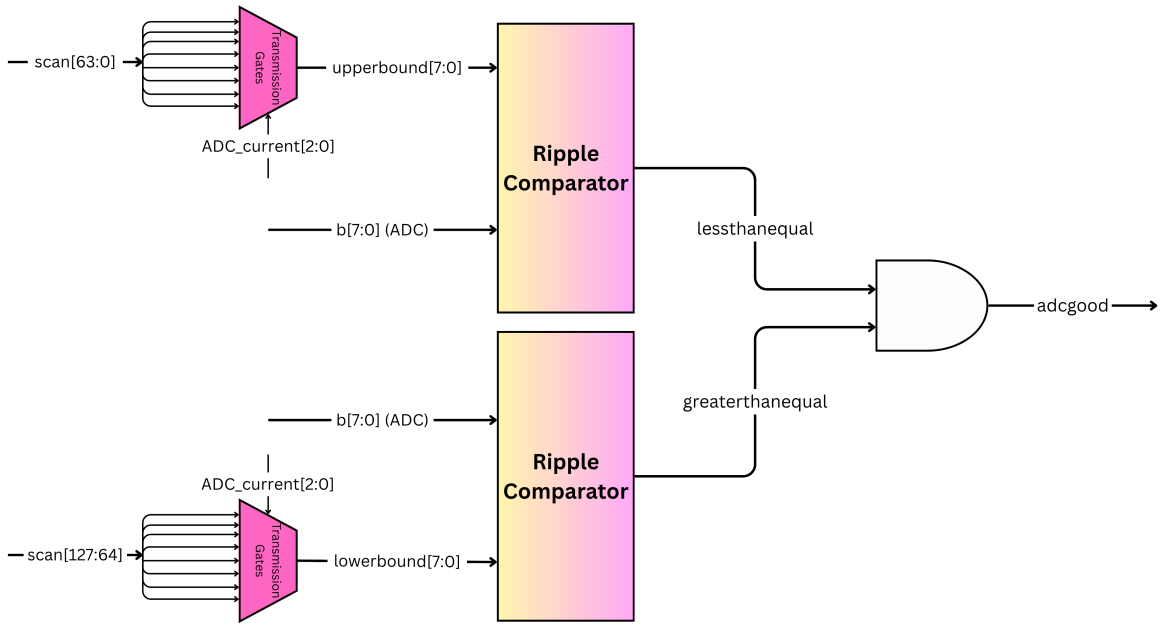

Figure 53: Watchdog Top Level

5.6.3 Watchdog Comparators

To perform the comparison, two ripple comparators were used, one to find whether the ADC value was equal to or below an upper bound, the schematic of which is shown in Figure 56, and the other to find if the ADC is equal to or above a lower bound, shown in Figure 55. As mentioned previously in Section 4.6.1 on page 13, the Ripple comparator finds greater than, less than, or equal to for one bit at a time, passing the information down to the next bit comparison. This results in a slower comparison than other options, but for our 30 kHz clock, the Ripple comparators are more than fast enough.

The comparator design uses eight identical bit comparison circuits, one for each bit, as seen in Figure 54. The upper NOR gate goes high when A is higher than B, the lower when B is higher than A, and the middle NOR when A = B. To find whether the entire number A is greater or less than the entire number B, NAND each of the previous middle gates with the current greater than or less than gate, doing so for each of the circuits. Then, NAND together all of those NAND gates.

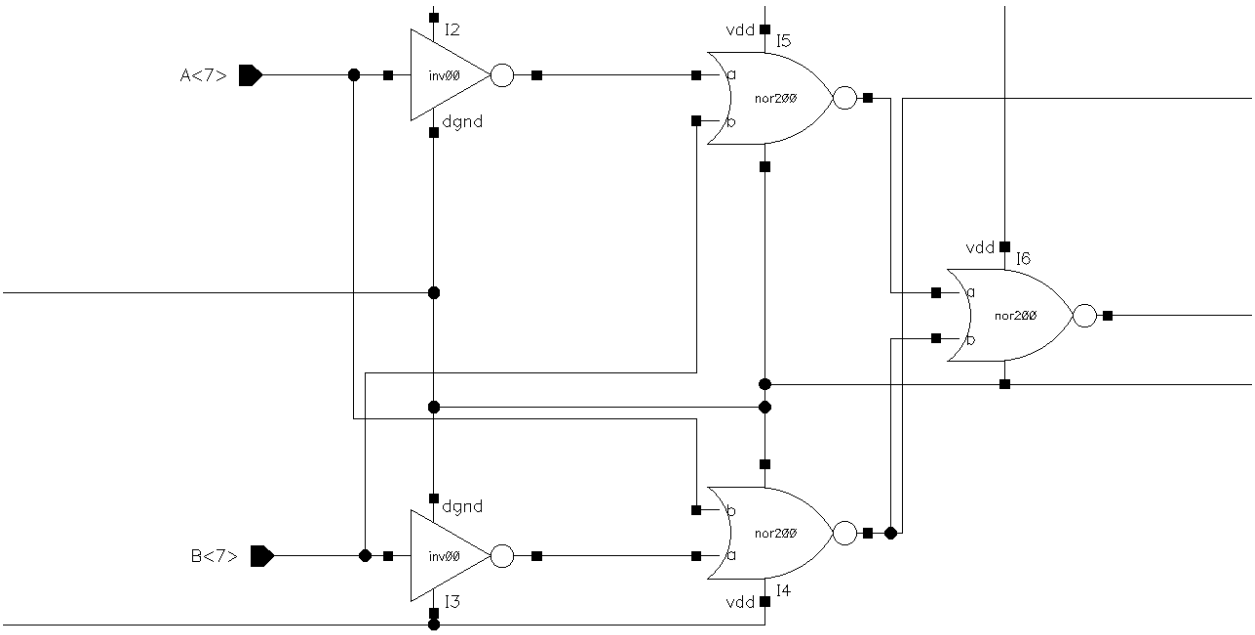

Figure 54: Comparison Circuit

The two comparators (Figures 55 and 56) are nearly identical, with one major wiring change: for the less than or equal to comparator (upper bound), all of the lower NOR gates in each bit comparison are NAND'd together; whereas for the greater than or equal to comparator (lower bound), all of the top NOR gates in each bit comparison are NAND'd together. For both comparators, this signal is OR'd with all of the equals signals AND'd together, calculated by the 8-input AND gate at the bottom of the diagram. This gives the final comparator output.

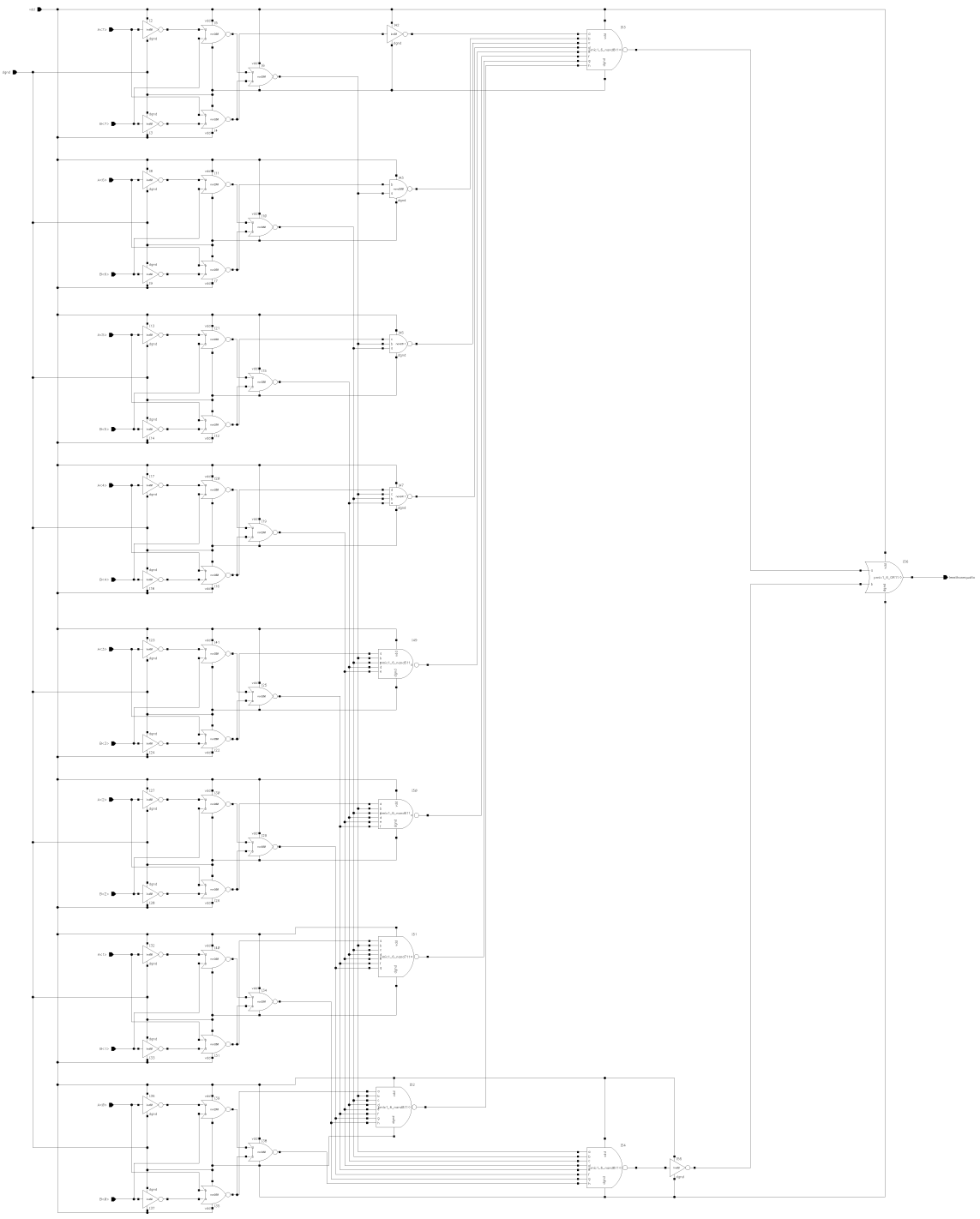

Figure 55: Less Than or Equal To Comparator (Upper Bound)

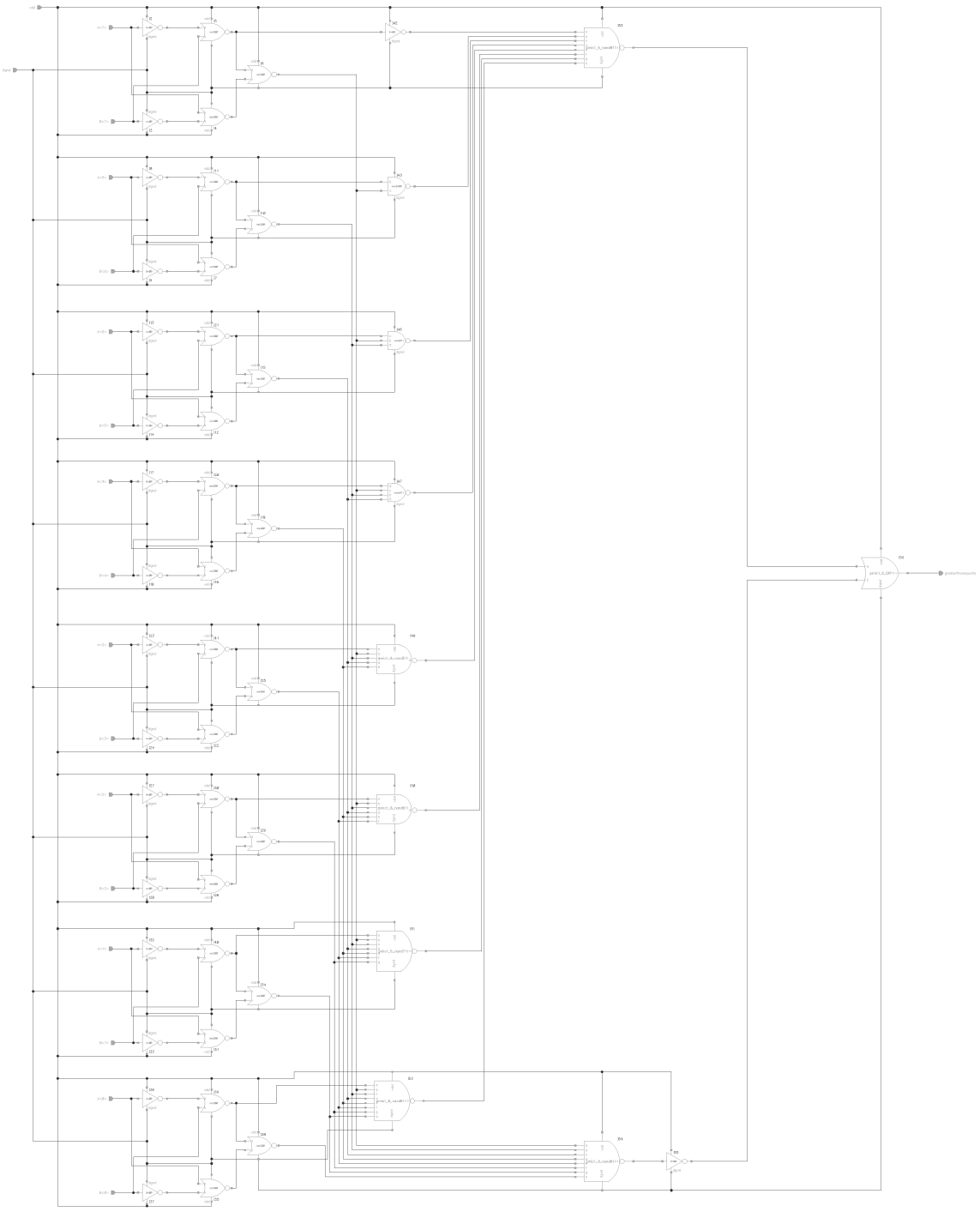

Figure 56: Greater Than or Equal To Comparator (Lower Bound)

5.6.4 ADC Controller

The ADC Controller exists to generate the start pulse needed by the ADC every 10 cycles, as well as to generate the control signals to decide which channel is currently being read (adccha 2-0), and to create the adcdone pulse for the State Machine to figure out when it can safely make a decision on the Watchdog output. For testing purposes, both the start signal and the ADC channel bits can be overridden with the scan chain.

At its core, it is two synchronous counters: the 10 cycle counter needed to generate the start pulse for the ADC, and a 3 bit counter that controls which ADC channel is being read. The 10 Cycle Counter, seen in Figure 57, counts 10 cycles for the 10 cycle operation of the ADC, setting a done signal high on the clock cycle the ADC has finished reading for use by the State Machine, and then the next cycle setting the start signal high, to re-engage the ADC and reset the counter.

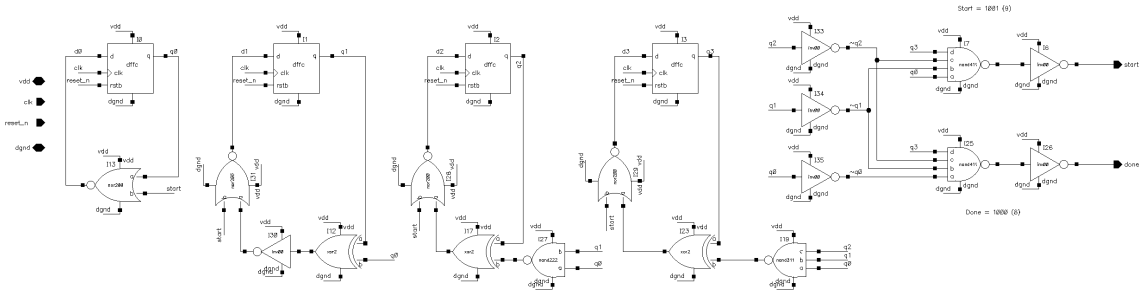

Figure 57: 10 Cycle Counter

The second synchronous counter, shown in Figure 58, only counts forward when start is asserted, as this counter controls which channel is currently being read by the ADC, and thus should only tick forward when starting a new reading. To create this behavior, MUXs with the start signal connected to select were used for switching between holding their current values and counting forward. The assignments of ADC channels to specific bits can be found in Table 7.

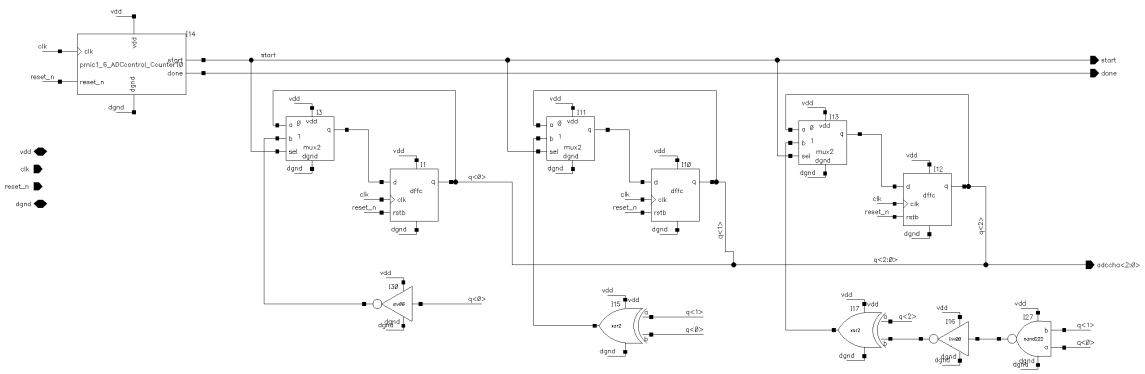

Figure 58: ADC Controller

Channel Name	adccha Bits
Output LDO Voltage	000
Boost Converter Voltage	001
Die Temperature Sensor	010
Backup Battery Voltage	011
Solar Generator Current	100
Solar Generator Voltage	101
Rechargeable Battery Current	110
Rechargeable Battery Voltage	111

Table 7: Channel Bit Assignments

The operation of the ADC controller can be seen below in Figure 59, with the ADC channel lines counting up once every 10 clock cycles, when the start signal triggers. The done signal can also be observed lagging behind the start signal by 1 clock cycle.

Figure 59: ADC Transient Simulation

5.6.5 State Machine

The State Machine is the brain of the chip, controlling block enables, timers, and power control. The top level design of the state machine can be seen in Figure 60. Combinational logic determines the next state based on ADC readings, the current state, power readings, timers, and more, passing it as an input to the state registers, which hold the state for the next cycle. Then, the output logic decides how the chip needs to operate based on the held state. Every output of the state machine, as well as the state registers themselves, have overrides through the scan chain that allow a potential tester to override enables or jump to specific states easily.

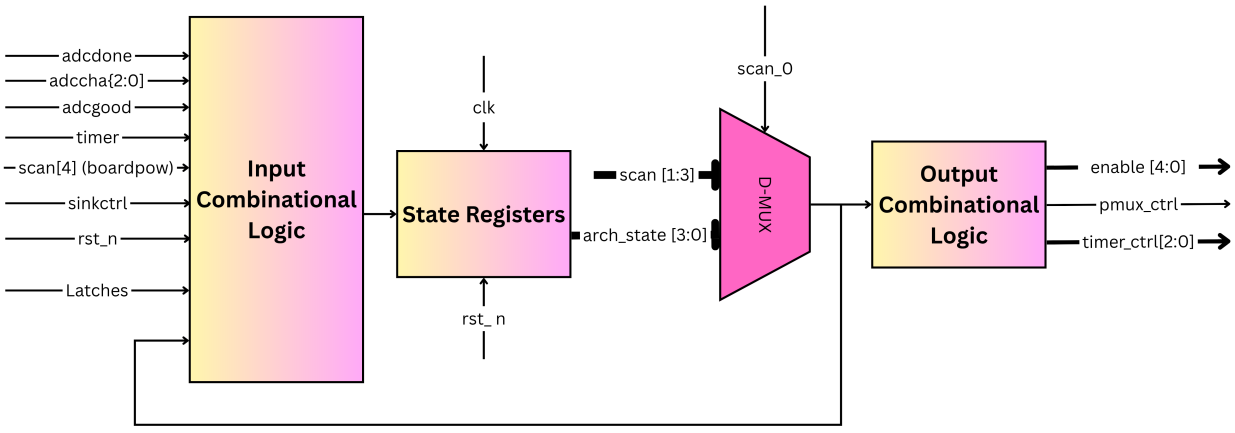

Figure 60: Full State Machine

The simplified movement between states can be seen in Figure 61.

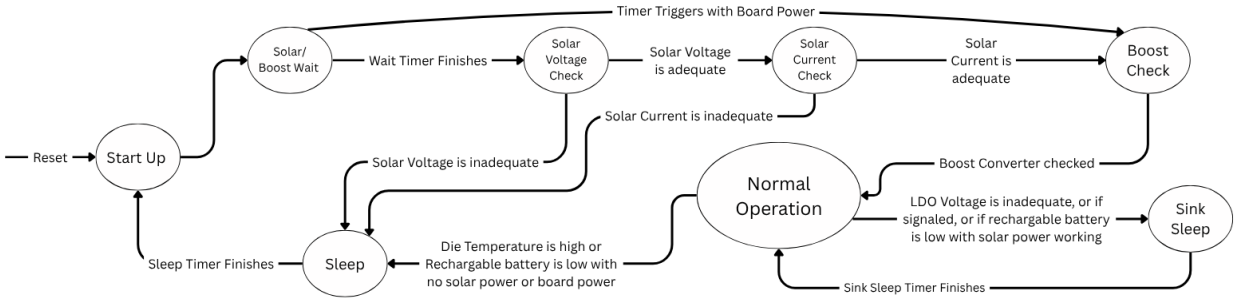

Figure 61: State Machine Transition Diagram

5.6.6 State Machine Next State Logic

Shown below in Figure 62 is the Next State Logic that determines the next state in sequence, depending on ADC inputs, timers, power status, the reset signal, sink control, and the current state. It has been proven to be optimally minimized by the Espresso-EXACT logic minimization algorithm, which has left it difficult for humans to follow. To carry information between clock cycles and states, the state machine uses two latches: boostg and solarg, which store whether the boost and solar generators are working. This needed to be done as the ADC values that indicate this can only be read every 80 clock cycles, so the values must be latched between readings.

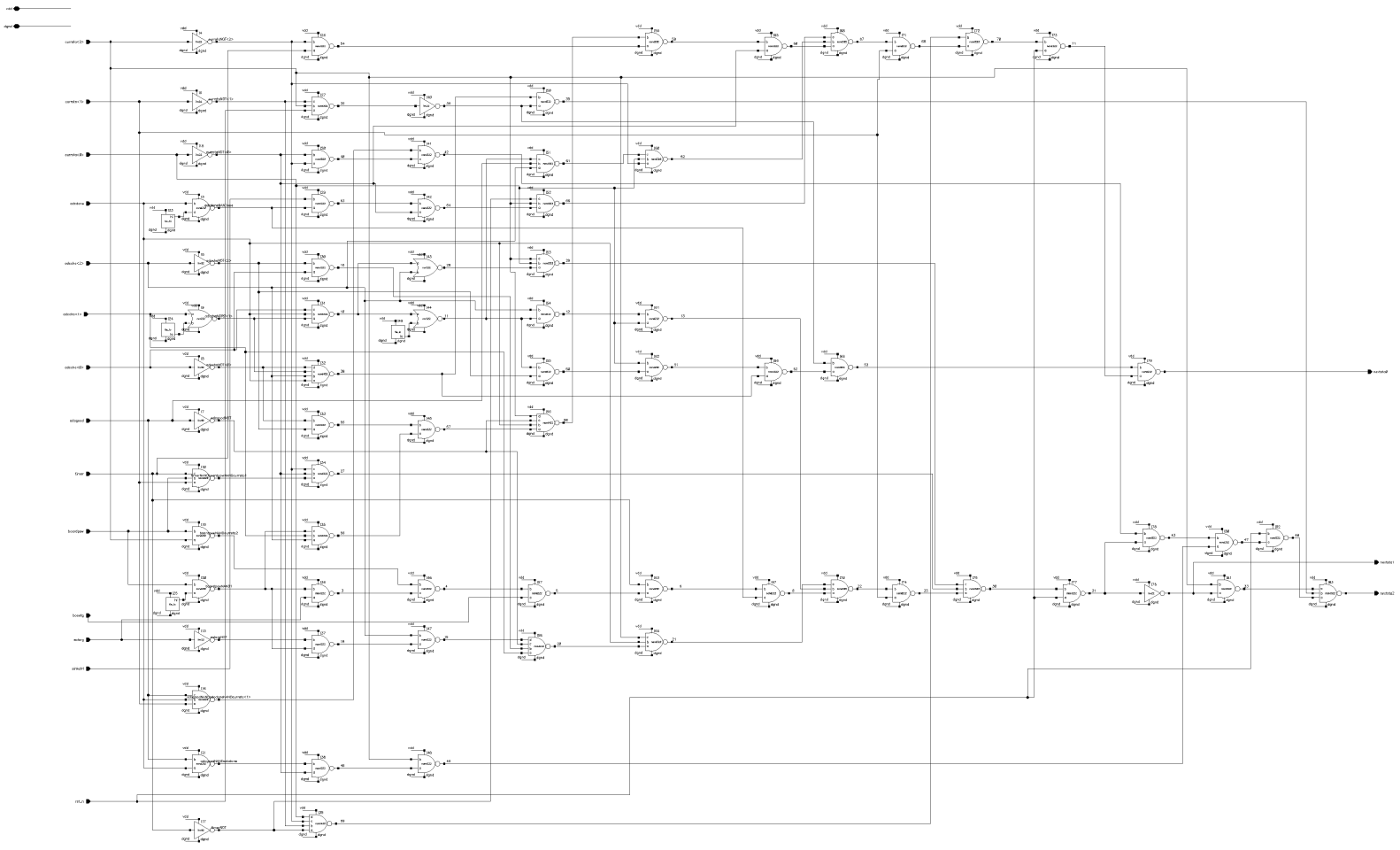

Figure 62: State Machine Next State Logic

The full list of signals that are inputs or outputs to the state machine are below in Table 8.

Name	**Function**	**Generated By**
currsta2-0	Current State, held by registers	State Machine
adcdone	ADC completed read signal	ADCControl
adccha2-0	Current ADC Channel	ADCControl
adcgood	Watchdog Signal testing ADC Value	Watchdog
timer	Timer for waiting set time periods	RTC/PLL
boardpow	Board Power Indicator	Scan Chain
boostg	Boost Good Indicator	State Machine Latch
solarg	Solar Good Indicator	State Machine Latch
sinkctrl	Signal from Sink to shut off output	Input Pin
rst_n	Reset signal active low	POR-PG
nextsta2-0	Next state signals, given to registers	State Machine
started_latch	Latch to hold whether startup has succeeded in the past	State Machine Latch
enable_X	Enables stated block	State Machine
timer_0	Turns on sleep timer	State Machine
timer_1	Turns on sink sleep timer	State Machine
timer_2	Turns on Solar/Boost startup timer	State Machine

Table 8: State Machine Signals

Current State	currsta2	currsta1	currsta0	adcdone	adccha2	adccha1	adccha0	adcgood	timer	boardpow	boostg	solarg	sinkctrl	rst_n	nextsta2	nextsta1	nextsta0	Next State	NOTES
X	X	X	X	X	X	X	X	X	X	X	X	X	X	0	0	0	0	Start_Up_Beginning (3'b000)	Reset
Start_Up_Beginning (3'b000)	0	0	0	X	X	X	X	X	X	X	X	X	X	1	0	1	0	Solar/Boost Wait (3'b010)	Exit Start Up Beginning
Sleep (3'b001)	0	0	1	X	X	X	X	X	0	X	X	X	X	1	0	0	1	Sleep (3'b001)	Stay in Sleep (Timer hasn't gone)
Sleep (3'b001)	0	0	1	X	X	X	X	X	1	X	X	X	X	1	0	0	0	Start_Up_Beginning (3'b000)	Leave Sleep
Solar/Boost Wait (3'b010)	0	1	0	X	X	X	X	X	0	X	X	X	X	1	0	1	0	Solar/Boost Wait (3'b010)	Stay in Wait (Timer hasn't gone)
Solar/Boost Wait (3'b010)	0	1	0	X	X	X	X	X	1	0	X	X	X	1	0	1	1	Solar Voltage Check (3'b011)	Leave Wait with NO BOARD POWER
Solar/Boost Wait (3'b010)	0	1	0	X	X	X	X	X	1	1	X	X	X	1	1	0	1	Boost Check (3'b101)	Leave Wait with BOARD POWER
Solar Voltage Check (3'b011)	0	1	1	X	0	0	0	X	X	X	X	X	X	1	0	1	1	Solar Voltage Check (3'b011)	Wrong ADC
Solar Voltage Check (3'b011)	0	1	1	X	0	0	1	X	X	X	X	X	X	1	0	1	1	Solar Voltage Check (3'b011)	Wrong ADC
Solar Voltage Check (3'b011)	0	1	1	X	0	1	0	X	X	X	X	X	X	1	0	1	1	Solar Voltage Check (3'b011)	Wrong ADC
Solar Voltage Check (3'b011)	0	1	1	X	0	1	1	X	X	X	X	X	X	1	0	1	1	Solar Voltage Check (3'b011)	Wrong ADC
Solar Voltage Check (3'b011)	0	1	1	X	1	0	0	X	X	X	X	X	X	1	0	1	1	Solar Voltage Check (3'b011)	Wrong ADC
Solar Voltage Check (3'b011)	0	1	1	0	1	0	1	X	X	X	X	X	X	1	0	1	1	Solar Voltage Check (3'b011)	Right ADC, Not Done
Solar Voltage Check (3'b011)	0	1	1	X	1	1	0	X	X	X	X	X	X	1	0	1	1	Solar Voltage Check (3'b011)	Wrong ADC
Solar Voltage Check (3'b011)	0	1	1	X	1	1	1	X	X	X	X	X	X	1	0	1	1	Solar Voltage Check (3'b011)	Wrong ADC
Solar Voltage Check (3'b011)	0	1	1	1	1	0	1	0	X	X	X	X	X	1	0	0	1	Sleep (3'b001)	Solar is Bad, go to sleep
Solar Voltage Check (3'b011)	0	1	1	1	1	0	1	1	X	X	X	X	X	1	1	0	0	Solar Current Check (3'b100)	Solar is good, continue
Solar Current Check (3'b100)	1	0	0	X	0	0	0	X	X	X	X	X	X	1	1	0	0	Solar Current Check (3'b100)	Wrong ADC
Solar Current Check (3'b100)	1	0	0	X	0	0	1	X	X	X	X	X	X	1	1	0	0	Solar Current Check (3'b100)	Wrong ADC
Solar Current Check (3'b100)	1	0	0	X	0	1	0	X	X	X	X	X	X	1	1	0	0	Solar Current Check (3'b100)	Wrong ADC
Solar Current Check (3'b100)	1	0	0	X	0	1	1	X	X	X	X	X	X	1	1	0	0	Solar Current Check (3'b100)	Wrong ADC
Solar Current Check (3'b100)	1	0	0	0	1	0	0	X	X	X	X	X	X	1	1	0	0	Solar Current Check (3'b100)	Right ADC, Not Done
Solar Current Check (3'b100)	1	0	0	X	1	0	1	X	X	X	X	X	X	1	1	0	0	Solar Current Check (3'b100)	Wrong ADC
Solar Current Check (3'b100)	1	0	0	X	1	1	0	X	X	X	X	X	X	1	1	0	0	Solar Current Check (3'b100)	Wrong ADC
Solar Current Check (3'b100)	1	0	0	X	1	1	1	X	X	X	X	X	X	1	1	0	0	Solar Current Check (3'b100)	Wrong ADC
Solar Current Check (3'b100)	1	0	0	1	1	0	0	0	X	X	X	X	X	1	0	0	1	Sleep (3'b001)	Solar is Bad, go to sleep
Solar Current Check (3'b100)	1	0	0	1	1	0	0	1	X	X	X	X	X	1	1	0	1	Boost Check (3'b101)	Solar is good, continue
Boost Check (3'b101)	1	0	1	X	0	0	0	X	X	X	X	X	X	1	1	0	1	Boost Check (3'b101)	Wrong ADC
Boost Check (3'b101)	1	0	1	0	0	0	1	X	X	X	X	X	X	1	1	0	1	Boost Check (3'b101)	Right ADC, Not Done
Boost Check (3'b101)	1	0	1	X	0	1	0	X	X	X	X	X	X	1	1	0	1	Boost Check (3'b101)	Wrong ADC
Boost Check (3'b101)	1	0	1	X	0	1	1	X	X	X	X	X	X	1	1	0	1	Boost Check (3'b101)	Wrong ADC
Boost Check (3'b101)	1	0	1	X	1	0	0	X	X	X	X	X	X	1	1	0	1	Boost Check (3'b101)	Wrong ADC
Boost Check (3'b101)	1	0	1	X	1	0	1	X	X	X	X	X	X	1	1	0	1	Boost Check (3'b101)	Wrong ADC
Boost Check (3'b101)	1	0	1	X	1	1	0	X	X	X	X	X	X	1	1	0	1	Boost Check (3'b101)	Wrong ADC
Boost Check (3'b101)	1	0	1	X	1	1	1	X	X	X	X	X	X	1	1	0	1	Boost Check (3'b101)	Wrong ADC
Boost Check (3'b101)	1	0	1	1	0	0	1	0	X	X	X	X	X	1	1	1	0	Normal Operation (3'b110)	Boost is bad
Boost Check (3'b101)	1	0	1	1	0	0	1	1	X	X	X	X	X	1	1	1	0	Normal Operation (3'b110)	Boost is good
Normal Operation (3'b110)	1	1	0	1	0	0	0	0	X	X	X	X	X	1	1	1	1	Sink Sleep (3'b111)	Output LDO out of operating range
Normal Operation (3'b110)	1	1	0	1	0	0	0	1	X	X	X	X	X	1	1	1	0	Normal Operation (3'b110)	Output LDO voltage in operating range
Normal Operation (3'b110)	1	1	0	1	0	0	1	0	X	X	X	X	X	1	1	1	0	Normal Operation (3'b110)	Boost Converter Output out of operating range
Normal Operation (3'b110)	1	1	0	1	0	0	1	1	X	X	X	X	X	1	1	1	0	Normal Operation (3'b110)	Boost Converter Output in operating range
Normal Operation (3'b110)	1	1	0	1	0	1	0	0	X	X	X	X	X	1	0	0	1	Sleep (3'b001)	Die Temp Sensor out of operating range
Normal Operation (3'b110)	1	1	0	1	0	1	0	1	X	X	X	X	X	1	1	1	0	Normal Operation (3'b110)	Die Temp Sensor in operating range
Normal Operation (3'b110)	1	1	0	1	0	1	1	0	X	X	X	X	X	1	1	1	0	Normal Operation (3'b110)	Backup Battery voltage out of operating range
Normal Operation (3'b110)	1	1	0	1	0	1	1	1	X	X	X	X	X	1	1	1	0	Normal Operation (3'b110)	Backup Battery voltage in operating range
Normal Operation (3'b110)	1	1	0	1	1	0	0	0	X	X	X	X	X	1	1	1	0	Normal Operation (3'b110)	Solar Current out of operating range
Normal Operation (3'b110)	1	1	0	1	1	0	0	1	X	X	X	X	X	1	1	1	0	Normal Operation (3'b110)	Solar Current in operating range
Normal Operation (3'b110)	1	1	0	1	1	0	1	0	X	X	X	X	X	1	1	1	0	Normal Operation (3'b110)	Solar Voltage out of operating range
Normal Operation (3'b110)	1	1	0	1	1	0	1	1	X	X	X	X	X	1	1	1	0	Normal Operation (3'b110)	Solar Voltage in operating range
Normal Operation (3'b110)	1	1	0	1	1	1	0	0	X	0	X	0	X	1	0	0	1	Sleep (3'b001)	Rechargeable Battery Current out of operating range NO BOARD POWER without solar
Normal Operation (3'b110)	1	1	0	1	1	1	0	0	X	0	X	1	X	1	1	1	1	Sink Sleep (3'b111)	Rechargeable Battery Current out of operating rangeNO BOARD POWER with solar
Normal Operation (3'b110)	1	1	0	1	1	1	0	0	X	1	X	X	X	1	1	1	0	Normal Operation (3'b110)	Rechargeable Battery Current out of operating range WITH BOARD POWER
Normal Operation (3'b110)	1	1	0	1	1	1	0	1	X	X	X	X	X	1	1	1	0	Normal Operation (3'b110)	Rechargeable Battery Current in operating range
Normal Operation (3'b110)	1	1	0	1	1	1	1	0	X	0	X	0	X	1	0	0	1	Sleep (3'b001)	Rechargeable Battery Voltage out of operating range NO BOARD POWER without solar
Normal Operation (3'b110)	1	1	0	1	1	1	1	0	X	0	X	1	X	1	1	1	1	Sink Sleep (3'b111)	Rechargeable Battery Voltage out of operating range NO BOARD POWER with solar
Normal Operation (3'b110)	1	1	0	1	1	1	1	0	X	1	X	X	X	1	1	1	0	Normal Operation (3'b110)	Rechargeable Battery Voltage out of operating range WITH BOARD POWER
Normal Operation (3'b110)	1	1	0	1	1	1	1	1	X	X	X	X	X	1	1	1	0	Normal Operation (3'b110)	Rechargeable Battery Voltage in operating range
Normal Operation (3'b110)	1	1	0	0	X	X	X	X	1	0	1	0	X	1	0	0	0	Start_Up_Beginning (3'b000)	Retry SOLAR
Normal Operation (3'b110)	1	1	0	0	X	X	X	X	1	X	0	1	X	1	0	0	0	Start_Up_Beginning (3'b000)	Retry BOOST
Normal Operation (3'b110)	1	1	0	0	X	X	X	X	1	X	0	0	X	1	0	0	0	Start_Up_Beginning (3'b000)	Retry SOLAR AND BOOST
Normal Operation (3'b110)	1	1	0	0	X	X	X	X	1	X	1	1	X	1	1	1	0	Normal Operation (3'b110)	Stay in Normal OP (both are good)
Normal Operation (3'b110)	1	1	0	0	X	X	X	X	1	1	1	0	X	1	1	1	0	Normal Operation (3'b110)	Stay in Normal OP (boost is good and bp is on
Normal Operation (3'b110)	1	1	0	0	X	X	X	X	0	X	X	X	1	1	1	1	1	Sink Sleep (3'b111)	SINK CONTROL SIGNAL
Normal Operation (3'b110)	1	1	0	0	X	X	X	X	0	X	X	X	0	1	1	1	0	Normal Operation (3'b110)	Stay in Normal OP
Sink Sleep (3'b111)	1	1	1	X	X	X	X	X	0	X	X	X	X	1	1	1	1	Sink Sleep (3'b111)	Stay in Sink Sleep waiting for timer
Sink Sleep (3'b111)	1	1	1	X	X	X	X	X	1	X	X	X	X	1	1	1	0	Normal Operation (3'b110)	Leave Sink Sleep

Figure 63: State Machine Transition Table

For reference, these are the minimized logical combinations for each next state bit, minimized from Figure 63.

Minimized:

```
nextsta2 = currsta2 currsta0 rst_n + currsta2 currsta1' adcdone' rst_n + currsta2
    currsta1' adccha1 rst_n + currsta2 adcdone adcgood rst_n + currsta2 adcdone'
    timer' rst_n + currsta2 currsta1 adcdone adccha1' rst_n + currsta2 adcdone
    adccha2' adccha0 rst_n + currsta2 adcdone' boostg solarg rst_n + currsta2'
    currsta1 currsta0' timer boardpow rst_n + currsta1 currsta0 adcdone adccha2
    adccha1' adccha0 adcgood rst_n + currsta2 adcdone adccha2' adccha1' rst_n +
    currsta2 adcdone adccha1' adccha0 rst_n + currsta2 adcdone adccha2 adccha1
    boardpow rst_n + currsta2 adcdone adccha2 adccha1 solarg rst_n + currsta1
    currsta0' adcdone' timer boardpow boostg rst_n;

nextsta1 = currsta2' currsta1' currsta0' rst_n + currsta1 currsta0 adcdone' rst_n
    + currsta1 currsta0 adccha2' rst_n + currsta1 currsta0 adccha1 rst_n + currsta1
     currsta0 adccha0' rst_n + currsta1 adcdone' timer' rst_n + currsta2 currsta1
    adcdone adccha1' rst_n + currsta2 currsta1 adcdone adcgood rst_n + currsta2
    currsta1 adcdone adccha2 boardpow rst_n + currsta2 currsta1 adcdone' boardpow
    boostg rst_n + currsta2 currsta0 adcdone adccha2' adccha1' adccha0 rst_n +
    currsta2' currsta0' timer' rst_n + currsta2' currsta0' boardpow' rst_n +
    currsta2 currsta1 adcdone adccha2' adccha0 rst_n + currsta1 adcdone' boardpow'
    boostg solarg rst_n + currsta1 adcdone adccha2 adccha1 boardpow' solarg rst_n;

nextsta0 = currsta2 currsta1' currsta0 adcdone' rst_n + currsta2 currsta1'
    currsta0 adccha2 rst_n + currsta2 currsta1' currsta0 adccha1 rst_n + currsta2
    currsta1' currsta0 adccha0' rst_n + currsta2' currsta1 currsta0' timer rst_n +
    currsta2 currsta1 adcdone' timer' sinkctrl rst_n + currsta2 currsta1' adcdone
    adccha2 adccha1' adccha0' rst_n + currsta2' currsta1 currsta0 adcdone' rst_n +
    currsta2' currsta1 currsta0 adccha2' rst_n + currsta2' currsta1 currsta0
    adccha1 rst_n + currsta2' currsta1 currsta0 adccha0' rst_n + currsta2' currsta1
     currsta0 adcgood' rst_n + currsta2 currsta1 currsta0 timer' rst_n + currsta2'
    currsta1' currsta0 timer' rst_n + currsta2 currsta1 currsta0' adcdone adccha2'
    adccha0' adcgood' rst_n + currsta2 currsta1 currsta0' adcdone adccha2 adccha1
    adcgood' boardpow' rst_n;
```

5.6.7 State Machine Output Logic

Using the state calculated in Next State logic, the outputs are decided. To allow for operations such as retrying the boost converter start up sequence without turning off the LDO output or switching back to the coin cell battery, another latch is used to store whether start up has previously happened: started_latch.

The full logic table for all outputs can be seen below in Table 64.

Current State	enable_ldo_out	enable_boost	enable_adc	enable_timer	enable_charger	enable_solar	timer0	timer1	timer2	powermux
Start_Up_Beginning (3'b000)	started_latch	1	1	0	0	1	0	0	0	started_latch
Sleep (3'b001)	0	0	0	1	0	0	1	0	0	0
Solar/Boost Wait (3'b010)	started_latch	1	1	1	0	1	0	0	1	started_latch
Solar Voltage Check (3'b011)	started_latch	1	1	0	0	1	0	0	0	started_latch
Solar Current Check (3'b100)	started_latch	1	1	0	0	1	0	0	0	started_latch
Boost Check (3'b101)	started_latch	1	1	0	0	1	0	0	0	started_latch
Normal Operation (3'b110)	1	boostg	1	0	1	solarg	0	0	0	1
Sink Sleep (3'b111)	0	0	0	1	1	1	0	1	0	1

Figure 64: State Machine Output Table

5.7 Programmable Active Filters Implementation

This section overviews the implementation of the three filter blocks designed for the CRIMSON IC. The ADC Anti-Aliasing Fitler, and the two versions of the general-purpose programmable filter blocks, one utilizing Gm-C filter cells, and the other utilizing Sallen-Key filter cells.

5.7.1 ADC Anti-Aliasing Filter Implementation

The ADC input filter was implemented using the Sallen-Key low-pass filter topology (Figure 9), with programmability using a selectable resistor block. To ensure the filter can handle the full 3.3V signal swing of the system, a rail-to-rail op-amp, sourced from previous CHIPS Design Studio projects, was used. The filter is configured for a Butterworth response with a critically damped quality factor of Q=0.707. This filter was designed to operate according to the ADC's default sampling rate of 3.2 Ksps, offering a 1.6 kHz cutoff frequency, but also accommodates a linear selection of user-programmable cutoff frequencies according to the ADC's maximum sampling rate of 100 Ksps, starting at 5 kHz up to 50 kHz in 5 kHz steps.

The top-level schematic of the ADC filter is shown below in Figure 65. It features its own digital control block, which translates the 4-bit binary into 11-bit 1-hot encoding used by the resistor blocks and also handles additional digital control for the circuit. This circuit block also integrates its own bias generator that takes a 1µA external current reference source to generate a low or high-power bias voltage for the op-amp depending on the selected cutoff frequency. In the default operating mode of 1.6 kHz cutoff frequency, the filter draws 8µW of quiescent power. When the filter is in any other configuration, the bias generator switches to a higher power mode, drawing about 90µW in total. This much power was required to give the filter enough full-power bandwidth to operate at full swing at its highest cutoff frequency of 50 kHz.

Several transmission gates are used in the circuit as well to direct signal flow. Two transmission gates are used to bypass the filter when the filter is disabled. The other two gates enable access from external pins on the input and output nodes, which are normally closed to isolate parasitics from the pins if they are not in use.

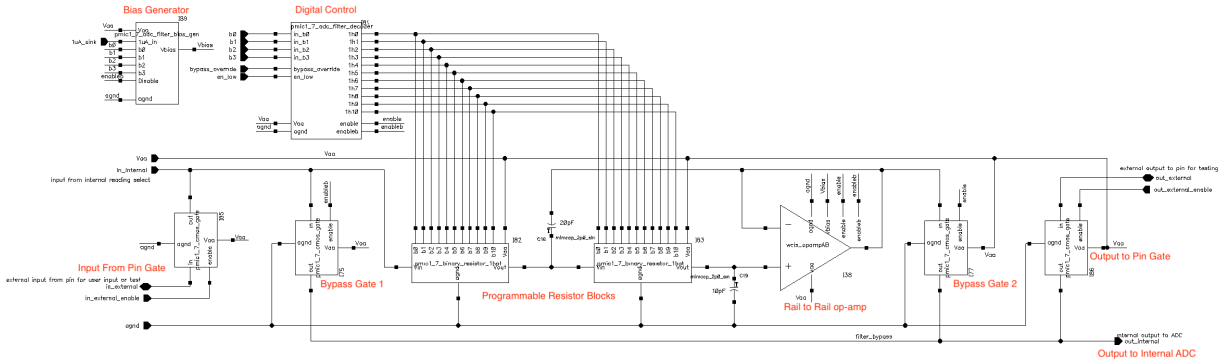

Figure 65: Programmable ADC Anti-Aliasing Filter Top-Level Schematic

Programmable Resistor Circuit

The filter's cutoff frequency is controlled by the programmable resistor circuits, the schematic of which is shown in Figure 66. This circuit uses a 1-hot selection scheme with 11 transmission gates connected to a series string of 11 resistors. In this configuration, only one transmission gate is active at a time, tapping into each node of the resistor string to select the total resistance. If the rightmost transmission gate is active, it would see all 11 resistors in series from input to output, corresponding to the lowest 1.6 kHz cutoff frequency. If, for example, the 5th gate from the left were active, it would see only the first 5 resistances, corresponding to a 30 kHz cutoff frequency, and so on.

This design was chosen over a binary-weighted resistor design to accommodate the initial non-linear steps from 1.6 kHz to 5 kHz, and subsequent 5 kHz increments. Furthermore, this approach offers improved area efficiency. In a binary-weighted implementation, the lowest frequency step would require a dedicated 7MΩ resistor, with additional parallel resistors totaling an additional 6.6MΩ needed for each subsequent linear step, totaling nearly 14MΩ total. By utilizing a shared series string where all elements contribute to the lowest frequency setting, the total resistance required is only about 7MΩ, reducing the required area by half compared to a binary-weighted equivalent.

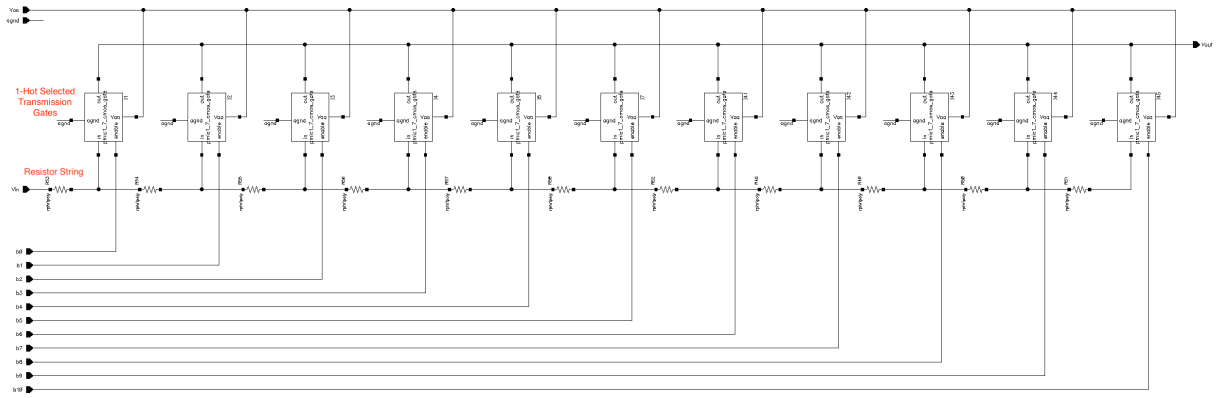

Figure 66: ADC Filter Programmable Resistor Block Subcircuit

Bias Selector Circuit

As mentioned previously, this filter block includes its own bias generator, shown in Figure 67, that takes a 1µA external current source, and mirrors it. The circuit chooses which mirror generates the bias voltage for the rail-to-rail op-amp. The first mirror is a 1 : 0.1 ratio, generating a 100nA bias current when the filter is in its default 1.6 kHz cutoff frequency. This biases the op-amp enough to achieve a full power bandwidth higher than its cutoff frequency and ensures that it runs at a low power of 8µW. When the filter is set to any other cutoff frequency configuration, which is used when the ADC is sampling external signals at higher sampling rates, the bias generator switches to using the second mirror, which is a 1 : 1.5 ratio, generating a 1.5µA bias current for the op-amp. This ensures that the filter can support a full power bandwidth over the highest cutoff frequency of 50 kHz but causes the filter to consume significantly more power of 90µW.

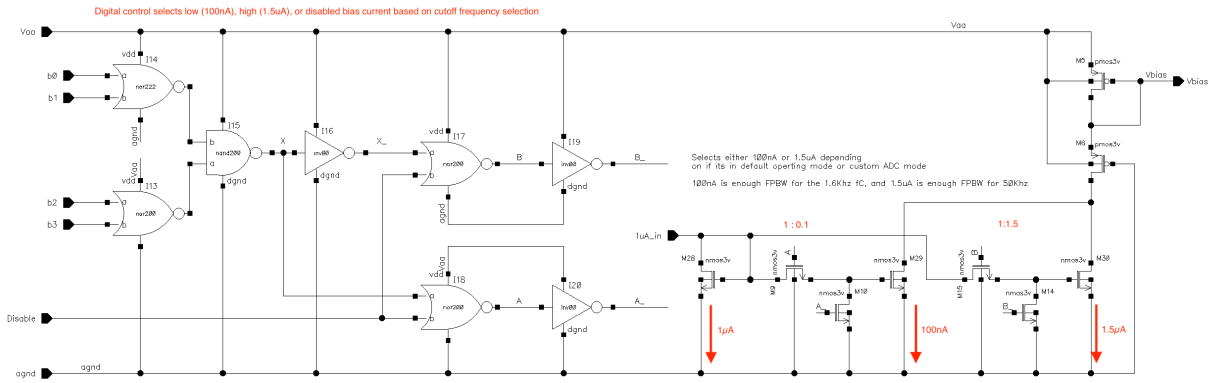

Figure 67: ADC Filter Bias Selector Subcircuit

ADC Filter Configurability and Programmability

The ADC filter circuit is controlled through 7 scan chain bits and 1 internal signal. The filter is enabled and disabled by the chip's internal control synchronously with the ADC. If the filter ever fails, to ensure that the ADC can still be used, scan bit 4 allows the user to bypass the filter. Scan bits 0 to 3 control the filter's cutoff frequency, where in default operating mode, all bits are 0, and the filter is set to a cutoff frequency of 1.6 kHz, which corresponds to the default sampling rate of the ADC. The filter can be set by the user to cutoff frequencies of 5 kHz to 50 kHz in 5 kHz steps to support when the ADC is using a custom clock frequency up to its maximum sampling rate of 100 ksps. Scan chain bits 5 and 6 correspond to input and output pin enable and disable. If the user wants to probe the node on the input and output of the filter or apply a custom signal to the filter input, then these bits can be set, otherwise by default, the pin will be isolated from the filter. Figure 68 shows the configurability layer diagram of this block with its control signals and I/O. Figure 69 outlines all possible configurations of this filter and its power consumption for each state.

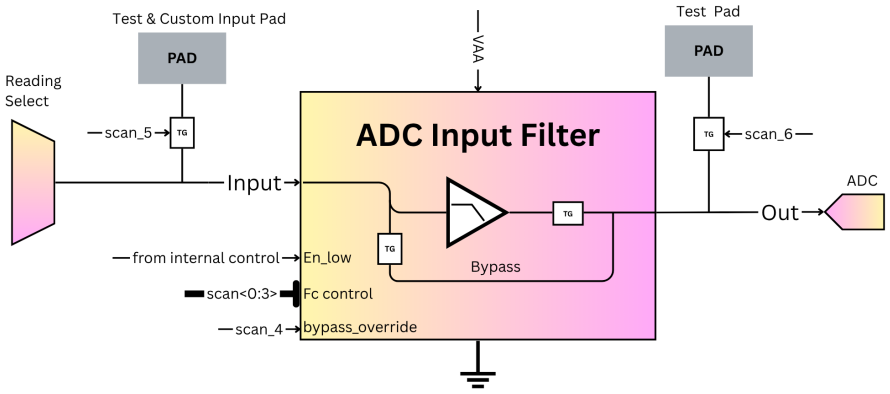

Figure 68: ADC Filter Configurability Layer Diagram

scan_4 || internal enable (bypass_override || en_low)	scan_3 (b3)	scan_2 (b2)	scan_1 (b1)	scan_0 (b0)	Cutoff Frequency	For Sampling Rate	Quiescent Power
0	0	0	0	0	1.6Khz (default)	3.2Ksps (default)	8uW
0	0	0	0	1	5Khz	10Ksps	90uW (higher power in custom mode to ensure high enough FPBW that no distortion occurs at high frequencies)
0	0	0	1	0	10Khz	20Ksps	
0	0	0	1	1	15Khz	30Ksps	
0	0	1	0	0	20Khz	40Ksps	
0	0	1	0	1	25Khz	50Ksps	
0	0	1	1	0	30Khz	60Ksps	
0	0	1	1	1	35Khz	70Ksps	
0	1	0	0	0	40Khz	80Ksps	
0	1	0	0	1	45Khz	90Ksps	
0	1	0	1	0	50Khz	100Ksps	
1	X	X	X	X	Filter disable and bypass	Any	3nW

	scan_5 (in_external_enable)	scan_6 (out_external_enable)
Function:	1= Enable input pin 0 = Isolate pin	1= Enable output pin 0 = Isolate pin

Figure 69: ADC Filter Programmability Table

The simulation results below in Figure 70 show the range and values of cutoff frequencies in pre-layout simulations. Additionally, this filter was verified in the time domain to ensure that a full-power bandwidth signal with a 3.3V amplitude will not face distortion.

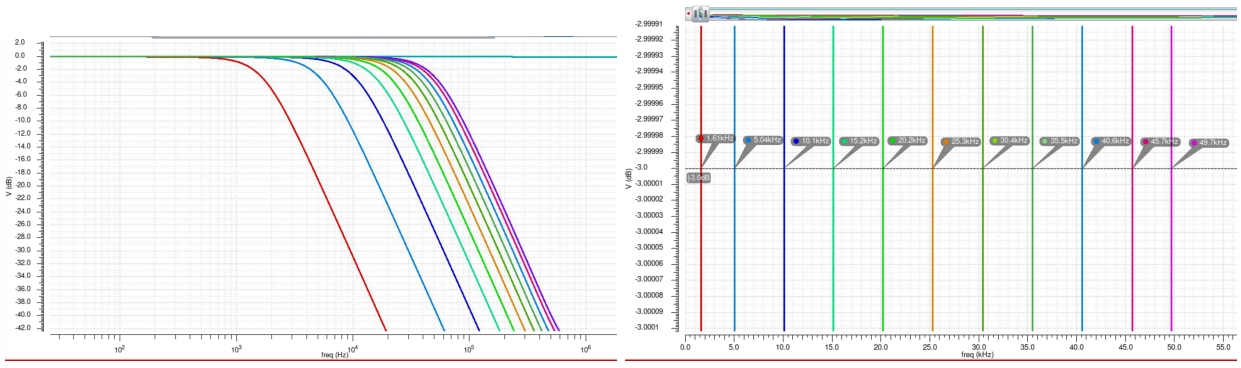

Figure 70: ADC Filter Cutoff Frequency Range (1.6kHz to 50Khz and Bypass)

5.7.2 General-Purpose Programmable Filter Blocks

Two general-purpose filter blocks are designed for the CRIMSON IC. One block utilizes Sallen-Key filter cells, and the other utilizes Gm-C filter cells. They are designed for a wide range of configurability, featuring user-programmable cutoff frequencies and the ability to cascade individual stages into higher-order low-pass, high-pass, band-stop, and band-pass configurations, allowing for a variety of external signals to be filtered as needed. For design simplicity and filtering performance, the band-stop and band-pass configurations are locked into 4th-order attenuation, whereas low-pass and high-pass can be selected as either 2nd order or 4th order.

These filters were designed to target initial provided specs of 10uW per block and a frequency target of 50Hz to 100Khz. To simplify design and integration, the supporting structure of these two programmable filter blocks was kept identical. Both blocks use identical digital control to select filter configuration (low-pass high-pass, bandpass etc.), and the signal routing scheme is shared across both blocks. The only usability difference between the two is the method of frequency programmability.

The top-level schematics of these two blocks are shown in Figure 71 for the Sallen-Key block and in Figure 78 for the Gm-C filter block. Both are composed of four filter cells, two second-order high-pass cells, and two second-order low-pass cells. To allow for configurability and cascadability between these filter cells, the signal path is routed through a network of CMOS transmission gates that switch according to the user-selected configuration.

To control the signal flow of these filters, it includes a dedicated digital controller, which interprets the 3-bit binary configuration code to manage the internal routing, output selection, and general state of the filter. It enables only the necessary sub-blocks for a given mode while disabling unused components to save power.

In the band-pass and band-stop configuration all four filter cells are active. For band-pass response, the signal is routed in series first through the two high-pass filters, and then cascaded through the two low pass-filters. In the band-stop response, the signal is passed through the high-pass filters and low-pass filters in parallel and is summed together using a non-inverting summing amplifier.

Sallen-Key Programmable Filter Block

The top-level schematic of the Sallen-Key configurable filter block is shown in Figure 71. The op-amps used in these filters are two-stage Miller-compensated op-amps. They use an NMOS diff-pair input

with an ICMR between 300mV and 3.25V, with a near rail-to-rail output swing. The op-amps are biased with a 4µA tail current. The filters can generally support a full power bandwidth of 1MHz with a $1V_{pp}$ signal. In higher order configurations, due to the additional stages and capacitive loading of each stage, full power bandwidth may be slightly degraded.

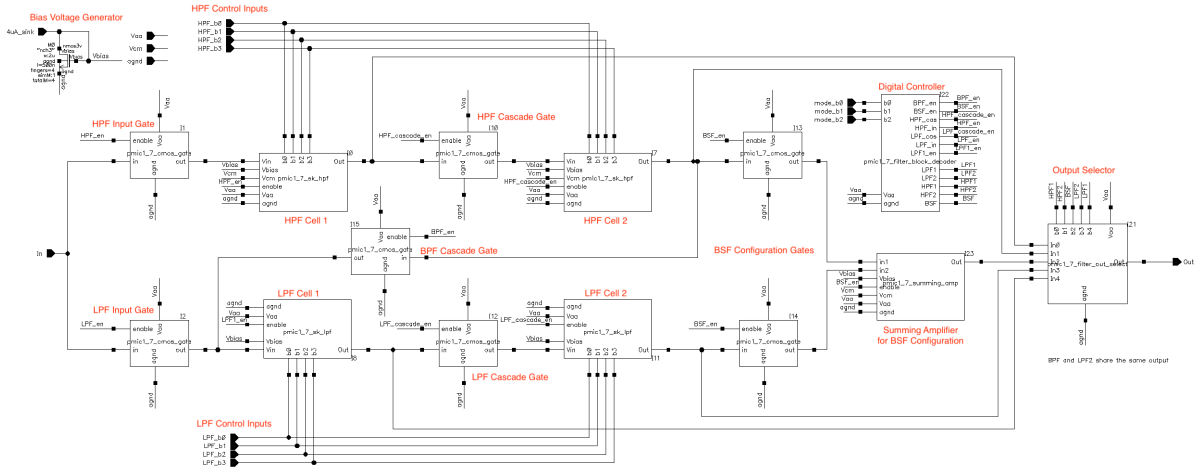

Figure 71: Sallen-Key General Purpose Filter Block

The low-pass and high-pass Sallen-Key filter cells are shown in Figure 72. Both filters are designed for a Butterworth response. They are both programmed using a 4-bit binary weighted resistor block to choose from 10 kHz up to 160kHz in 10kHz steps.

The LPF uses equal-sized resistors ($R_1 = R_2 = 2.25M\Omega$) with a capacitor ratio of ($C_1 = 2 \cdot C_2$) with C_2 sized as 10 pF and C_1 as 5 pF. This achieves a Q=0.707 for a Butterworth low-pass response. The HPF uses equal sized capacitors ($C_1 = C_2 = 10pF$) with a resistor ratio of ($R_2 = 2 \cdot R_1$) with R_2 sized as 2.25MΩ and R_1 at 1.125MΩ. This achieves a Q=0.707 for a Butterworth high-pass response. (See Figure 9 for the component designators of each filter.) The resistance values correspond to the base resistance of the programmable resistance blocks (R_{base}).

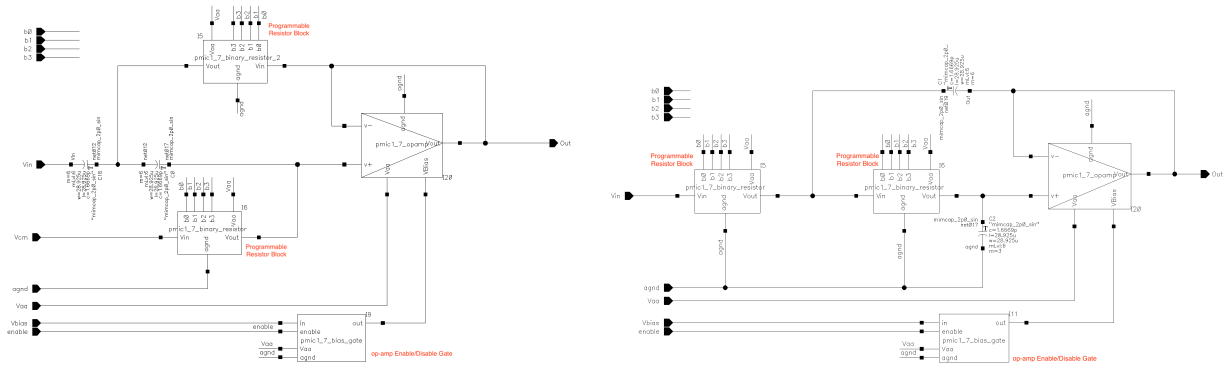

Figure 72: Sallen-Key high-pass (left) and low-pass (right) Filter Cells

Digitally Programmable Resistor Block

The binary-weighted programmable resistor blocks, shown in Figure 73, are used to digitally set the filter's cutoff frequency. The network is designed in a parallel configuration to achieve linear steps in the frequency domain.

The Most Significant Bit (MSB) is represented by the leftmost resistor with a base resistance of R_{base}, which corresponds to the minimum cutoff frequency of 10 kHz. The second branch provides an additional R_{base}, and the third branch utilizes $R_{base}/2$, then $R_{base}/4$, and lastly $R_{base}/8$. Unlike the resistor block used for the ADC filter, these are binary selectable, allowing for 16 total configurations with 4 bits. Due to the much smaller value of resistors for these filters compared to the ADC filter, area was not as large of a concern, and this binary-weighted design was viable.

Because of this parallel arrangement, resistance does not scale linearly as the binary input changes, though it produces linear increments in the cutoff frequency. The 16 discrete resistance steps of this block follow a power regression with the equation of $R_{eq} = R_{base} \cdot x^{-1}$. By substituting x with 1 through 16 corresponding to each step, the effective resistance can be calculated. Using the cutoff frequency formula for the Sallen-Key topology $f_c = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}}$, and substituting the resistance equation into both R_1 and R_2 with the appropriate capacitance and R_{base} values, will result in an exactly linear function of the cutoff frequency.

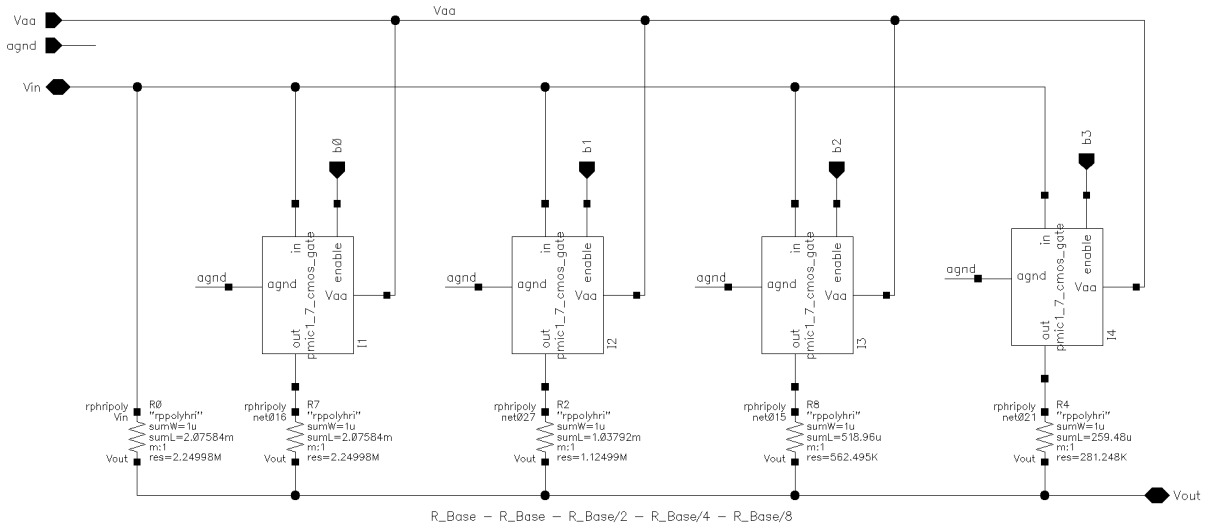

Figure 73: Binary Weighted Programmable Resistor Block

Sallen-Key Filter Block Configurability and Programmability

This filter block uses 11 scan chain bits for control. 4 bits for high-pass frequency selection, 4 bits for low-pass frequency selection, and 3 bits for mode selection. If the mode selection bits are all set to 0 (000), the block is automatically disabled. When a mode is selected, the block enables itself. Each op-amp consumes 40µW, meaning that the power consumption of the block ranges from 40µW up to 200µW depending on the configuration selected.

In band-pass and band-stop modes, the high-pass and low-pass frequencies are still independenly controllable, meaning any width or position band or notch within the programmability range is possible.

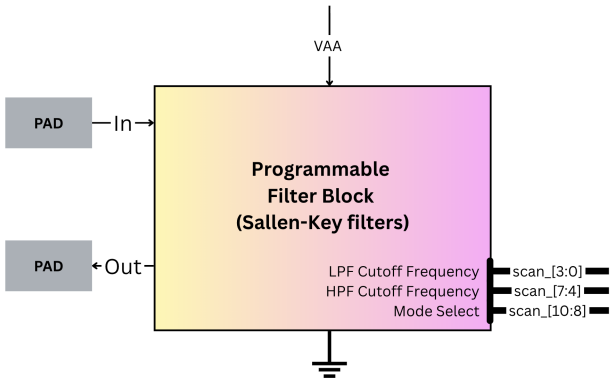

Figure 74: Sallen-Key Filter Block Configurability Layer

The table below in Figure 75 represents all configurations of this filter block, it shows which binary codes correspond to each cutoff frequency and operating mode.

Sallen Key Configurability

LPF Cutoff Frequency Scan_[3:0] (LPF_b[3:0]) **HPF Cutoff Frequency** Scan_[7:4] (HPF_b[3:0])	
Code	**Fc (kHz)**
0000	10
0001	20
0010	30
0011	40
0100	50
0101	60
0110	70
0111	80
1000	90
1001	100
1010	110
1011	120
1100	130
1101	140
1110	150
1111	160

Mode Selection Scan_[10:8] (mode_b[2:0])		
Code	**Configuration**	**Power**
000	Disabled	3nw
001	2nd Order LPF	40µW
010	4th Order LPF	80µW
011	2nd Order HPF	40µW
100	4th Order HPF	80µW
101	BSF	160µW
110	BPF	200µW
111		

Figure 75: Sallen-Key Filter Block Configurability Tables

The simulation results below in Figure 76 show the range of cutoff frequencies for both low-pass and high-pass filters cells.

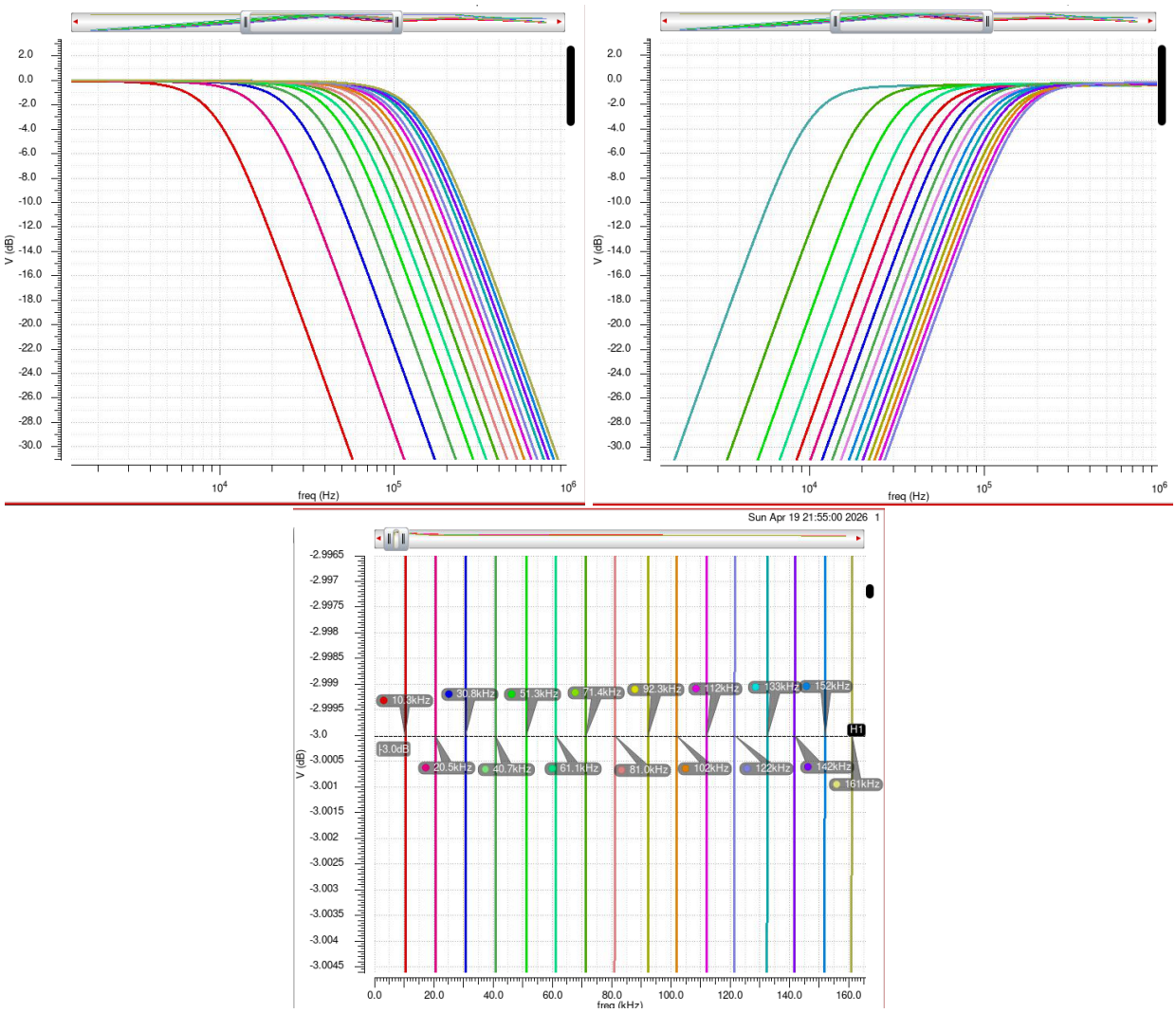

Figure 76: Cutoff Frequency Range of LPF and HPF (10Khz to 160Khz in 10Khz Steps)

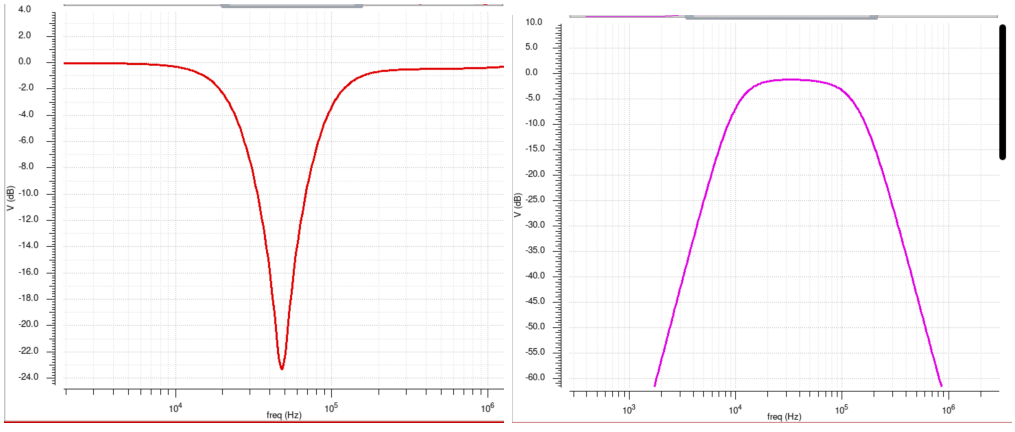

Figure 77: Example of The Filter in Band-Stop and Band-Pass Configuration

Gm-C Programmable Filter Block

The top-level schematic of the Gm-C configurable filter block is shown in Figure 78. The OTAs used in each filter cell are simple single-stage OTAs designed for low Gm. They use an NMOS diff-pair input with an ICMR between 600mV and 3.1V, with an output swing of 300mV to 3.2V.

These filters support a much lower signal amplitude compared to the Sallen-Key filters but support a much higher range of cutoff frequencies. These filters can safely support a signal amplitude of about 200mV_{pp} throughout all cutoff frequency settings before distortion occurs, and the maximum bandwidth of the high-pass filters is roughly 5Mhz. These filters can achieve a cutoff range from roughly 1.5 kHz up to 260 kHz.

The programmability requires I-DACs that are not included within the top-level filter schematic but are part of the chip's bias generator block. There will be two sets of 5-bit I-DACs, one set for high-pass configuraiton and another set of two DACs for low-pass configuration. One I-DAC will provide a fine 10 nA bias current step, from 10 nA up to 310 nA, resulting in about 1.5 kHz up to 36kHz This will allow for fine, low-frequency selection. The other will provide a coarser 100nA current step ranging from 100nA up to 3.1µA, resulting in about 13kHz up to 260kHz. This will allow for coarse, high-frequency selection. These two DACs are mutually exclusive, meaning that only one of the DACs is in use at a time, and they are not used together.

The filter cells have been verified to operate at bias currents at and lower than 1 nA, corresponding to a 150Hz cutoff frequency, but a 10nA minimum was chosen to ensure reliability. However, should a higher-resolution I-DAC be implemented, the architecture is capable of supporting it to achieve lower cutoff frequencies.

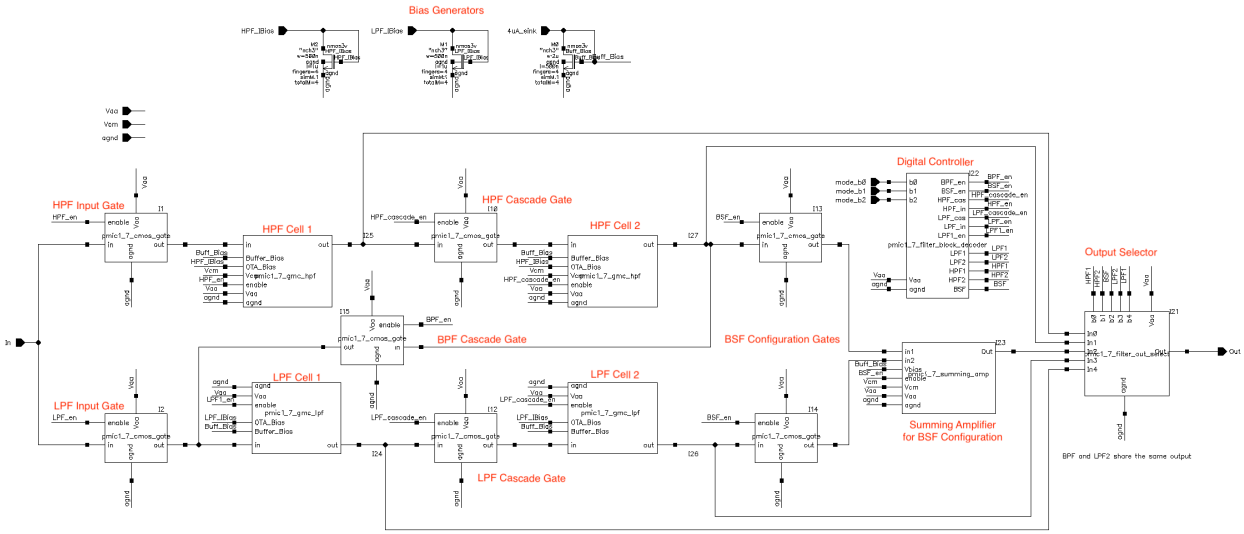

Figure 78: Gm-C General Purpose Filter Block

The Gm-C filter topologies outlined in the design options section were used with an additional op-amp buffer to ensure that each stage could be properly driven. The requirement of buffers unfortunately cancels out any of the power consumption benefits of the Gm-C filters themselves, but without the buffers, significant signal attenuation and signal quality issues arise due to the OTAs not being able to drive their own capacitances in addition to the next stage's input capacitance when they are all cascaded together.

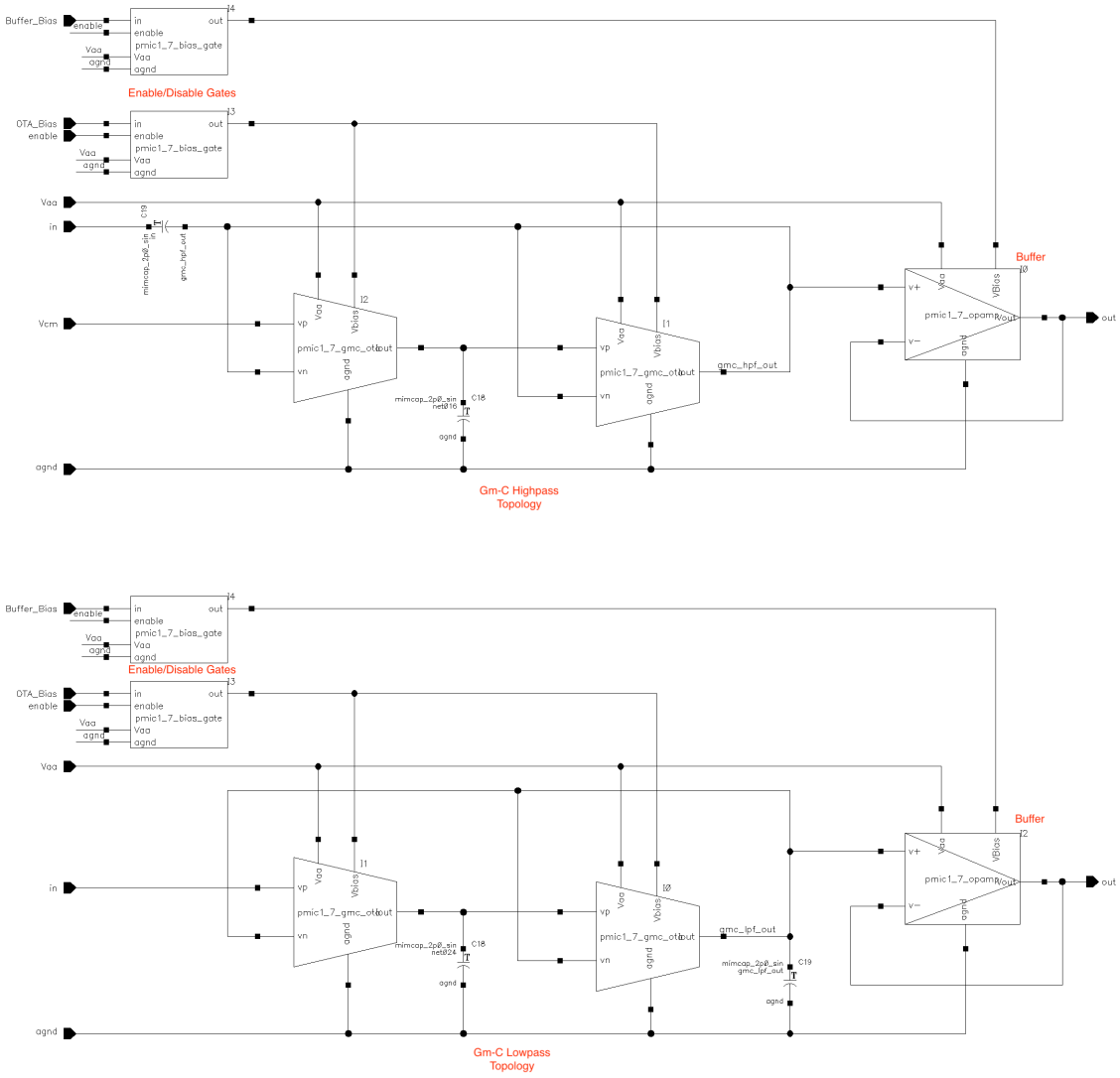

Figure 79: Gm-C High-Pass (top) and Low-Pass (bottom) Filter Cells

Gm-C Filter Block Configurability and Programmability

The Gm-C programmable filter block utilizes 3 internal scan chain bits for mode selection, which operates identically to the Sallen-Key filter block. An external I-DAC block will be used to generate the bias currents for the Gm-C filter cells, which set the filter's cutoff frequency. 5 bits are used to select the high pass I-DAC value, and 5 bits are used to select the low-pass I-DAC value. 2 bits are allocated to enable the fine or coarse I-DAC selection, 1 bit for low-pass and 1 for high-pass. Like the Sallen-Key block, in band-pass and band-stop modes, the high-pass and low-pass frequencies are independently controllable, meaning any width or position band or notch within the programmability range is possible.

Power consumption in each mode is more complicated than in the Sallen-Key block, each op-amp consumes 40µW, meaning that the power consumption of the block ranges from 40µW up to 200µW depending on the configuration selected, but with the addition of the power consumed by the OTAs within each filter. So power consumption in each mode is estimated by a simple equation (shown in the configurability tables) that adds the base power consumption from the op-amp buffers in each cell with the power consumption of each OTA at the selected bias current.

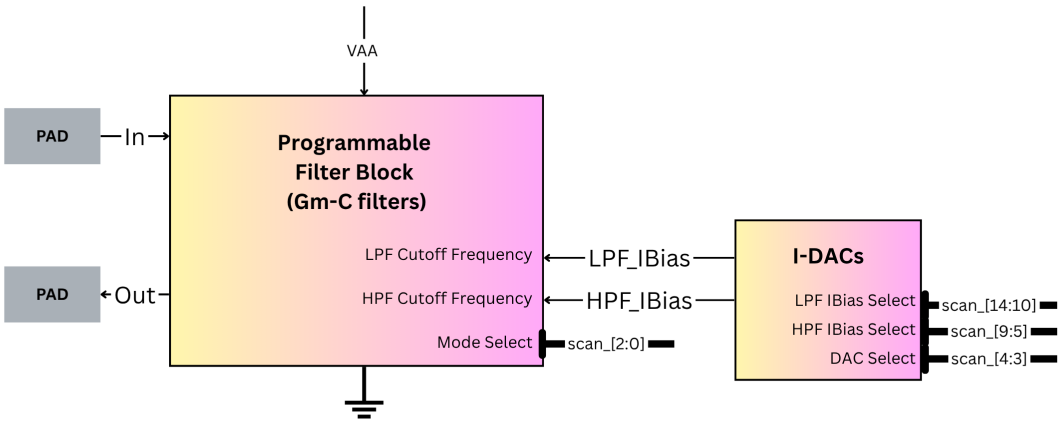

Figure 80: Gm-C Filter Block Configurability Layer

The tables below in Figure 81 overviews the configurability of this block. The left table overviews the DAC and frequency programmability. The table on the right overviews the operation mode selection. The cutoff frequencies of both high-pass and low-pass filters are identical.

Gm-C Configurability

I-DAC and Frequency Select				
Type	**Fine I-DAC (sel=0)**		**Coarse I-DAC (sel=1)**	
Resolution	**5-bit**		**5-bit**	
Step Size	**10nA**	**~1.1kHz avg**	**100nA**	**~8kHz avg**
Minimum	**10nA (00001)**	**1.5KHz**	**100nA (00001)**	**13 kHz**
Maximum	**310nA (11111)**	**36KHz**	**3.1uA (11111)**	**260 kHz**

Due to slight non-linear Fc tuning, Fc steps get slightly smaller as frequency increases, table shows average step size

Mode Selection Scan_[2:0] (mode_b[2:0])		
Code	**Configuration**	**Power (Base Power + Gm-C Power)** Gm-C filter power consumption depends on selected bias current Power (µW) = Base + (Vsupply * Num OTAs enabled * Bias Current)
000	**Disabled**	**3nw**
001	**2nd Order LPF**	**40 + (3.3 × 2 × LPF_Ibias) µW**
010	**4th Order LPF**	**80 + (3.3 × 4 × LPF_Ibias) µW**
011	**2nd Order HPF**	**40 + (3.3 × 2 ×HPF_Ibias) µW**
100	**4th Order HPF**	**80 + (3.3 × 4 × HPF_Ibias) µW**
101	**BSF**	**160 + (3.3 × 4 × (HPF_Ibias + LPF_Ibias)) µW**
110	**BPF**	**200 + (3.3 × 4 × (HPF_Ibias + LPF_Ibias)) µW**
111		

Figure 81: Gm-C Filter Configurability Table

In the simulation results in Figure 82 below, the range of each cutoff frequency can be seen. The cutoff frequencies remain mostly linear, with some slight drift in linearity, and decrease in step size as frequency increases.

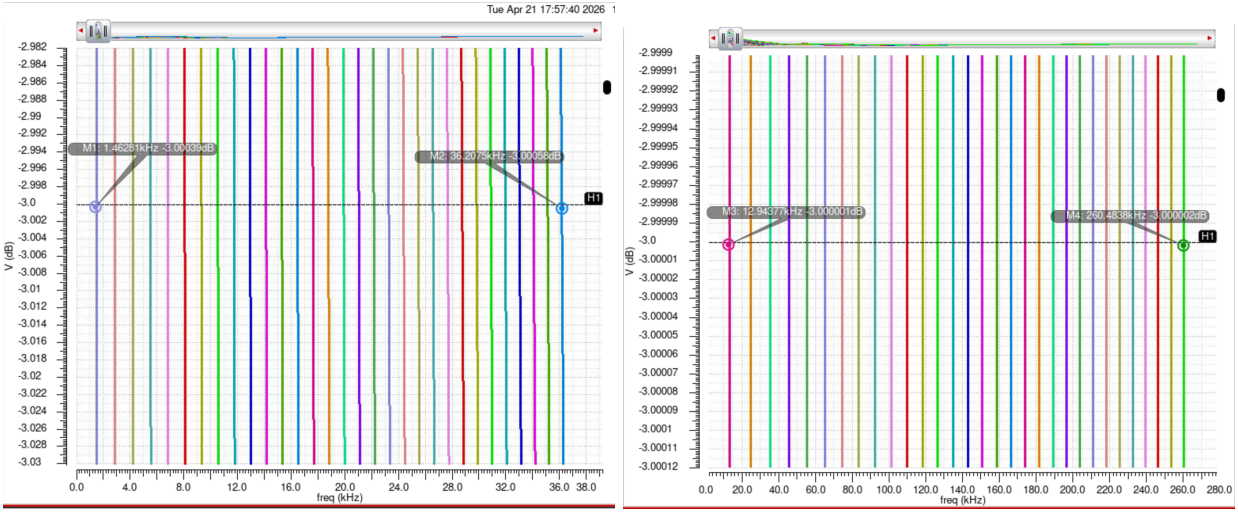

Figure 82: Simulation Results of the fine DAC-controlled cutoff frequencies (left) and the coarse DAC-controlled cutoff frequencies (right)

5.8 Buck-Boost Converter Implementation

The boost converter is part of the power architecture, which takes in an external input voltage, and outputs a programmable output voltage. Programmability is done by a 6-bit DAC, and a 32 kHz clock is used to store and release current from an off-chip inductor.

5.8.1 Buck-Boost Converter: Architectural Integration

The configurability layer is shown on figure 83. Here, the programmability aspect of the boost converter is shown through the scan chain bits of the DAC, along with scan chain enable bits for the clock, and the biasing voltages. There are 6 scan chain bits for the DAC, corresponding to it being a 6-bit programmable input. This makes it so the boost converter output can be programmed from 0-3 V, in increments of 50 mV. The enable signals make it so that if something were to go wrong in the boost converter, it would be shut off by disabling it.

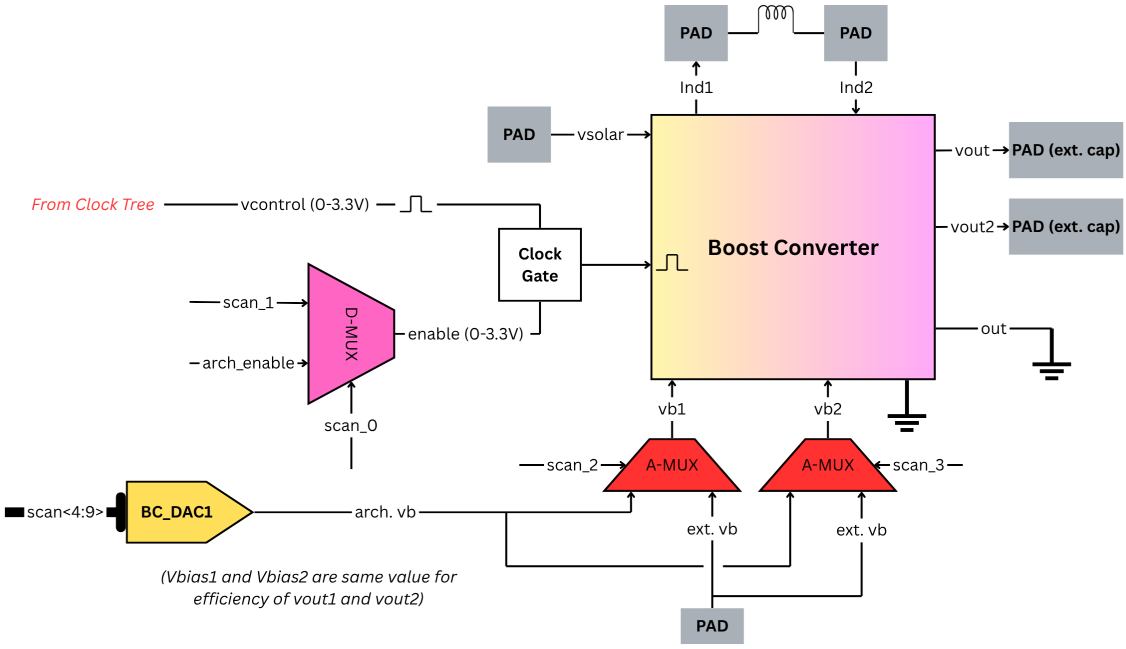

Figure 83: Configurability Layer for the Boost Converter

5.8.2 Buck-Boost Converter: Design

Figure 84 shows the selected boost converter inspired topology selected, as mentioned previously, shows the selected boost converter inspired topology presented by Kanago et al. [10]. This topology has an input NMOS switching transistor, in which the gate is connected to the clock, the source is connected to a

pad leading to an external input voltage source, and the drain is connected to a pad connecting to an off-clip inductor. The next two transistors are an NMOS transistor and a PMOS transistor connected in parallel that act as a diode. The off-chip inductor is connected to a pad on the other side to go back to the chip, and that side connects to the drain of the NMOS and source of the PMOS transistors. The gates of both of these transistors are connected to the same clock as the input switching transistor. The source of the NMOS is connected to ground. The way this transistor pair acts as a diode is that when the clock is on, the input switching transistor is turned on, letting current enter the inductor, which charges up the inductor. Here, the NMOS transistor is on, and the PMOS transistor is off, which ensures that the inductor is getting charged, as the current cannot go to the output, as that is the side of the PMOS. When the clock is off, the input switching transistor and the NMOS transistor are off, and the PMOS transistor is on, which makes the current stored in the inductor is released to the outputs, charging up off-chip capacitors, in which the drain of the diode PMOS is connected to the first capacitor stage. In a conventional boost converter, the same thing happens, when the clock is on, the inductor stores current from the source, and when the clock is off, the inductor released this current, charging up a load capacitor to get a new voltage.

The output stage is where this selected boost converter topology differs from a conventional boost converter. The output voltage of a conventional boost converter depends on the input voltage and the duty cycle of the clock switching the input transistor. The equation V(o)=V(s)/(1-D) shows this, where V(o) is the output voltage, V(s) is the input voltage, and D is the duty cycle. For the selected topology being used, the output voltage is roughly equal to Vdac+1.15*Vthp, where Vthp is the threshold voltage of output PMOS transistors, and Vdac is a programmable input voltage from a DAC. Vthp here has been measured to be roughly 0.735 V, and Vdac can be a value between 0-3.2 V, with 50 mV steps. Because there is no direct dependency of the input voltage on the output, this DC-DC converter topology can also have bucking capabilities, if the input voltage is high enough, and the output voltage is set to a low value. The input voltage mostly affects how quickly the output capacitors get charged.

The first output stage PMOS transistor has its source connected to the pad connection of the first output capacitor, the gate connected to the DAC, and the drain is connected to the pad of the second output capacitor. The purpose of the second output is to waste any excess charge that may not have been stored by the first output capacitor. The source of this output PMOS transistor is connected to the pad connection of the second output capacitor. The gate is connected to the same DAC as the first output. Once the first output capacitor is charged, the excess charge will enter the second output capacitor until that capacitor is charged. Any excess charge from this capacitor will go to the ground connected drain.

For the bulk connections, the 2 NMOS transistors have the bulks connected to ground. For the

PMOS transistors, the diode PMOS is connected to Vaa, while the output PMOS transistors have their outputs connected to their source. This is because connecting these bulks to the highest potential, in this case Vaa, drastically affects the behavior of the converter. The minimum voltage goes from being about 1.15*Vthp to a bit over 2*Vthp. This makes the minimum voltage that can be reached by the converter much greater, which makes it unable to meet spec. On top of a higher minimum voltage, this different bulk connection also makes it so that the voltage being imputed from the DAC does not proportionally affect the output, however, only about 0.68*Vdac is effective.

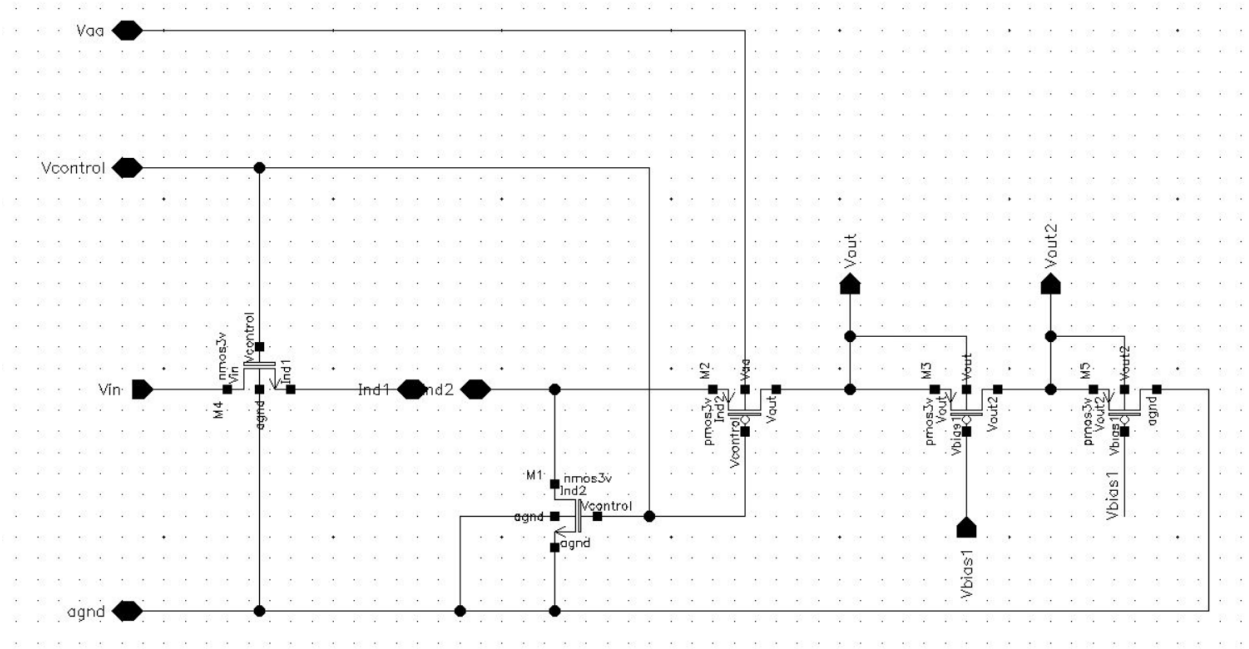

Figure 84: Boost Converter Schematic

5.8.3 Buck-Boost Converter: Results

Figure 85 shows a parametric sweep of Vbias from 0-3 V, with 500 mV steps, for Vin on 1.5 V. Here, for a bias of 0 V, the output voltage for the first output is 827 mV, and for the second output it is 771 mV. With no biasing, the first output voltage is about 1.15*Vthp, as stated before. The reason the minimum voltage for this circuit is 1.15*Vthp instead of Vthp, as shown on the design from Kanago et al, is because the biasing DAC is connected to both output translators, instead of each output transistor having its own DAC. This was made to reduce power consumption by adding a second DAC, as the purpose of the second output is to use any potential wasted power from the first output. Because the same DAC is used for both outputs, both outputs should have a similar value, but the second output should be slightly lower, due to transistor potential properties, with how voltage at the source of a PMOS having to be greater than

the drain, which is exactly what is being observed here. The maximum voltage observed here is 3.68 V, for a biasing voltage of 3 V. This is slightly lower than the expected 1.15*Vthp+Vbias, which would lead to a voltage of about 3.85 V, but this equation is an estimate. Also, because of the limitations caused by the semiconductor manufacturing process, any voltage value greater than 3.6 V could destroy the transistors, so using a DAC voltage this high is impractical. This plot shows that this boost converter can have an output voltage range of at least 0.9-3.3 V, as shown on the spec.

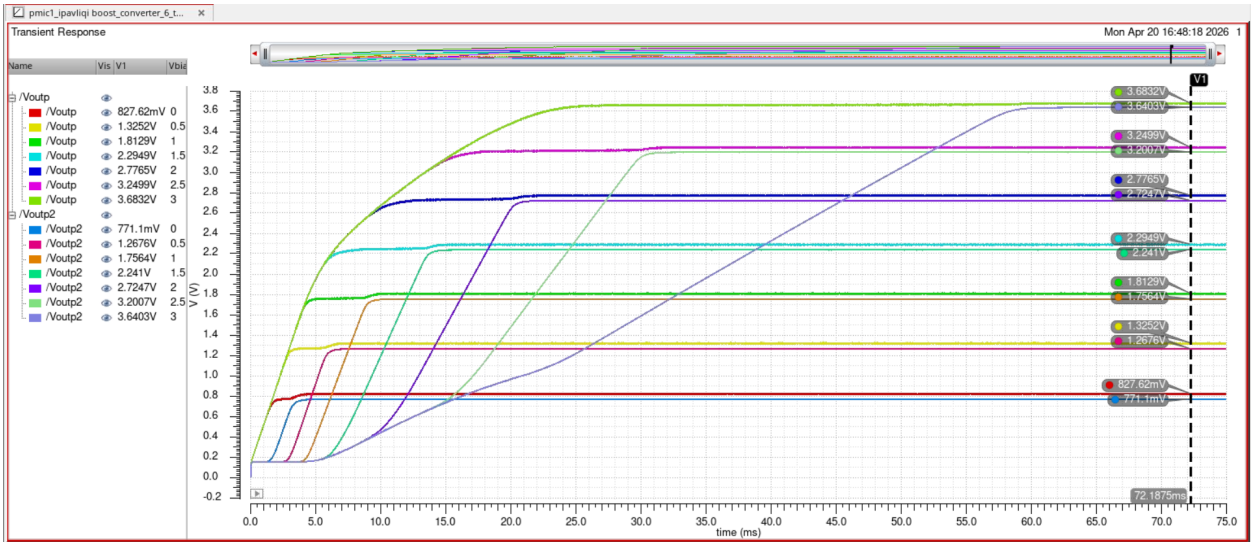

Figure 85: Parametric Sweep of the Biasing Voltage from 0-3 V, with Vin=1.5 V

Figure 86 shows a similar parametric sweep, but for Vin=2 V. Here, the two most noticeable things that have been affected are capacitor output charge time and output voltage. Because there is more voltage entering the circuit, the output capacitors charge faster, so the output reaches steady state faster. The output voltage here is slightly higher, by about 50 mV, which shows that input voltage does have an effect on output voltage, but relatively low, at least for higher input voltages.

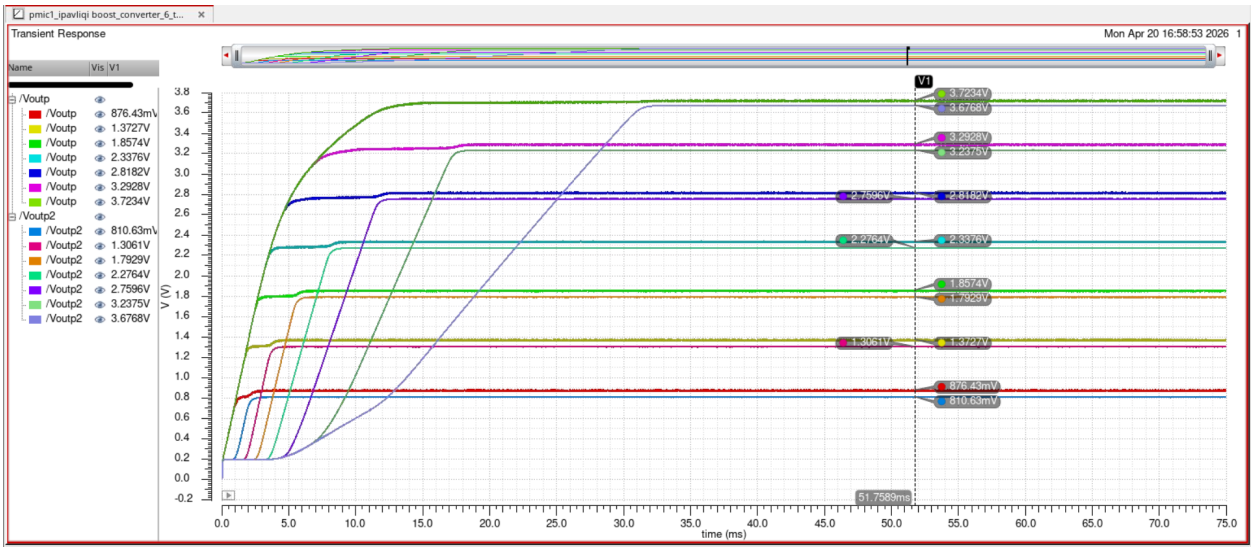

Figure 86: Parametric Sweep of the Biasing Voltage from 0-3 V, with Vin=2 V

There are some limitations with this topology. The first one is supply voltage dependency. Because this topology needs a voltage from a DAC to set the output, it is limited by what the DAC can output with a given supply voltage. From testing, the maximum optimal DAC voltage is around Vaa-0.3 V. Anything above this, and a combination of diminishing returns, along with improper circuit function were observed. With this, the absolute minimum Vaa where this boost converter can have an output voltage of 3.3 V is about 2.8 V, but that is not for all input voltages, and this is limited by another factor of the absolute maximum voltage that the boost converter. This maximum value is Vaa+0.5 V. From testing, this has been observed to be a hard cap to the output voltage. For example, a Vaa value of 3.3 V will have an absolute maximum value of 3.8 V, and will not get any higher than that, even if Vbias is changed to obtain an output voltage greater than it. This max output voltage is even more limited by process and temperature constraints. At the SS corner, at 70°C, the maximum output voltage is about Vaa+0.35 V, so the minimum value that Vaa must be for the boost converter to be able to reach the full output voltage spec is about 3 V.

Another limitation is the output voltage given a lower input voltage signal. As shown before, input voltage does have a slight effect on the output voltage, and this effect becomes more apparent as the input voltage becomes less than 1 V, and even more apparent when it goes below 500 mV. For the worst case, of Vin=0.2, the output was more than 300 mV lower than the expected output, and took a significantly longer amount of time to reach steady state. This behavior is expected, as having an output become 5 to 15 times greater than the input voltage is a limitation.

The other major limitation is power consumption and efficiency. From testing, the highest efficiency

that was observed is 53%, much less than the desired 90%, and the power consumption was about 1 mW, which is much greater than the desired 10 μW. This may be due to an inadequate testing setup, as it is difficult to perfectly simulate a realistic low power input voltage source. Due to this also being schematic testing, it does not show the physical layout limitations that come from testing this circuit. Because this is a schematic, this circuit can draw as much current as it needs to. However, in reality, the layout limits the amount of current that can flow through the circuit. Also, certain simulations, to properly observe the behavior of this circuit, would be very computationally heavy and impractical to perform.

Spec	**Target Output**	**Achieved Output**	**Spec Met**
Input Voltage Range	0.2-2.5 V	0.2-3 V	Yes
Output Voltage Range	0.9-3.3 V	0.85-3.6 V	Yes
Temperature Range	-10-70°C	-10-70°C	Yes
Supply Voltage Range	2.5-3.6 V	3-3.6 V	No
Power Consumption	>10 μW	1 mW	No
Efficiency	<90%	53%	No
Layout Size	N/A	99.905 μm X 81.75 μm	N/A

Table 9: Boost Converter Results

5.8.4 Buck-Boost Converter: Testing Overview

The testing apparatus of the boost converter includes connecting all of the off chip components of the boost converter block to the boost converter through wpi_IOANAN2P pads. The input voltage DC source is connected to a pad, and then enters the boost converter through the Vin pin, with the value being set as a parameter of Vin. The off chip inductor is connected between the Ind1 and Ind2 pins, with each side of the inductor connected to the pad. Here, a capacitor is connected in parallel before the inductor and connected to ground to prevent any voltage spikes The Vout and Vout2 ponds connect to pads that lead to off chip capacitors where the output voltage is stored. The two gates of the biasing transistors are set to a DC voltage source with the parameter of Vbias. To see the output voltages, the node between the output pads and the output capacitors are selected, and currents are measured similarly. The parameters that can be changed while testing are Vin, and Vbias. For more testing, changing the capacitor and inductor values also works. For testing capacitor values, it is recommended to not exceed 1 uF, as higher capacitor values lead to much longer simulation times. Using lower capacitor values is also not recommended, as lower capacitor values lead to higher ripple voltages, and simulation inaccuracies that come with that. The values

of capacitors that were used during testing were 1 uF, and the inductor value was 10 mH. This high inductor value is due to the relatively low clock frequency of 32 kHz, with a fixed duty cycle of 50%. This is due to the inverse relationship between frequency and necessary inductance value for a boost converter.

5.9 Current Sensor Implementation

The schematic of the current sensor is shown below in Figure 87. It follows the design presented in Design 10 of the Analog Devices AN-105 Application Note [18]. This design utilizes an external 4-terminal Kelvin sensing resistor for the best accuracy, which is currently modeled as an ideal resistor R_{sense} inside the schematic, which generates a voltage across it based on the current flowing through, which the instrumentation amplifier reads. The current-to-voltage output formula is given as $V_{out} = I_{load} \cdot R_{sense} \cdot (\frac{R_2}{R_1})$, and the design is currently configured to scale 0-1mA of current to 0-3.0V linearly, though this gain will be changed depending on the final current draw specifications of the chip once final integration is completed. Currently $R_1 = 10K\Omega$, $R_2 = 3M\Omega$, and $R_{sense} = 10\Omega$.

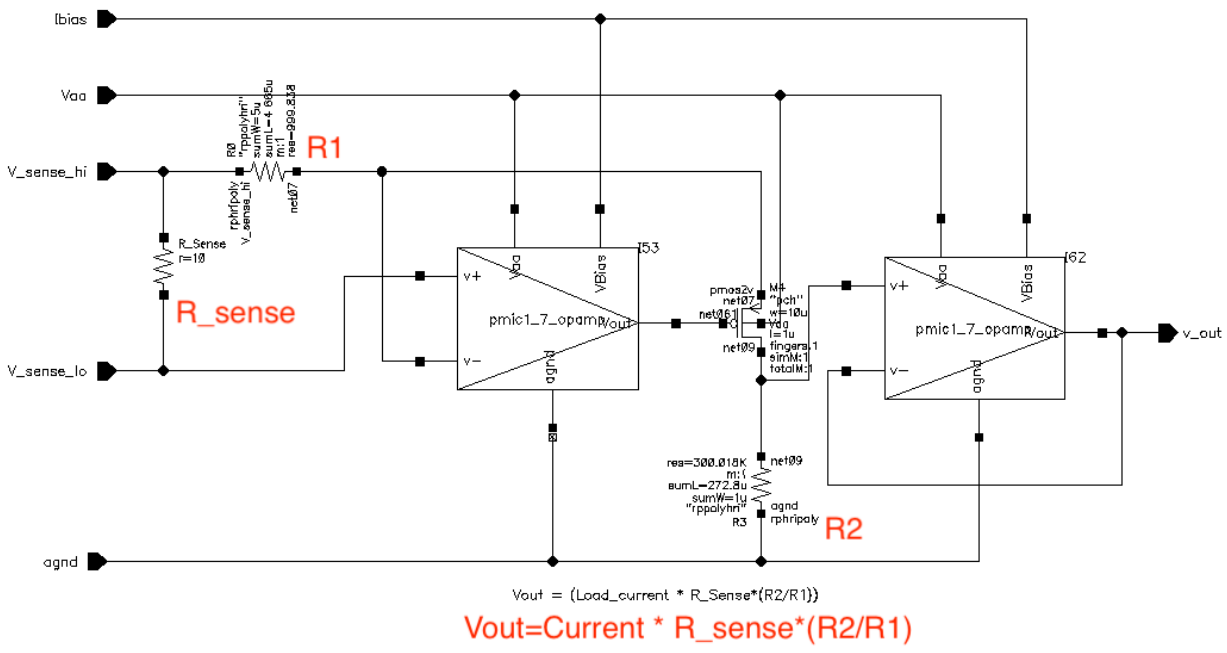

Figure 87: Current Sensor Schematic

In its current configuration, with a 1mA current through the 10 ohm sense resistor, the sense resistor dissipates 10µW of power, and the sensor circuit itself dissipates 15µW.

The simulation below in Figure 88 shows the scaling from the 0-1 mA current input to the 0-3 V voltage output. Due to the output swing limitations of the op-amp buffer. The minimum sensible current in this configuration is roughly 60µW.

The sensor achieves a full-gain bandwidth of 10 kHz and a slew rate of 1V/µS, ensuring quick enough response time to fluctuations in current.

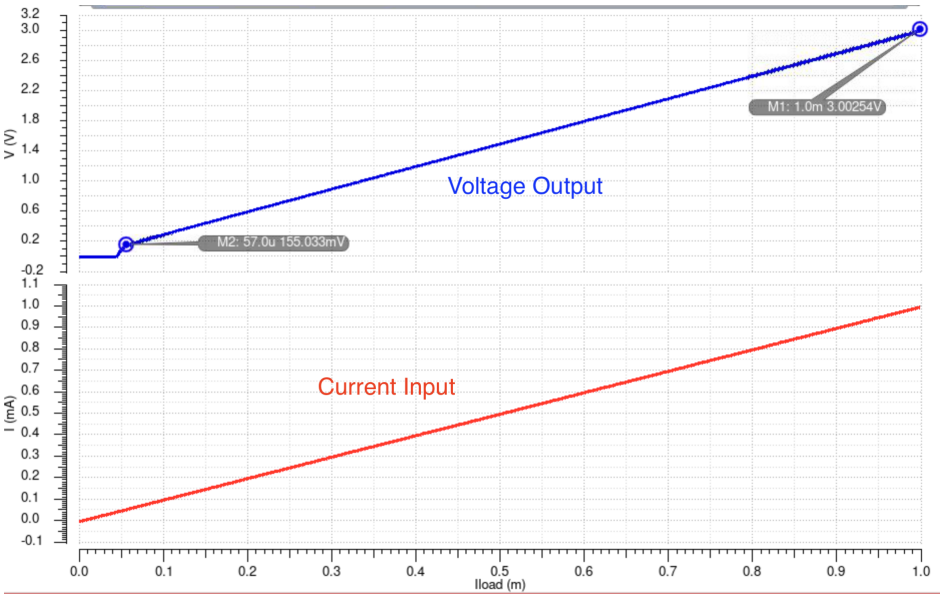

Figure 88: Current Sensor DC Sweep

Figure 89 shows the high-level implementation of the block. 4 pins will be required, 2 pins are placed very close to the sense resistor to instrument the voltage across, and two pins are used to carry current through the resistor.

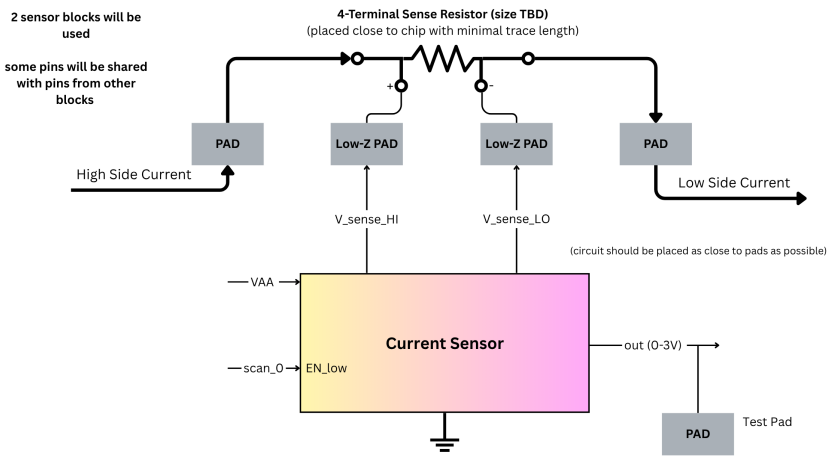

Figure 89: Current Sensor Block Diagram

5.10 Die Temperature Sensor Implementation

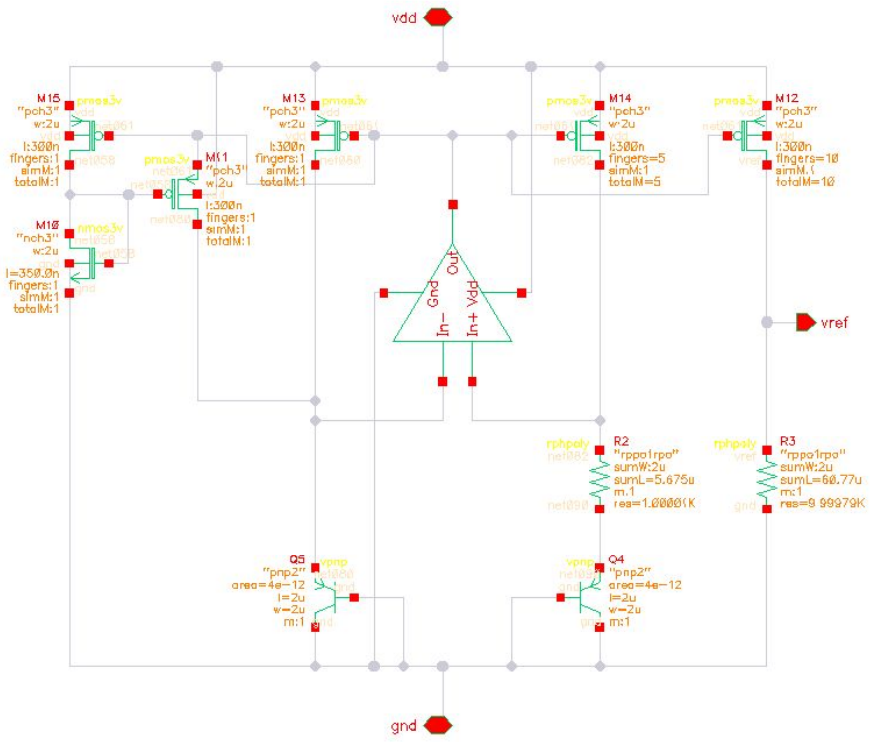

Figure 90: BJT PTAT Bandgap Reference

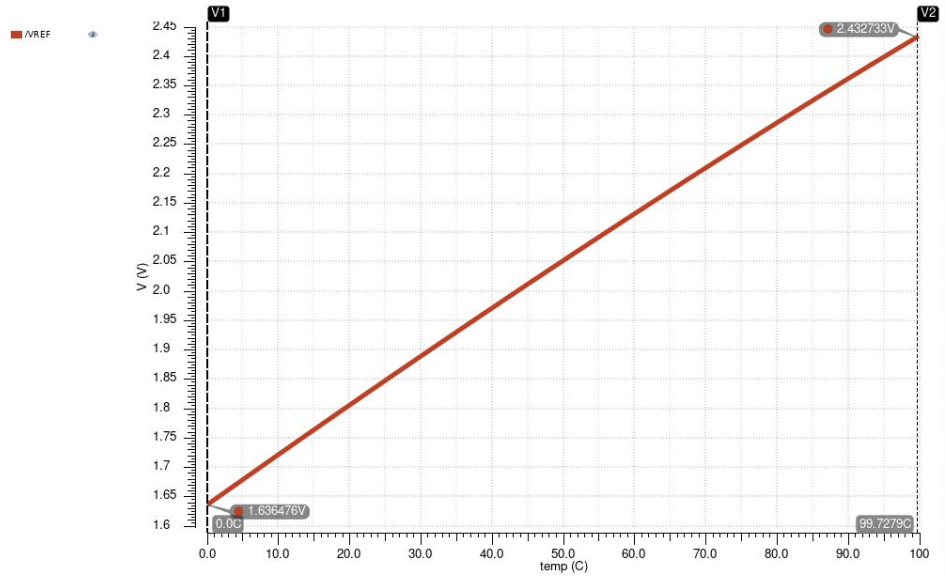

Figure 91: Temp Sensor DC Sweep

The temperature sensor shown is based on a proportional-to-absolute-temperature (PTAT) bandgap reference using BJTs. This architecture takes advantage of the temperature dependence of the base–emitter voltage of a BJT, which decreases with increasing temperature. By operating two BJTs at different current densities, a voltage difference is generated that is proportional to absolute temperature. This PTAT voltage is then amplified and used to produce an output voltage that varies with temperature. The implemented circuit produces an output ranging from approximately 1.73- 2.57V over the measured temperature range of 0-100°C. One advantage of this design is that the PTAT behavior provides a predictable and nearly linear temperature response, making it well suited for integrated temperature sensing. Additionally, bandgap-based circuits are relatively stable across supply variations, allowing the circuit to operate reliably over the specified supply range. However, the circuit consumes significantly more power than desired, reaching nearly 1.4mW, which exceeds the target of 100μW. The output voltage range is also narrower than the desired specification, limiting the available dynamic range. Despite these limitations, the design demonstrates a functional PTAT temperature sensor capable of operating over a wide temperature range.

Specification	**Desired**	**Achieved**
Output Voltage	0.1-3.3V	1.73-2.57V
Temperature Range	-10° - 70°	0° - 100°C
Power	$< 100\mu$W	989μW - 1.4mW
VAA Range	2.5-3.3V	Achieved

Table 10: The Temp Sensor Specs Comparison

6 Discussion

This chapter discusses the external factors and considerations that were referenced during CRIMSON's design.

6.1 Consideration of Public Health and Other Factors

6.1.1 Public Health

Although CRIMSON was not designed specifically as a healthcare device, its power regulation and monitoring capabilities could support future public health technologies. Many medical and wellness systems, such as wearable sensors or remote patient monitoring devices, require low power operation and dependable battery management. By improving efficiency and power reliability, CRIMSON could contribute to the long-term operation of health devices.

6.1.2 Public Safety

Public safety was not the direct focus of the design, but dependable power management can indirectly improve the safety of electronics used in remote or critical environments. CRIMSON includes monitoring and regulation features that help prevent under voltage conditions, unstable operation, and unnecessary power loss.

6.1.3 Public Welfare

CRIMSON was designed to improve the reliability and efficiency of low-power electronic systems, which can positively support public welfare in a variety of ways. Devices used for environmental sensing, infrastructure monitoring, and communications networks all depend on stable power delivery. By providing dependable regulation and monitoring, CRIMSON can help improve the performance and lifespan of systems that serve the public.

6.1.4 Global Considerations

Many regions of the world rely on low-power electronics in environments where access to stable electrical infrastructure may be limited. CRIMSON's support for battery systems, solar energy inputs, and efficient

regulation makes it suitable for remote or off-grid applications. These capabilities could assist global efforts in agriculture, environmental monitoring, or transportation.

6.1.5 Cultural Considerations

CRIMSON focuses primarily on technical power-management functions and does not directly involve culturally specific content, language, localization, or user behavior customization. However, its configurable architecture allows the chip to be adapted for many different system needs and priorities. This may support use in a wide range of communities and industries.

6.1.6 Social Considerations

Reliable electronics play an important role in communication, connectivity, and access to information. By improving power management for IoT and edge devices, CRIMSON can indirectly support technologies that help people remain connected in everyday life. Applications such as portable sensors, monitoring systems, and smart devices may improve convenience and accessibility.

6.1.7 Economic Considerations

Efficient power management can reduce operating costs by extending battery life, lowering maintenance requirements, and improving device longevity. CRIMSON's integrated architecture may also reduce the need for multiple separate components in certain systems, which could lower implementation costs. Additionally, CRIMSON helped prepare students for careers in the semiconductor industry, supporting long-term economic growth through workforce development.

6.2 Recognizing Ethical and Professional Responsibilities

6.2.1 Recognizing Ethical and Professional Responsibilities

The CRIMSON project demonstrates awareness of ethical and professional responsibilities through its emphasis on reliable power management, transparent system operation, and future usability. Because many low-power electronic systems depend on stable power delivery, careful design practices were used to improve regulation, monitoring accuracy, and configurability. The inclusion of testing features such as scan-chain access and block-level isolation also supports responsible verification, debugging, and long-term maintainability of the chip.

6.2.2 Consideration of the Impact of Engineering Solutions of this MQP in Global, Economic, Environmental, and Social Contexts

Global

CRIMSON was developed for IoT and edge-device applications that may be deployed worldwide. Its support for battery systems, solar inputs, and configurable operation allows the design to be adapted for use in regions with varying infrastructure and energy availability.

Economic

By integrating multiple power management functions into a single chip, CRIMSON may reduce the need for several separate components, lowering implementation cost and improving efficiency for future systems.

Environmental

The design emphasizes efficient regulation, battery management, and energy harvesting support. These features can help reduce wasted power and improve the useful life of battery-powered devices.

Social

Reliable electronics support communication, sensing, and access to information in modern society. CRIMSON can indirectly support technologies such as smart sensors, portable devices, and monitoring systems that improve convenience and connectivity.

6.3 Appropriate Incorporation of Engineering Standards

The CRIMSON project incorporates relevant engineering standards and professional design practices in several technical areas.

TSMC 180nm CMOS Process Design Rules

This chip was designed for fabrication using the TSMC 180nm CMOS 3v3/1v8 1P6M process. All schematics, layout, and verification work were completed while following foundry device models and layout design rules.

Cadence Verification Standards

Industry-standard verification methods were used through Cadence Virtuoso, including schematic simulation, layout verification, design rule checking (DRC), and layout-versus-schematic (LVS) comparison to ensure functional accuracy.

Digital Logic Design Practices

Logic simulations and digital schematic development were performed using AMD Vivado before implementation in Cadence. This reflects common professional workflows for mixed-signal IC design.

Power Management Design Methodologies

This project follows accepted power-integrated circuit design principles such as voltage regulation, battery charging control, monitoring architectures, and efficient power conversion for low-power systems.

Scan Chain and Testability Standards

CRIMSON includes 116 scan-chain control bits and extensive pin access for testing and characterization. Designing for testability is an important professional standard in semiconductor development that improves validation and future usability.

7 Conclusion

The goal of this Major Qualifiying Project was to provide hands-on experience in integrated circuit design through the development of a fully configurable power management integrated circuit capable of supporting modern low-power electronic systems. The team successfully designed, CRIMSON, the Conversion and Regulation Integrated Management SolutiON, as an intelligent PMIC intended for IoT and edge-device applications where reliable power delivery is needed. This project allowed the team to apply industry level analog and digital IC design practices while contributing a platform for future work at WPI.

On the hardware side, the team developed a chip architecture that integrates power conversion, voltage regulation, battery charging, monitoring circuits, digital control logic, and programmable interconnects into a single device. CRIMSON was designed to operate across a useful voltage range while supporting both on and off chip loads. The inclusion of boost conversion, solar management capability, and battery interfacing allows the chip to function in systems that have to harvest, store, and regulate energy in remote environments. Simulations and verification were completed using Cadence Virtuoso and AMD Vivado, demonstrating the feasibility of the overall design prior to fabrication.

A major contribution of the project was the Configurability Layer, which allows users of the chip to selectively enable, disable, and reroute functional blocks through scan-chain controlled multiplexing. This feature gives CRIMSON flexibility for several different applications or for future student testing. Along with this, the monitoring allows the chip to measure internal operating conditions without relying entirely on an external microcontroller. This adds value for edge systems with limited computational resources.

Beyond the technical design itself, the project also strengthens the long term capabilities of the WPI CHIPS Design Studio through expanding its portfolio of reusable circuit blocks and student developed silicon systems. CRIMSON demonstrates that undergraduate engineering teams can successful complete sophisticated mixed-signal IC projects while navigating real fabrication constraints and professional design flows. The experience gained through this effort prepares future students for careers in semiconductor design and other related fields.

Future work includes tapeout submission, fabrication, and post-silicon testing of each of the subsystems. Additional improvements may include expanded monitoring functions, higher efficiency conversion stages and new configurabiliy control modes for more specialized applications. Overall, the CRIMSON project successfully met its objects by creating a practical, intelligent PMIC while providing valuable educational experience in advanced chip design, system integration and collaborative engineering.

Appendices

A Authorized Personnel Acknowledgment Forms

Security C - TSMC Secret

Contract No: Security C - TSMC Secret

APPENDIX A

Authorized Personnel Acknowledgement Form

To the Undersigned:

As part of your engagement with Worcester Polytechnic Institute ("University"), you may have access to highly confidential information of TSMC, including, for example, 65nn, 180nm and 0.35nm technology nodes' PDKs, SPICE models, RC tech files, and front end views of the following: standard cell, I/O, memory, eFuse, and ESD libraries that you learn from TSMC's or Muse Semiconductor LLC ("Company A")'s employees, contractors, or agents ("***TSMC Confidential Information***"). TSMC, Company A and University are parties to a Nondisclosure Agreement dated March 19, 2018, as fully amended ("***NDA***") that govern the disclosure and use of TSMC Confidential Information.

You agree that during and after your engagement with University, you will hold the TSMC Confidential Information in strictest confidence and only use the TSMC Confidential Information in accordance with the NDA. Upon completion of your services, you will immediately return to Company A, or upon TSMC's request, destroy all tangible TSMC Confidential Information, including but not limited to all electronic files, documentation, notes, plans, and drawings.

You understand that unauthorized disclosure or use of TSMC Confidential Information during or after your engagement with University could cause irreparable harm and significant injury to TSMC. **You agree that in addition to any disciplinary or legal actions that Company A may seek against you, TSMC may seek and obtain immediate legal remedies against you, including claims for damages and criminal liabilities where applicable.**

Please acknowledge your agreement with the above terms by signing below.

Taiwan Semiconductor Manufacturing Company, Ltd.

Acknowledged and agreed.

Signature: _____

Name: Leo Borsari

Title: Student

University: Worcester Polytechnic Institute

Date: 8/26/2025

University email: lfborsari@wpi.edu

1

Figure 92: Authorized personnel acknowledgment form signed by Leo Borsari

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Please acknowledge your agreement with the above terms by signing below.

Taiwan Semiconductor Manufacturing Company, Ltd.

Acknowledged and agreed.

Signature: Diego Canovas

Name: Diego Canovas

Title: Student

University: Worcester Polytechnic Institute

Date: 8/29/2025

University email: dcanovas@wpi.edu

1

Figure 93: Authorized personnel acknowledgment form signed by Diego Canovas

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You understand that unauthorized disclosure or use of TSMC Confidential Information during or after your engagement with University could cause irreparable harm and significant injury to TSMC. **You agree that in addition to any disciplinary or legal actions that Company A may seek against you, TSMC may seek and obtain immediate legal remedies against you, including claims for damages and criminal liabilities where applicable.**

Please acknowledge your agreement with the above terms by signing below.

Taiwan Semiconductor Manufacturing Company, Ltd.

Acknowledged and agreed.

Signature: Peter Danise

Name: Peter Danise

Title: Student

University: Worcester Polytechnic Institute

Date: 08/26/2025

University email: pjdanise@wpi.edu

1

Figure 94: Authorized personnel acknowledgment form signed by Peter Danise

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You agree that during and after your engagement with University, you will hold the TSMC Confidential Information in strictest confidence and only use the TSMC Confidential Information in accordance with the NDA. Upon completion of your services, you will immediately return to Company A, or upon TSMC's request, destroy all tangible TSMC Confidential Information, including but not limited to all electronic files, documentation, notes, plans, and drawings.

You understand that unauthorized disclosure or use of TSMC Confidential Information during or after your engagement with University could cause irreparable harm and significant injury to TSMC. **You agree that in addition to any disciplinary or legal actions that Company A may seek against you, TSMC may seek and obtain immediate legal remedies against you, including claims for damages and criminal liabilities where applicable.**

Please acknowledge your agreement with the above terms by signing below.

Taiwan Semiconductor Manufacturing Company, Ltd.

Acknowledged and agreed.

Signature: Dennis Garvey

Name: Dennis Garvey

Title: Student

University: Worcester Polytechnic Institute

Date: 8/22/25

University email: dlgarvey@wpi.edu

1

Figure 95: Authorized personnel acknowledgment form signed by Dennis Garvey

Contract No: Security C - TSMC Secret

APPENDIX A

Authorized Personnel Acknowledgement Form

To the Undersigned:

You understand that unauthorized disclosure

As part of your engagement with Worcester Polytechnic Institute ("University"), you may have access to highly confidential information of TSMC, including, for example, 65nn, 180nm and 0.35nm technology nodes' PDKs, SPICE models, RC tech files, and front end views of the following: standard cell, I/O, memory, eFuse, and ESD libraries that you learn from TSMC's or Muse Semiconductor LLC ("Company A")'s employees, contractors, or agents ("***TSMC Confidential Information***"). TSMC, Company A and University are parties to a Nondisclosure Agreement dated March 19, 2018, as fully amended ("***NDA***") that govern the disclosure and use of TSMC Confidential Information.

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Please acknowledge your agreement with the above terms by signing below.

Taiwan Semiconductor Manufacturing Company, Ltd.

Acknowledged and agreed.

Signature: Ilia Pavliqi

Name: Ilia Pavliqi

Title: Student

University: Worcester Polytechnic Institute

Date: 8/27/25

University email: ipavliqi@wpi.edu

Figure 96: Authorized personnel acknowledgment form signed by Ilia Pavliqi

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Contract No: Security C - TSMC Secret

APPENDIX A

Authorized Personnel Acknowledgement Form

To the Undersigned:

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Please acknowledge your agreement with the above terms by signing below.

Taiwan Semiconductor Manufacturing Company, Ltd.

Acknowledged and agreed.

Signature: Luke Ramsay

Name: Luke Ramsay

Title: Student

University: Worcester Polytechnic Institute

Date: 8/25/2025

University email: LGRAMSAY@WPI.EDU

Figure 97: Authorized personnel acknowledgment form signed by Luke Ramsay

Contract No: Security C - TSMC Secret

APPENDIX A

Authorized Personnel Acknowledgement Form

To the Undersigned:

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Please acknowledge your agreement with the above terms by signing below.

Taiwan Semiconductor Manufacturing Company, Ltd.

Acknowledged and agreed.

Signature: Luca Tropea (Aug 20, 2025 12:58:25 EDT)

Name: Luca Tropea

Title: Student

University: Worcester Polytechnic Institute

Date: Aug 20, 2025

University email: lntropea@wpi.edu

1

Figure 98: Authorized personnel acknowledgment form signed by Luca Tropea

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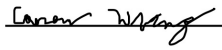
Please acknowledge your agreement with the above terms by signing below.

Taiwan Semiconductor Manufacturing Company, Ltd.

Acknowledged and agreed.

Signature: 

Name: Cameron Wilding

Title: Student

University: Worcester Polytechnic Institute

Date: 08/25/2025

University email: cwilding@wpi.edu

Figure 99: Authorized personnel acknowledgment form signed by Cameron Wilding

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